Geological Report on Preliminary Exploration (G3) for Bauxite in Madua Pat – I Block, District – Lohardaga, State – Jharkhand, India.

SALIENT FEATURES, TEXT & ANNEXURES



Prepared For

National Mineral Exploration Trust (NMET)

Ministry of Mines
Room No. 327, A-Wing
Shastri Bhawan,
Dr. Rajendra Prasad Road
New Delhi- 110001

Prepared by

Regional Institute – III,
Central Mine Planning & Design Institute Limited
(A Subsidiary of Coal India Limited / Govt. of India PSU),
Gondwana Place, Kanke Road, Ranchi - 834 031, Jharkhand.

August, 2023

GEOLOGICAL REPORT ON PRELIMINARY EXPLORATION (G–3 LEVEL) FOR BAUXITE IN MADUA PAT-I BLOCK DISTRICT- LOHARDAGA, STATE – JHARKHAND.

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PART–IV A OF SCHEDULE ANNEXED TO THE MINERALS

(EVIDENCE OF MINERAL CONTENTS) RULES, 2015

REPORTING OF MINERAL RESOURCES.

<table>
<tr><th>Contents</th><th>Explanation</th></tr>
<tr><td>Title & Ownership</td><td>
Title of Report: GEOLOGICAL REPORT ON PRELIMINARY EXPLORATION (G-3 LEVEL) FOR BAUXITE IN MADUA PAT-I BLOCK, DIST: LOHARDAGA JHARKHAND.

Ownership: Directorate of Geology and Mining, Govt. of Jharkhand.

Name of Prospector: Central Mine Planning & Design Institute Limited, CMPDI (A Mini Ratna Company), A Subsidiary of Coal India Limited.

Address of Prospector: Gondwana Place, Kanke Road, Ranchi – 834008, State - Jharkhand.

E-mail of Prospector: cmd.cmpdi@coalindia.in ; gmexpl.cmpdi@coalindia.in

Telephone numbers of Prospector: +91651-2232637, +91-89877 89059 (CUG).

Period of prospecting: March -2022 to July – 2022.

Exploration agency: Central Mine Planning and Designing Institute Limited. (A Subsidiary of Coal India Limited, Kolkata).

Qualification: M. Sc. (Geology), M.Sc. Tech (Geology), M. Tech. (Geology),Doctorate in Geology.

Experience: Year 1975 (Since Inception).
</td></tr>
<tr><td></td><td>Village(s) : Madua Pat.</td></tr>
<tr><td></td><td>District : Lohardaga, State- Jharkhand.</td></tr>
<tr><td></td><td>Survey of India Toposheet No : 73 A/10.</td></tr>
<tr><td></td><td>Forest : Partly.

Mineral under investigation: Bauxite.</td></tr>
<tr><td></td><td>
<table>
<tr><th colspan="3">BOUNDING CO-ORDINATES OF BLOCK</th></tr>
<tr><th>POINT</th><th>NORTHING</th><th>EASTING</th></tr>
<tr><td>A</td><td>2610786.617</td><td>258810.537</td></tr>
<tr><td>B</td><td>2611769.969</td><td>257506.474</td></tr>
<tr><td>C</td><td>2610570.943</td><td>258465.214</td></tr>
<tr><td>D</td><td>2610987.141</td><td>259131.602</td></tr>
<tr><td>E</td><td>2612553.658</td><td>258189.146</td></tr>
</table>
</td></tr>
<tr><td>Infrastructure &</td><td>The Maduapat-I block is located at a distance of 16 kms from Kisko block office. Kisko is</td></tr>
</table>

<table>
<tr><th>Contents</th><th>Explanation</th></tr>
<tr><td>Environment</td><td>connected to Lohardaga district HQ with metal road of about 10 kms. The nearest railway station is Lohardaga, located at a distance of 26 Kms. from the block. The distance from Lohardaga town to Ranchi the state capital is 75 km, connected by an all-weather metalled road. Thus Maduapat-I Block is located at a distance of about 101 km from Ranchi.
• Historical site: Nil
• Forest: Little.
• Sanctuaries: No
• National Park: No
The district experience warm humid climate with three well defined seasons i.e. summer, winter and Monsoon. The winters commence from middle November and extend up to middle of March with December being the coldest month. During winter the temperature goes down to 4˚C. Summer starts from middle of March and continues up to middle of June, when the temperature shoots up to 42˚C.</td></tr>
<tr><td>Previous Exploration</td><td>The prospecting in Madua Pat-I block by the GSI, Jharkhand during 1963 revealed that principal rock formation of this area comprises what has been earlier designated as the Chotanagpur granite gneiss comprising the various types of granite gneiss, porphyritic granite, granulitic rocks, quartz veins and pegmatites, which is a composite rock type. Laterite and bauxite in this area rest on the older Chotanagpur granite gneiss rock. Bauxite occurs as pockets of various dimensions and is Massive, compact and bouldery in nature.</td></tr>
<tr><td>Geology of the Block</td><td>Bauxite occurs as segregated sheets and lenses within the laterite capping over the Archean gneisses. The gneisses contain inclusions of calc-silicate rocks, amphibolites, quartzite of varying dimensions aligned in general parallel to the foliation. The laterite crust is underlain by variegated clay containing void of white clay. The zone between the highly colored clay and the gneisses is covered. The clays are soft and moderately plastic.
<table>
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<tr><td>Recent</td><td></td><td>Alluvium, conglomerate and carbonaceous shale</td></tr>
</table></td></tr>
</table>

Contents	Explanation		
	Tertiary to Recent		Laterite, Bauxite and lithomarge
	Cretaceous	Lameta	Limestone and basalt ferruginous grit
	Proterozoic		Quartz vein
	Archean to Proterozoic	Chhotanagpur Granite Gneissic complex	Diorite, granite, granite gneiss
	Archean to Paleoproterozoic	Unclassified Metamorphics	Limestone, calc-silicate rock, quartzite,phyllite, slate, mica schist, hornblende schist, amphibolite, talc schist

The principal rock formation of this area comprises what has been earlier designated as the Chotanagpur granite gneiss, which is a composite rock type. Laterite and bauxite in this area rest on the older Chotanagpur granite gneiss comprising the various types of granite gneiss, porphyritic granite, granulitic rocks, quartz veins and pegmatites. The major part of the blocks is covered by lateritic soil/laterite and sometime with morrum except the scarp and edges of the plateau where bauxite and aluminous/ferruginous laterite are exposed. Laterite along with Bauxite occurs at plateau top in the block. Laterite overlies lithomargic clay which in turn unconformably overlies Chotanagpur granite. The bauxite/laterite profile is outcome of prolonged weathering of the granitic rocks. Bauxite occurs as segregations and patches/pockets/lenses with various dimensions within laterite capping over the plateau. Bauxite is pisolitic to massive in nature and having various colours like pale yellow, cream to pink.

In Madua Pat-I block, bauxite zones have been delineated as desired by DGM, Jharkhand. as per End Use Specification of IBM and resources are Categorized, based on polygonal method as below:

A total of 1.397 million tonnes net in-situ bauxite resource with an average grade of 33.46% Al_2O_3, 5.62% SiO_2, 32.74% Fe_2O_3, 8.01% TiO_2 & 19.53% LOI have been estimated in the block in an area of 0.38 sq. km under the cut-off value of 7% SiO_2. A total of 1.814 million tonnes net in-situ bauxite resource with an average grade of 33.14% Al_2O_3, 6.33% SiO_2, 32.59% Fe_2O_3, 7.90% TiO_2 & 19.41% LOI have been estimated in the block in an area of 0.38 sq. km under the cut-off value of 10% SiO_2.

A total of 11.666 million tonnes net in-situ aluminous Laterite resource with an average grade of 34.46% Al_2O_3, 18.82% SiO_2 have been estimated in the block in an area of 1.33 sq. km.

Ore resource is classified as “Inferred Category” as per the United Nations Frame Work

Contents	Explanation
	Classification (UNFC) and assigned code 333.
Technological Investigation	During the present exploration work (G-3 level), 107.00 m has been drilled in 5 boreholes up to a vertical depth of about 20.00 m to establish resource as per UNFC norms. All the boreholes were drilled (G-3 level) and analyzed. As desired by DGM (Jharkhand), the data of 05 boreholes drilled by CMPDI within this block boundary has been considered for entire assessment.
Location of data points	The Co-ordinates and Reduced level (in meter) of boreholes have been given using DGPS and Total Station instrument and topographical map prepared at 2 m contour interval on 1:2000 scale. Topographic contouring has been done at 2 m interval and plotted on map of on 1:2000 scale. The Topographical Survey has been carried out by fixing new triangulation stations by DGPS instrument as per requirement. On the basis of survey stations, the block was taken into control by re-fixing their co-ordinates and reduced level using DGPS and total station instrument.
SamplingTechnique	A total of 10 Nos of surface/outcrop samples have been collected during G-3 level randomly. Samples have been subjected to chemical analysis for radical results are furnished as annexure IIIA. The drill core samples of the bauxite were split into two equal halves by using core splitter for preparation of primary samples. One half was grounded to fine powder and the other half was kept for further studies. The powdered sample was passed through (-) 120 mesh size sieve. By mixing, coning and quartering of the sieved powder desired representative samples were drawn. Out of these representative samples, 100 gm samples were sent to Chemical Laboratory for primary analysis and remaining samples were preserved as duplicate samples. A total of 49 primary core samples have been prepared and sent to lab for analysis of 5 radicals i.e. Al_2O_3, Fe_2O_3, SiO_2, TiO_2, & LOI. These are the basic and most essential chemical constituents to specify the grade of bauxite. In addition to 49 primary samples 3 numbers primary samples (around 5% of primary samples) have been analyzed for Al_2O_3, Fe_2O_3, SiO_2, TiO_2, & LOI as internal and external check samples to check the sampling bias.
Drilling technique & drill sampling employed.	The entire core drilling has been done by wire line method in NQ borehole size using diamond bits & TC bits. Core recovery of more than 90% has been maintained in bauxite using both dry drilling. Details of boreholes drilled by CMPDI in Madua Pat-I Block at G-3 level are provided in

Contents	Explanation
	Geological Report as Annexure-IA. The entire core recovered by drilling of CMPDI boreholes was logged systematically in detail describing different lithological units. The details of lithology, its grain size, colour, structural features etc. were recorded. All the cores were kept and preserved properly in the core boxes following 'Book pattern".
Sub-sampling techniques and sample preparation	**Composite Sample Analysis:** After receiving the chemical analysis for primary samples of the boreholes, the thickness of bauxite zone was delineated for each of the bauxite horizons intersected in the boreholes at 30% Al_2O_3 & 7% SiO_2, 30% Al_2O_3 & 10% SiO_2, 20% Al_2O_3, composite samples were prepared by combining all the constituent primary samples of the zone in proportion of their lengths they represent. A total of 4 composite samples have been analyzed for 14 radicals i.e. Al_2O_3, SiO_2, Fe_2O_3,TiO_2, CaO, MgO, SO_3, P_2O_5, V_2O_5, MnO, K_2O, Na_2O, LOI & Organic carbon. A total of 10 nos of composite samples were analyzed at G-3 level of exploration for 7 radicals i.e. Al_2O_3, SiO_2, Fe_2O_3, TiO_2, P_2O_5, & LOI. The details of analytical results of 7 & 14 radicals of composite samples for G-3 level are given as Annexure-III/B and III/C. **THA, MHA and Reactive Silica studies:** The THA and MHA alumina and reactive silica studies on 4 composite samples (G-3) have been carried out in Chemical Laboratory. The samples are selected in such a manner that the entire bauxite bearing areas are represented by them. **X-Ray Diffraction Studies:** 4 nos. of composite samples from selected zones of the boreholes were subjected to X-ray diffraction to determine the individual composition and to identify major and minor constituents in the Bauxite profile. **Petrographic and Mineragraphic Studies:** A total of 4 Nos. of specimen from outcrop were subjected to petrographic studies. **REE Studies:** Rare Earth Element studies on 19 no. primary clay samples from all the 5 boreholes have been carried out using ICP-MS methods. The elemental ranges of REE in 19 no. of primary clay samples in respect La, Ce and Nd are i. La ranges from 1.85 ppm to 20.74 ppm, ii. Ce ranges from 5.46 ppm to 73.61 ppm and iii. Nd ranges from 1.09 ppm to 30.35 ppm. Other elements such as Eu, Gd and Dy are ranges from 0.07 ppm to 2.18 ppm, 0.35 to 4.63 ppm and 0.34 to 3.69 ppm respectively. Twenty (20) nos of samples from 1 entire borehole have been studied for 33 suite of elements (REE, Trace Elements) ICP-AES/ICPMS. The detailed results are tabulated in annexure – VI/B. Among trace elements **Ga** ranges from 33ppm to 96ppm, **Zr** ranges from 397ppm to 851ppm. The concentration of Li is very low and <5ppm. Among REE, Ce ranges from 21.50ppm

Contents	Explanation
	to 155.25 ppm, La ranges from 55.01ppm to 103.75ppm. Length of primary sample was kept at 0.50-1.5 m during sampling of CMPDI borehole cores and it varies according to variation in Lithology and core
Specific Gravity	- 5 nos. Bauxite samples were subjected to specific gravity determination. The average specific gravity has been calculated as 2.33 and considered for calculation of resources. Details of specific gravity of bauxite sample are provided in Geological Report as Annexure- VIII/A.
Resource Estimation technique	- The resources and grade have been estimated by polygonal method. - The thickness and grade of the bauxite zone at a given sample point was considered to hold good, up to the half distance between the point and the adjacent ones. This method, utilizes the 'Rule of the nearest points'. Each sample point has an area of influence in the form of rectangle, or a square or any other geometrical shape depending on the pattern of grid used for exploration. In this way, the area of influence of a given point was delineated on the plan (Drawing no. VI). In case of peripheral borehole, the plateau boundary has been considered as influence area. The volume of the ore available in that area was arrived by the formula. ***R= P_A X Th X Sp. Gravity*** Where, **P_A** = Area of Polygon. **R**= Resource/ Tonnage. **Th**= Thickness of Bauxite. **Sp. Gravity**= Specific Gravity. The tonnage has been calculated by multiplying volume with average specific gravity of 2.33. All resources have been categorized under Inferred (333) category as per UNFC classification. Resource have been estimated as desired by DGM (Jharkhand) on the basis of **IBM's End Use Grade Classification**.
Resource Estimation	A total of **1.397 million tonnes net in-situ bauxite resource** with an average grade of 33.46% Al_2O_3, 5.62% SiO_2, 32.74% Fe_2O_3, 8.01% TiO_2 & 19.53% LOI have been estimated in the block in an area of 0.38 sq. km under the cut-off value of 7% SiO_2**.** A total of **1.814 million tonnes net in-situ bauxite resource** with an average grade of 33.14% Al_2O_3, 6.33% SiO_2, 32.59% Fe_2O_3, 7.90% TiO_2 & 19.41% LOI have been estimated in the block in an area of 0.38 sq. km under the cut-off value of 10% SiO_2. A total of **11.666 million tonnes net in-situ aluminous Laterite resource** with an

Contents	Explanation
	average grade of 34.56% Al_2O_3, 18.15% SiO_2 have been estimated in the block in an area of 1.33 sq. km**.** Measurement of ore resources based on TiO_2 content has been taken up (≥5% TiO_2) within potential 'Bauxite zone' as well as 'Aluminous Laterite zone'. The resources remains same as of Bauxite in the 'Bauxite zone' whereas, in Aluminous Laterite zone a total of **9.551 million tonnes** net in-situ ore containing **TiO2 ≥5%** containing 35.47% Al_2O_3, 15.48% SiO_2, 24.10% Fe_2O_3, 7.04% TiO_2 & 17.22% LOI have been estimated in the block and belongs to "Inferred Category" (333).

GEOLOGICAL REPORT ON PRELIMINARY EXPLORATION (G–3 LEVEL) FOR BAUXITE IN MADUA PAT – I BLOCK, DISTRICT – LOHARDAGA, STATE – JHARKHAND.

SALIENT FEATURES

Particulars	Details
Name of the block	Madua Pat - I
Tehsil	Kisko.
District	Lohardaga
Mineral	Bauxite
Area	1.50 sq. km.
Elevation	Varies from 1015.364 meter to 998.00 meter.
Period of Exploration	March -2022 to July - 2022
Meterage drilled by CMPDI	107.00 meter in 5 boreholes.
Thickness of Bauxite zone	

BH No.	Depth Details (m)		Thickness (m)	Lithology	Cut-off Grade
	From	To			
CMMD-02	11.50	12.50	1.00	Siliceous Bauxite	Max 10% SiO_2, >=30% Al_2O_3
CMMD-02	12.50	13.00	0.50	Bauxite	
CMMD-02	13.00	13.50	0.50	Fe Bauxite	
CMMD-02	13.50	14.00	0.50	Fe Bauxite	
CMMD-02	14.00	15.00	1.00	Bauxite	
CMMD-02	15.00	15.50	0.50	Bauxite	
CMMD-03	18.00	19.30	1.30	Bauxite	

BH No.	Depth Details (m)		Thickness (m)	Lithology	Cut-off Grade
	From	To			
CMMD-02	12.50	13.00	0.50	Bauxite	Max 7% SiO_2, >=30% Al_2O_3
CMMD-02	13.00	13.50	0.50	Fe Bauxite	
CMMD-02	13.50	14.00	0.50	Fe Bauxite	
CMMD-02	14.00	15.00	1.00	Bauxite	
CMMD-02	15.00	15.50	0.50	Bauxite	
CMMD-03	18.00	19.30	1.30	Bauxite	

Resources and Avg. Grade**:**

STATEMENT SHOWING RESOURCES OF BAUXITE ZONES AT VARIOUS CUT- OFF AS END USE OF BAUXITE RECOMMENDED BY IBM, MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

Cut off Value – Max 7% SiO_2, >=30% Al_2O_3.

Specific Gravity (gm/cc)	Reserves (Tonne)	Reserves (million tonne)	Net Reserve (million tonne)	Weighted SiO_2%	Weighted Al_2O_3%	Weighted Fe_2O_3%	Weighted TiO_2%	Weighted LOI%
	Gross Reserve	Gross Reserve						
2.33	2075165.739	2.075	1.245	5.67	33.20	32.88	8.04	19.58
2.33	243727.369	0.244	0.146	5.25	35.69	31.55	7.71	19.13
		2.328	**1.397**	**5.62**	**33.46**	**32.74**	**8.01**	**19.53**

Cut off Value – Max 10% SiO_2, >=30% Al_2O_3.

Specific Gravity (gm/cc)	Reserves (Tonne)	Reserves (million tonne)	Net Reserve (million tonne)	Weighted SiO_2%	Weighted Al_2O_3%	Weighted Fe_2O_3%	Weighted TiO_2%	Weighted LOI%
	Gross Reserve	Gross Reserve						
2.33	2766887.652	2.767	1.660	6.42	32.91	32.68	7.91	19.44
2.33	243727.3694	0.244	0.146	5.25	35.69	31.55	7.71	19.13
		3.023	**1.814**	**6.33**	**33.14**	**32.59**	**7.90**	**19.41**

UNFC Category 333 (Inferred Category Resources).

Report Submission August, 2023.

STATEMENT SHOWING RESOURCES OF ALUMINOUS BAUXITE ZONES CUT- OFF AS END USE OF LATERITE RECOMMENDED BY IBM, MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

BH No (Polygon No)	Area (in Sq. meter)	From (meter)	To (meter)	Thickness (meter)	Specific Gravity (gm/cc)	Reserves (Tonne)	Reserves (million tonne)	Net Reserve (million tonne)	Weighted SiO_2%	Weighted Al_2O_3%	Weighted Fe_2O_3%	Weighted TiO_2%	Weighted LOI%
						Gross Reserve	Gross Reserve						
CMMD-01	313418.2361	1.00	4.00	3.00	2.33	2190793.47	2.1908	1.3145	20.25	35.97	19.92	4.82	18.29
CMMD-01	313418.2361	8.50	9.00	0.50	2.33	365132.2451	0.3651	0.2191	8.9	35.95	31.44	7.02	16.02
Sub Total in CMMD-01 Polygon							**2.5559**	**1.5336**	**18.63**	**35.97**	**21.57**	**5.13**	**17.97**
CMMD-02	298155.997	2.50	3.00	0.50	2.33	347351.7365	0.3474	0.2084	28.03	28.77	26.56	2.20	13.73
CMMD-02	298155.997	9.50	11.50	2.00	2.33	1389406.946	1.3894	0.8336	11.635	31.24	34.87	5.45	16.09
CMMD-02	298155.997	15.50	16.50	1.00	2.33	694703.473	0.6947	0.4168	9.68	34.42	30.09	6.15	19.03
Sub Total in CMMD-02 Polygon							**2.4315**	**1.4589**	**13.42**	**31.80**	**32.32**	**10.46**	**16.59**
CMMD-03	80464.6317	3.50	10.00	6.50	2.33	1218636.847	1.2186	0.7312	23.44	34.59	20.19	4.62	16.37
Sub Total in CMMD-03 Polygon							**1.2186**	**0.7312**	**23.44**	**34.59**	**20.19**	**4.62**	**16.37**
CMMD-04	371560.0251	4.00	10.40	6.40	2.33	5540703.094	5.5407	3.3244	20.03	36.02	20.26	7.13	15.86
Sub Total in CMMD-04 Polygon							**5.5407**	**3.3244**	**20.03**	**36.02**	**20.26**	**7.13**	**15.86**
CMMD-05	272993	0.30	3.00	2.70	2.33	1717398.963	1.7174	1.0304	31.00	29.36	22.80	2.68	13.35
CMMD-05	272993	3.60	5.40	1.80	2.33	1144932.642	1.1449	0.6870	19.38	32.92	26.38	5.51	15.17
CMMD-05	272993	6.10	10.30	4.20	2.33	2671509.498	2.6715	1.6029	14.35	37.16	23.60	7.10	17.11
CMMD-05	272993	12.70	16.10	3.40	2.33	2162650.546	2.1627	1.2976	8.97	34.03	28.43	7.65	20.19
Sub Total in CMMD-05 Polygon							**7.6965**	**4.6179**	**17.30**	**33.91**	**22.92**	**6.03**	**16.85**
							19.4432	**11.6659 (~11.666)**	**18.15**	**34.56**	**22.99**	**6.69**	**16.65**

UNFC Category 333 (Inferred Category Resources).

Report Submission August, 2023.

STATEMENT SHOWING ORE RESOURCES CONTAINING TiO_2 CONTENT ≥5% IN ALUMINOUS LATERITE ZONES, MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

BH No (Polygon No)	Area (in Sq. meter)	From (meter)	To (meter)	Thickness (meter)	Specific Gravity (gm/cc)	Reserves (Tonne)	Reserves (million tonne)	Net Reserve (million tonne)	Weighted SiO2%	Weighted Al2O3%	Weighted Fe2O3%	Weighted TiO2%	Weighted LOI%
						Gross Reserve	Gross Reserve						
CMMD-01	313418.24	2.00	3.00	1.00	2.33	730264.4901	0.7303	0.4382	15.49	38.01	20.40	5.40	20.02
CMMD-01	313418.24	8.50	9.00	0.50	2.33	365132.2451	0.3651	0.2191	8.9	35.95	31.44	7.02	16.02
Sub Total in CMMD-01 Polygon							**1.0954**	**0.6572**	**13.29**	**37.32**	**24.08**	**5.94**	**18.68**
CMMD-02	298156.00	9.50	11.50	2.00	2.33	1389406.946	1.3894	0.8336	11.635	31.24	34.87	5.45	16.09
CMMD-02	298156.00	15.50	16.50	1.00	2.33	694703.473	0.6947	0.4168	9.68	34.42	30.09	6.15	19.03
Sub Total in CMMD-02 Polygon							**2.0841**	**1.2505**	**10.98**	**32.30**	**33.28**	**5.68**	**17.07**
CMMD-03	80464.63	6.10	10.00	6.50	2.33	1218636.847	1.2186	0.7312	17.79	38.25	18.90	5.82	18.54
Sub Total in CMMD-03 Polygon							**1.2186**	**0.7312**	**17.79**	**38.25**	**18.90**	**5.82**	**18.54**
CMMD-04	371560.03	4.00	10.40	6.40	2.33	5540703.094	5.5407	3.3244	20.03	36.02	20.26	7.13	15.86
Sub Total in CMMD-04 Polygon							**5.5407**	**3.3244**	**20.03**	**36.02**	**20.26**	**7.13**	**15.86**
CMMD-05	272993.00	3.60	5.40	1.80	2.33	1144932.642	1.1449	0.6870	19.38	32.92	26.38	5.51	15.17
CMMD-05	272993.00	6.10	10.30	4.20	2.33	2671509.498	2.6715	1.6029	14.35	37.16	23.60	7.10	17.11
CMMD-05	272993.00	12.70	16.10	3.40	2.33	2162650.546	2.1627	1.2976	8.97	34.03	28.43	7.65	20.19
Sub Total in CMMD-05 Polygon							**5.9791**	**3.5875**	**13.37**	**35.21**	**25.88**	**7.00**	**17.85**
							15.9179	**9.551**	**15.48**	**35.47**	**24.10**	**7.04**	**17.22**

Note: Measurement of ore resources based on TiO_2 content has been taken up (≥5% TiO_2) with-in potential 'Bauxite zone' as well as 'Aluminous Laterite zone'. The resources values in 'Bauxite zone' remain unaltered compared to 'Bauxite resources' considering the same (≥5% TiO_2 content) whereas the resources values of Aluminous Laterite zone changes and tabulated above.

UNFC Category 333 (Inferred Category Resources).

Report Submission August, 2023.

GEOLOGICAL REPORT ON PRELIMINARY EXPLORATION (G–3) FOR BAUXITE IN MADUA PAT-I BLOCK DISTRICT- LOHARDAGA, STATE – JHARKHAND.

CHAPTER-I

1.0.0 INTRODUCTION

1.1.0 GENERAL

Bauxite is a naturally occurring, heterogeneous material composed primarily of one or more aluminum hydroxide minerals, plus various mixtures of silica, iron oxide, titania, alumino silicate, and other impurities in minor or trace amounts. The principal aluminum hydroxide minerals found in varying proportions with bauxites are gibbsite and the polymorphs of boehmite and diaspore. Bauxites are typically classified according to their intended commercial applications: abrasive, cement, chemical, metallurgical, refractory, etc. The bulk of world bauxite production (approximately 85%) is used as feed for the manufacture of alumina through wet chemical caustic leach method commonly known as the Bayer process. Subsequently, the majority of the resulting alumina produced from this refining process is in turn employed as the feedstock for the production of aluminum metal by the electrolytic reduction of alumina in a molten bath of natural or synthetic cryolite (Na_3AlF_6), the Hall-Héroult process.

Aluminium is a soft and lightweight metal, the properties of metallic aluminium of being good conductor and amenable to get moulded in various shapes and sizes make the metal very useful for various domestic and industrial use. The use of aluminium exceeds that of any other metal except iron as it forms alloys with many elements. Aluminium is used in many industries to make millions of different products and is very important to world economy. It has wide application in various areas such as transport, building, architectural sectors, machinery and equipment, solar panels, lithographic (offset) plates as required in printing technology, car industry and so on.

The aluminium industry in India is strategically well-placed and is one of the largest producers in the world with discernible growth plans and prospects for the future. The overall total annual installed capacity of aluminium in the country has remained same at 41.00 lakh tonnes during 2015-16 & 2016-17. The production of bauxite at 22,313 thousand tonnes in 2017-18 decreased by 10% as compared to the previous year (as per Indian mineral year book-2018 by IBM).

Bauxite does not have specific composition. It is a mixture of hydrated aluminium

oxides, aluminum hydroxides in the form of diaspore and Boehemite, Al_2O_3, H_2O (Al_2O_3-85%, Al- 45%), gibbsite or hydragillite $Al_2O_3.3H_2O$ (Al_2O_3-65.4%, Al-34.6%), clay minerals and insoluble minerals like iron oxides, free quartz and titania as leucoxene or rutile in various proportions.

Reserves/Resources of bauxite in the country as on 1.4.2015, as per NMI database, based on UNFC system have been placed at 3,896 million tonnes. These resources include 656 million tonnes Reserves and 3,240 million tonnes remaining of these, about 77% resources are of Metallurgical grade. The resources of Refractory and Chemical grades are limited and together account for about 4%. By States, Odisha alone accounts for 51% of country'sresources of bauxite followed by Andhra Pradesh (16%), Gujarat (9%), Jharkhand (6%), Maharashtra (5%) and Madhya Pradesh & Jharkhand (4% each). Major bauxite resources are concentrated in the East Coast bauxite deposits in Odisha and Andhra Pradesh (as per Indian mineral year book-2020 by IBM).

According to the data from the United States Geological Survey (USGS), the global bauxite resources left to humans are 55-75 billion tons, and the basic reserves are 30 billion tons. India occupies nineth position in terms of bauxite reserve with deposit of about 580 million tonnes. with a share of 2.00% of world reserves. Odisha and Andhra Pradesh accounts for more than 90% of country's metallurgical grade resources. While gibbsitic bauxite resources in the world are depleting, vast gibbsitic deposits in India assume particular interest because of its ease in processing.

In 2019-20, the consumption of bauxite estimated at 24.03 million tonnes increased marginally by 8% as compared to 22.17 million tonnes in the previous year. Alumina/Aluminium Industry was the principal consumer of bauxite and accounted for 89% consumption in 2019-20 followed by Cement (8%) and Calcination (2%).

1.2.0 DETAILS OF PROJECT AND INVESTIGATING AGENCY

- The Preliminary Exploration (G-3) is to demarcate the various grades of bauxite and resource estimation under inferred category (333) in Madua Pat-I block, Dist: Lohardaga, State - Jharkhandh, which will facilitate for auction by DGM, State Government.
- The investigation agency for the block is **Central Mine Planning and Designing Limited** (CMPDI), Ranchi and the project has been implemented through NMET funding.

1.3.0 OBJECTIVES OF INVESTIGATION

The Exploration was proposed with the following objectives:

- The Preliminary Exploration (G-3) is to demarcate the various grades of bauxite zone(s) along the longer axis as well as along a direction at right angle to the longer axis.
- To assess the bauxite resources of the block both quantitatively and qualitatively.
- To carry out exploration as per Mineral (Evidence and Mineral contents) Rule-2015, Mineral Auction Rule-2015 and MMDR Amendment Act-2015 in turn to facilitate the State Govt (Jharkhand) for auctioning of the block under composite licensing.

1.4.0 BASIS FOR TAKING UP INVESTIGATION

C.S. Fox of the Geological Survey of India studied extensively the lateritic bauxite occurrences in India. Earlier the bauxite of Chotanagpur plateau was described by T.H. Holland and L.L. Fermor. More recently the subject has been dealt with by J.B. Auden, A.K. Dey and M.K, Roy Chowdhury, of the Geological Survey of India. Amongst the non-official geologists H.L. Chibber described bauxites deposits near the Lohardaga area.

Later Bauxite deposits of the area around Lohardaga, Ranchi district, Bihar, were investigated during the field-season 1961-62 with a view to estimate the reserves of various grades of ore in the unleased areas. All the bauxite bearing areas in the vicinity were also investigated and plotted on a Degree Sheet (scale 1:2,53,440) with a view to assess the regional reserves of bauxite. As a part of this program, Madua Pat was also investigated and found that about 1.5m- 3m of bauxite of various grades is exposed in the quarries worked by Minerals & Minerals Ltd. The bauxite is pinkish or grey in colour and massive or bouldery in nature. Sometimes the bauxite is found to grade gradually into aluminous Laterite and Laterite. Quarries on the northern scarp show development of the best quality bauxite.

On enactment of MMDR Amendment Act- 2015, Mineral (Evidence of Mineral Contents) Rule- 2015 and Mineral Auction Rule - 2015, Govt. of India directed State Governments to speed up exploration work for different Mineral Commodities especially Notified Minerals i.e., Iron ore, Manganese, Bauxite and Limestone in the respective states for auctioning of the blocks. An MOU has been signed between Department of Mines and Geology (DMG), Government of Jharkhand and Central Mine Planning & Design Institute Limited (CMPDI) for exploration of various mineral blocks by CMPDI in the state of Jharkhand. DMG, Government of Jharkhand

requested CMPDI to take up exploration work of Bauxite in 2 Blocks of Lohardaga district in Jharkhand. On request of CMPDI DMG, Jharkhand provided information required for preparation of exploration proposals pertaining to Bauxite deposit vide letter No भूिनअंवे– 13/200-983 dt. 25/06/2021. Accordingly, CMPDI has prepared the proposal for G-3 Level exploration for Bauxite in the proposed block viz. Madua Pat - 1, District - Lohardaga, Jharkhand and put up for consent of DMG, Jharkhand for taking up exploration work under NMET funding.

The Exploration Program for G3 stage of Exploration was approved by TCC of NMET during its 33rd Meeting on dated 24.09.2021. Later the executive committee of NMET in its 22nd meeting held on 22.12.2021 approved the proposal for upgrading to G-3 level. The administrative / financial sanction for exploration for Madua Pat-I block at estimated cost of 42.98 lakhs INR was received from Ministry of Mines, Govt of India, National Mineral Exploration Trust, New Delhi vide F.No.23/228/2022-NMET/17, dated, 02.05.2022.

On receipt of final approval of EC, NMET, the exploration (G-3 level) in Madua Pat-I block commenced. The exploration work comprised of Geological Mapping, drilling of 107.00 meter in 5 boreholes, collection of outcrop samples and Topographic and DGPS survey. The Draft Geological Report on Preliminary survey (G-3) has been submitted in July, 2023. At +30% Al_2O_3 (threshold value) bauxite zones (7% SiO_2) and at +20% Al_2O_3 (threshold value) Aluminous Laterite zone were identified with resources estimated as 1.397 million tonnes and 11.666 million tonnes respectively.

An attempt of measurement of ore resources based on TiO_2 content has been taken up (≥5% TiO_2) with-in potential 'Bauxite zone' as well as 'Aluminous Laterite zone'. The resources remains same as of Bauxite in the 'Bauxite zone' whereas, in Aluminous Laterite zone a total of 9.551 million tonnes net in-situ ore containing TiO_2 ≥5% containing 35.47% Al_2O_3, 15.48% SiO_2, 24.10% Fe_2O_3, 7.04% TiO_2 & 17.22% LOI have been estimated in the block and belongs to "Inferred Category" (333).

The details of the work done in G3 exploration by CMPDI in the Madua Pat-I Block is given in the Table: 1

1.5.0 DETAILS, NATURE AND QUANTUM OF WORK PROPOSED VS ACHIEVEMENT (G-3)

TABLE- 1

DETAILS OF WORK DONE IN MADUA PAT – I BLOCK (G-3), DISTRICT – LOHARDAGA, STATE – JHARKHAND

Sl. No.	Item of Work	Unit	Quantum of Proposed Work	Quantum of work done
I	**Topographic Survey (with contour interval of 5m.)**	sq. km.	1.50	1.50
II	**Geological mapping on 1:4000 scale**	sq. km.	1.50	1.50
III	**Core drilling**	meters.	107.00 (5 BHs)	107.00 (5 BHs)
IV	**Borehole Pillaring - Construction of Con -Pillar**	Nos.	5	5
V	**Approach road making drill site (Hilly terrain)**	kms.	2.00	1.00
VI	**Bore hole Fixation and determination of co-ordinates & Reduced Level and Demarcation of lease hold boundary**	Nos.	10	10
VII	**Pitting for determination of bulk density (1m x1m x1m)**	Nos.	2	2
VIII	**Laboratory Studies**			
	A. Chemical Analysis			
	a) Primary core Samples for 5 radicals viz. Al_2O_3, SiO_2, Fe_2O_3, TiO_2, LOI	Nos.	49	49
	b) Internal (5%) & External (10%) Check samples of 5 radicals Al_2O_3, SiO_2,Fe_2O_3, TiO_2, LOI.		3	3
	c) Composite samples for 7 radicals viz. Al2O3, SiO2 , Fe2O3, TiO2, V2O5, P2O5 & LOI	Nos.	5	4
	d) Composite samples for 14 radicals viz. Al2O3 , SiO2 , Fe2O3 , TiO2 , V2O5 , P2O5 , LOI, MnO, CaO, MgO, K2O, Na2O, SO3 & Organic carbon	Nos.	5	4
	e) ICP-MS studies (REE analysis of Saprolite zone)	Nos.	50	19

Sl. No.	Item of Work	Unit	Quantum of Proposed Work	Quantum of work done
	f) Combined determination of THA, MHA & Reactive silica	Nos.	5	4
	g) Package of 34 elements by ICPMS in 1 borehole	Nos.	20	20
	B. Physical Analysis			
	a) X-RD Studies on composite	Nos.	5	4
	b) Bond Work Index	Nos.	5	3
	c) Determination of in-situ bulk density	Nos.	2	2
	d) Specific Gravity of Rock samples	Nos.	5	5
	C. Complete Petrographic studies	Nos.	5	5
	D. Complete Mineragraphic study report	Nos.	5	5
IX.	**Geological Report Preparation**	No.	1	1

1.6.0 ACKNOWLEDGEMENT

Central Mine Planning and Designing Institute Limited (CMPDIL) expresses its sincere gratitude to the Ministry of Mines, National Mineral Exploration Trust (NMET), Govt. of India, Geological Survey of India (GSI) and Directorate of Geology and Mining (DGM), Jharkhand on whose behalf, the exploration work in Madua Pat-I block has been executed.

The local administration / authorities of Lohardaga District, Jharkhand also deserve a special mention with thanks for extending all necessary helps and guidance in completing the exploration work successfully.

Last but not least, CMPDIL express its earnest thankfulness to the local villagers/gram panchayat for their co-operation at the time of execution.

1.7.0 LIST OF PERSONNEL ASSOCIATED

TABLE – 2

PERSONAL INVOLVED WITH PRELIMINARY EXPLORATION (G3) IN MADUA PAT-I BLOCK, DIST: LOHARDAGA, JHARKHAND.

Responsibility Assigned	Person Involved
Overall Co-operation.	Shri Jayant Chakroborty, Regional Director, RI3, CMPDI
	Shri Bratindra Nath Basu, HOD (Exploration), RI3, CMPDI
	Shri Surendra Kumar Gupta, Ex HOD (Exploration), RI3, CMPDI.
Overall Technical Supervision	Shri Jayant Kumar, Advisor (Non-COAL), CMPDI HQ.
	Dr. Ragwendra Pratap Singh, Senior Manager (Geology).
Overall Co-ordination	Shri Arijit Saha, Senior Manager (Geology).
Preparation of Exploration Scheme	Shri Saurabh Singh, Senior Manager (Geology).
Project Presentation to NMET	Dr. Ragwendra Pratap Singh, Senior Manager (Geology).
Geological Report Coordinator	Shri Arijit Saha, Senior Manager (Geology).
Field Activity Coordinator	Shri Arijit Saha, Senior Manager (Geology).
Team Members	Shri Arijit Saha, Senior Manager (Geology).
	Shri Saurabh Singh, Senior Manager (Geology).
	Shri Abhishek Mukhopadhyay, Manager (Geology).
Field Geology	Shri Arijit Saha, Senior Manager (Geology).
	Shri Abhishek Mukhopadhyay, Manager (Geology).
Drilling	Shri Rahul Rajput, Manager (Drilling), MECL.
Survey	Shri Manoj Kumar Sinha, Sr. Officer (Survey).
	Shri B.B.Majumdar, Surveyor (C).
	Shri Kingshuk Roy, Surveyor (C).
	Shri D. Singh, Surveyor (M).
	Shri F. Sheikh, Surveyor (M).
ArcGIS work	Shri Arijit Saha, Senior Manager (Geology).
	Shri Saurabh Singh, Senior Manager (Geology).

Responsibility Assigned	Person Involved
	Shri Abhishek Mukhopadhyay, Manager (Geology).
Peer Review and Comment	Shri D. K. Rai, Dy. Director General (Retd.), GSI.
Documentation and Report Preparation	Shri Arijit Saha, Senior Manager (Geology).
	Shri Abhishek Mukhopadhyay, Manager (Geology).

CHAPTER – II

2.0.0 PROPERTY DESCRIPTION

2.1.0 LOCATION OF THE BLOCK

The Maduapat-I Block falls in Survey of India Toposheet No.73 A/10 and is bounded by the following Co-ordinates under Maduapat plateau which is located in Lohardaga district of Jharkhand. The Bounding coordinates are given in table -3:

Table – 3

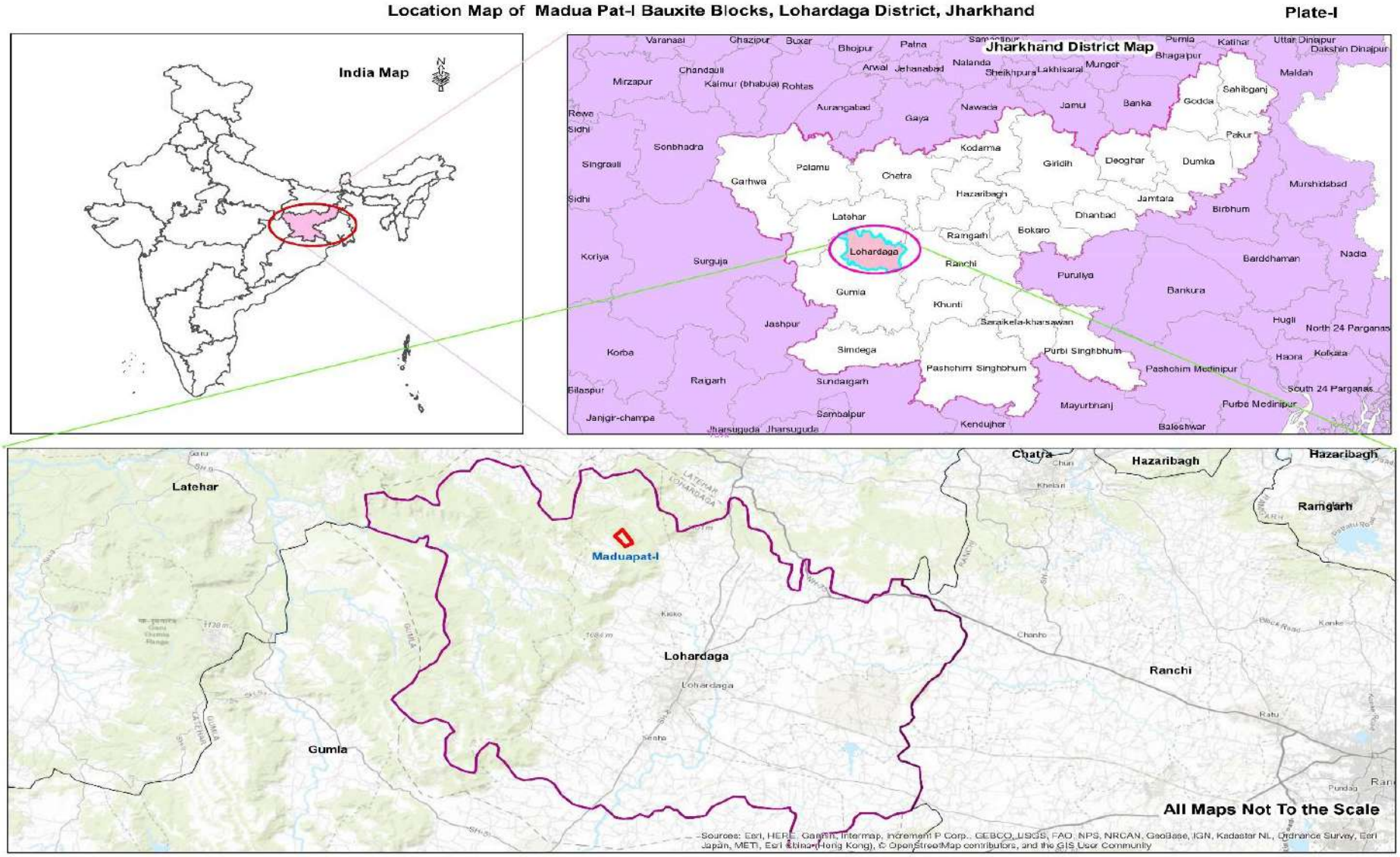
Co-ordinates of Bounding Points of the block boundary of Madua Pat-I,

Dist – Lohardaga.

Bounding Co-ordinates (DMS Datum WGS-1984)		
Sl No.	**Longitude**	**Latitude**
A	84˚37’25.12”	23˚35’53.95”
B	84˚37’48.73”	23˚36’19.78”
C	84˚38’22.87”	23˚35’29.39”
D	84˚37’59.62”	23˚35’15.51”

The Maduapat-I block is located at a distance of 16 kms from Kisko block office. Kisko is connected to Lohardaga district HQ with metal road of about 10 kms The nearest railway station is Lohardaga, located at a distance of 26 kms. from the block. The distance from Lohardaga town to Ranchi the state capital is 75 km, connected by an all-weather metaled road. Thus Maduapat-I Block is located at a distance of about 101 km from Ranchi. The nearest police station of the block is situated at Kisko at a distance of 15 kms from the block. But other infrastructures like Bank facility, Hospital, workshop facility, Bus stand, Railway Station etc. are located al Lohardaga Town, i.e the district town at a distance of 32 kms. The nearest Airport is situated at State capital Ranchi at a distance of 90 kms. The location map is furnished as Plate No. I.

LOCATION MAP



2.2.0 PHYSIOGRAPHY & DRAINAGE

The Lohardaga district covers the south - western part of Chhotanagpur plateau. Physiographically this area can be divided into two parts; the tablelands, about 975m high, and the valleys about 600-650m in elevation. In general, the area comprises a series of flat topped ridges, detached from each other by narrow valleys. Small nalas and rivulets originating from these ridges join up and flow west-wards and eastwards into the North Koel and South Koel rivers respectively.

Madua Pat Plateau is tableland located in Kisko block of Lohardaga District. The bauxite deposits under reference are characterized by the extensive blanket of plateau Laterite on an almost flat topography, locally known as “pat”. The flatness of the plateau together with well-defined steep scarp faces gives distinct feature to the topography of the deposits with gentle slope towards south-west. The RL of the block varies from 1064.00 meter to 1022.00 meter and sloping towards NE and SW mainly from the central.

2.3.0 CLIMATE AND RAINFALL

The area forms a part of the Lohardaga plateau. The area experience warm humid climate with three well defined seasons i.e. summer, winter and monsoon. The winters commence from middle November and extend up to middle of March with December being the coldest month. During winter the temperature goes down to 4˚C. Summer starts from middle of March and continues up to middle of June, when the temperature shoots up to 42˚C.

2.4.0 FLORA AND FAUNA

The most important tree is 'Sal' (Shorea robusta) which occurs along with a number of other species such as Mahua (Bassia latifolia), Jamun (Eugenia jambolana), Piar (Buchanania latifolia), Khair (Acacia latechu), Bamboo (Dendrocalamus striota), Mango (Mangifera indica), Pipal, Pakar, Asan, Karma, Karanj, Neem, Sisam, Jackfruit etc.

The wild animals are rare in the area. However, in the top and dense forests in surrounding, several types of animal viz. Dog, Cat, Cow, Buffalo, Goat, Jackals, Hare, etc. are seen rarely. The jungle hounds and several types of birds are commonly found.

Photograph – 1

Southwestern part of the block showing Escarpment,

Madua Pat-I Block, Dist – Lohardaga, Jharkhand.



Photograph – 2

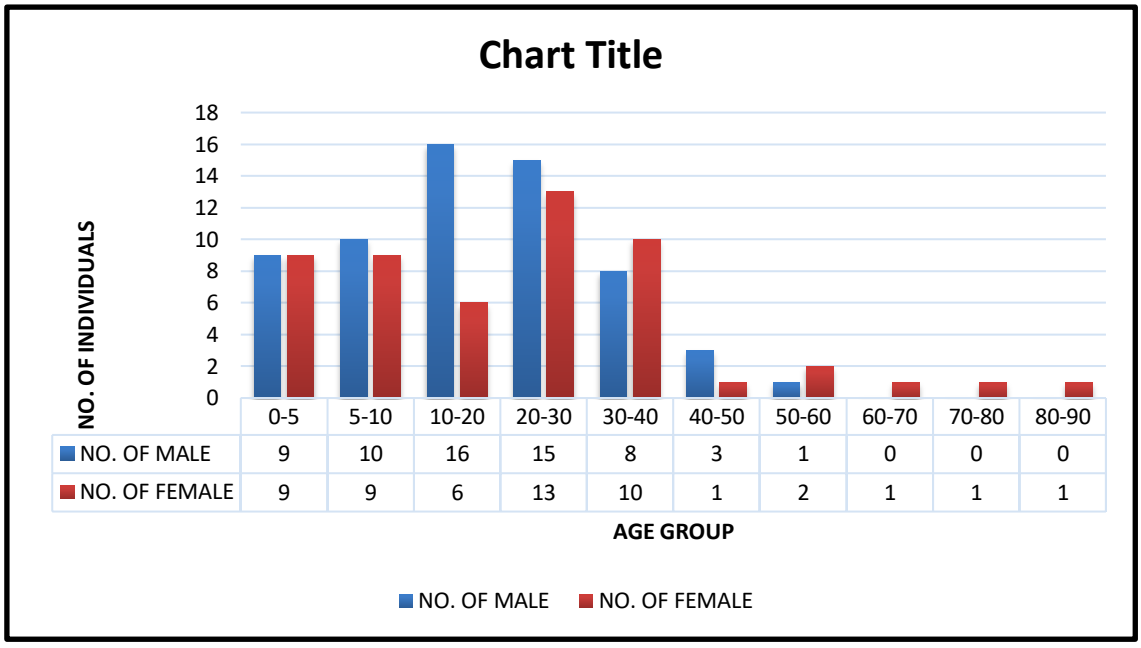
Panoramic view of the Madua Pat-I block, Dist – Lohardaga, Jharkhand.



2.5.0 DEMOGRAPHIC DATA OF THE BLOCK.

During Exploration work, the demographic data has been collected from the local Madua Pat village. The data collected from the 24 families are mainly the age group, male-female ratio etc. The data are given below:

- Total human being – 115.
- Total Male – 62.
- Total Female – 53.



	0-5	5-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
NO. OF MALE	9	10	16	15	8	3	1	0	0	0
NO. OF FEMALE	9	9	6	13	10	1	2	1	1	1

2.6.0 FOREST TYPE

The area on the Pat is almost devoid of Forest Area. Some areas along the flank of the boundary comes under open forest and scrub types.

2.7.0 LAND USE

Most of the area is being used for Agricultural purpose and this is the only livelihood being used by locals. Several area over the pat area are excavated/mined out in nature and presently acting as pond for the locals.

CHAPTER-III

3.0.0 PREVIOUS WORK

3.1.0 C.S. Fox of the Geological Survey of India studied extensively the lateritic bauxite occurrences in India. Earlier the bauxite of Chotanagpur plateau was described by T.H. Holland and L.L. Fermor. More recently the subject has been dealt with by J.B. Auden, A.K. Dey and M.K, Roy Chowdhury, of the Geological Survey of India. Amongst the non-official geologists H.L. Chibber described bauxites deposits near the Lohardaga area.

Later Bauxite deposits of the area around Lohardaga, Ranchi district, Bihar, were investigated during the field-season 1961-62 with a view to estimate the resources of various grades of ore in the unleased areas. Madua Pat was also investigated and found that about 1.5m- 3m of bauxite of various grades is exposed in the quarries worked by Minerals & Minerals Ltd.

In view of insufficient data DGM, Jharkhand requested CMPDI as part of existing MoU for taking up G3 stage of exploration in Madua Pat-I block for ease of auction in later phase. Accordingly, the Exploration Program for G3 stage was submitted at NMET and approved by TCC of NMET during its 33rd Meeting on dated 24.09.2021. Later the executive committee of NMET in its 22nd meeting held on 22.12.2021 approved the proposal.

A total of 1.397 million tones net in situ bauxite resource with an average grade of 33.46% Al_2O_3, 5.62% SiO_2, 32.74% Fe_2O_3, 8.01% TiO_2 and 19.53% LOI and a total of 1.814 M.Tonne net in situ bauxite resource with an average grade of 33.14% Al_2O_3, 6.33% SiO_2, 32.59% Fe_2O_3, 7.90% TiO_2 , 19.41% LOI have been estimated in block.

A total of 11.666 million tonnes net in-situ aluminous Laterite resource with an average grade of 34.56% Al_2O_3, 18.15% SiO_2 have been estimated in the block in an area of 1.33 sq. km.

Measurement of ore resources based on TiO_2 content has been taken up (≥5% TiO_2) with-in potential 'Bauxite zone' as well as 'Aluminous Laterite zone'. The resources of remains same as of Bauxite in the 'Bauxite zone' whereas, in Aluminous Laterite zone a total of 9.551 million tonnes net in-situ ore containing TiO_2 ≥5% containing 35.47% Al_2O_3, 15.48% SiO_2, 24.10% Fe_2O_3, 7.04% TiO_2 & 17.22% LOI have been estimated in the block and belongs to "Inferred Category" (333).

CHAPTER-IV

4.0.0. GEOLOGY OF THE AREA

4.1.0 REGIONAL GEOLOGICAL SET UP OF THE AREA

The bauxite bearing plateaus form a part of the Pre-Cambrian shield of the Indian Peninsula. They consist mainly of Chhotanagpur Granite Gneiss, enclaves of quartzites and meta basic rocks belonging to the Older Metamorphic Group are occasionally met with. The laterite profiles on the 'pats' are underlain by the clayey and ferruginous sediments of Upper Gondwana age.

The Regional Geological succession of the area after Roy Chowdhury (1958) is as given in table below:

Table – 4

Regional Geological sequences of the area.

General Stratigraphic Sequence of the Area		
Age	**Group/ Formation**	**Litho-units**
Recent		Alluvium, conglomerate andcarbonaceous shale
Tertiary toRecent		Laterite. Bauxite and lithomarge
Cretaceous	Lameta	Limestone and basalt ferruginous grit
Proterozoic		Quartz vein
Archean to Proterozoic	Chhotanagpur Granite Gneissic complex	Diorite, granite, granite gneiss
Archean to Paleoproterozoic	Unclassified Metamorphics	Limestone, calc-silicate rock, quartzite,phyllite, slate, mica schist, hornblende schist, amphibolite, talc schist

4.2.0 GEOLOGY OF THE BLOCK:

The study area lies in the western part of the Chotonagppur Gneissic Complex which exposes majorly the Granite Gneiss and Porphyritic Granite, Granulitic rocks, quartz veins and Pegmatites. Bauxite forms the capping which may be divided into different zones –

Bauxite, Aluminous Laterite, Ferruginous Laterite etc. which varies in thickness from place to place. Thick soil profile is developed at the central part of the plateau. At places Bauxite is exposed forming irregular outcrop which have been mapped accordingly in 1:4000 scale.

During Geological Mapping carried out in 1:4000 scale, different lithological units like Bauxite, Aluminous Laterite, Massive Laterite, Laterite, Saprolite Zone and Granite Gneiss have been identified on field and plotted in Geological Map. (Refer Plate – 3).

The central portion of the block i.e the most elevated portion of the block is dominated by Soil Profile. Towards the boundary of the block with a reduction of RL values, the exposure of the laterite or aluminous laterite have been seen. At places exposure of Bauxite has been reported in the block. Down the slope, along the flank of proposed boundary saprolite zone or clay zone has been reported. The exposure of Aluminous Laterite and Bauxite along with laterite are also reported in mined out area situated within the block area. The occurrence of all the lithologies indicated above are highly correlated with the topographical contour (Refer plate – 4).

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CHAPTER – 5

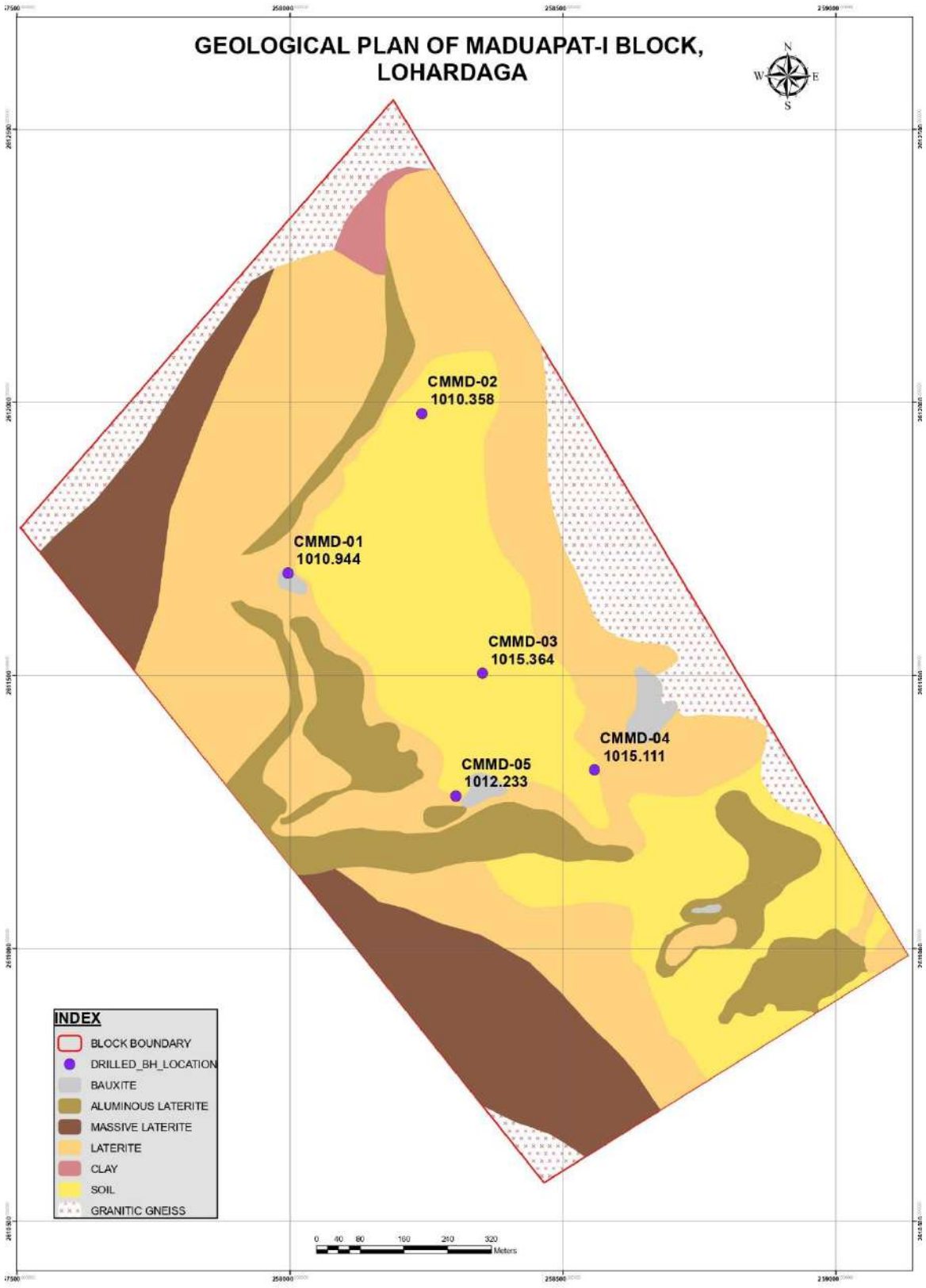
5.0.0 ACTIVITY DURING THE PERIOD (GEOSCIENCE INVESTIGATION)

5.1.0 TOPOGRAPHIC SURVEY

Topographical surveying triangulation network has been laid down during G-3 stage of exploration in the proposed study area with the help of DGPS. Boreholes have been fixed on the ground. Surface contouring was done at 2.0 m contour interval throughout the block. Actual plateau boundary was surveyed by DGPS and Total Station in WGS-84 Datum (Plate-IIIA). List of Cardinal Points from A to E of the block along with the co-ordinates are given in Annexure – I/B. The available surface features like roads, villages, mined out area etc. were surveyed and presented along with drilled exploratory boreholes in the Topographical Map on 1:2,000 scale (Plate No –II). All the boreholes have been connected with the survey triangulation network. RL and co-ordinates of survey stations and Boreholes have been determined and the same is furnished in Annexure- I/A.

5.2.0 GEOLOGICAL MAPPING

The geological mapping was carried out in details in 1:4000 scale during G-3 stage preliminary exploration activities. The available rock exposures of bauxite, aluminous laterite, lateritic bauxite and saprolite clay have been mapped and updated on Geological Plan (Plate – II). Geological mapping was carried out in 1.50 sq. km area by taking traverses with the help of handheld GPS. The exposures of bauxite have been observed on the surface generally embedded within aluminous laterite and along the margin of plateau there is occurrence of massive laterite (Photograph- 03 and 04). Although in the central region, exposures are scanty as the plateau is almost covered by soil. Geological features, have been recorded and plotted on 1:4,000 scale (ref: Geological plan – Plate-3).


GEOLOGICAL PLAN OF MADUAPAT-I BLOCK, LOHARDAGA
N
W
E
S
CMMD-02
1010.358
CMMD-01
1010.944
CMMD-03
1015.364
CMMD-04
1015.111
CMMD-05
1012.233
INDEX
BLOCK BOUNDARY
DRILLED_BH_LOCATION
BAUXITE
ALUMINOUS LATERITE
MASSIVE LATERITE
LATERITE
CLAY
SOIL
GRANITIC GNEISS
0 40 80 160 240 320
Meters

Photograph- 03

Bauxite Exposure on surface at Madua Pat-I Block, Dist – Lohardaga, Jharkhand.



Photograph- 04

Bauxite Exposure on surface at Madua Pat-I Block, Dist – Lohardaga, Jharkhand.



5.3.0 OUTCROP SAMPLING

A total of 09 (Nine) Nos. of surface/outcrop samples of the various litho types (bauxite, aluminous laterite and laterite) have been collected for petrographic studies (PS-05,07,08,09,10), minerograhic studies (PS-01,02,03,06,09) and three samples for Bond Work Index analysis (BW-01,02,03) during the exploration from the total area of the block, wherever bauxite/laterite outcrops are exposed. However due to the occurrence of the scanty outcrops of bauxite/laterite, the samples were collected randomly instead of grid pattern. Locations of the outcrop samples are shown on the geological map (Plate- 4).

Table – 5

Details of outcrop sampling for different analysis, Madua Pat-I Block, Dist – Lohardaga District, Jharkhand.

Analysis Name	No of Samples
Bond work Index	3
Petrographic Analysis	5
Mineragraphic Analysis	5

5.4.0 GEOLOGY OF THE EXPLORATION BLOCK

5.4.1 Bauxite occurs as segregated sheets and lenses within the laterite capping over the gneisses of Chotanagpur Granite Gneisses (CGG). The gneisses contain enclaves of amphibolites, quartzite of varying dimensions aligned along the regional foliation/Gneissosity. The laterite crust is underlain by variegated clay. The clays are soft and moderately plastic.

5.4.2 The rocks of this area can be organized into the following geological succession:

Age	Formation
Recent	Alluvium
Lower Tertiary	Laterite, Bauxite and Clay
Archean to Proterozoic	Granite gneiss, Calc silicate rocks, Quartzite and Amphibolite.

5.4.3 Bauxite and laterite occur as fine grained, massive rock. Pisolitic variety are also common. The pisolites range in size from 0.3cm to 1 cm in diameter. Laterite is brown to red-brown in color. Greyish tinge in the lower part of the laterite crust is noted due to the presence of goethite.

5.4.4 The bauxite is grey to ash grey in colour and is generally concentrated towards the upper part of the laterite profile. Bauxite is missing from a large part of the area where only laterite is exposed. Towards the base, the laterites show vesicular or cavernous texture and the caverns are at times lined with goethite.

5.4.5 The bauxite bands may at times be persistent but very often it tapers out abruptly within a short distance. The contact between the bauxite and laterite in general is sharp and can be easily delineated in the field.

5.5.0 DESCRIPTION OF THE FORMATIONS

The rock types intersected in the boreholes drilled in the block during exploration activity are as follows:

5.5.1 SOIL

Incidence of soil is generally confined to the central portion of the block. The soil is mostly clayey and occurs as a thin layer varying in thickness from 0.30 m. (CMMD-05) to 2.50 m. (CMMD-03) as intersected in drilled boreholes. The contact of the soil and the underlying morrum / laterite is sharp.

5.5.2 LATERITE

The laterite in this area falls under the category of plateau laterite. It occurs as extensive blanket on the high plateaus. It is rich in hematite, limonite, goethite etc. and are dark grey, dark brown, pale yellowish to yellowish brown in colour, spongy and pisolitic in nature. Ferruginous Laterite, wherever occurs, is dark brown, brick-red to cherry-red having different physical characteristics. This laterite is highly vesicular and vermicular. It has been found from borehole data and observations on scarps that the lower horizons are comparatively more ferruginous and vesicles are irregular whereas in the upper horizons there are pockets of less ferruginous laterite with the vesicles and cavities. The thickness of laterite intersected in the boreholes varies from 6.10 m. (CMMD-05) to 11.40 m. (CMMD-02).

The aluminous laterite horizon though highly variable in thickness, occurs near the bauxite zones. It is reddish brown to pink in colour, massive, pisolitic and partly porous. This pisolitic variety contains pisolites (0.2 to 0.5 cm.) of aluminous matter and the matrix of ferruginous material. Nevertheless, it has a general appearance of bauxite, though darker in colour, more porous and apparently poorer in quality. It also shows colloform structure. At other places, it is massive and hard. The cavities in the laterite may be filled with secondary clay or iron minerals. The chemical change of laterite to aluminous laterite and bauxite is generally gradational, however, laterite to clay is generally sharp. The thickness of aluminous laterite intersected in the boreholes varies from 3.50 m. (CMMD-02) to 11.10m. (CMMD-05).

5.5.3 BAUXITE

There are frequent segregations of bauxite within the laterite. The exposures of bauxite are present alongside the margin of plateau mostly in massive form. In the central region, exposures are scanty as the plateau is almost covered by soil (Photograph 3 and

4).

Bauxite zone delineated at (+) 30% Al_2O_3 and (-) 7% SiO_2 cut-off has been intersected in two boreholes i.e. CMMD-02, and 03 dominantly in the central part and north eastern part of the block.

Bauxite zone delineated at (+) 30% Al_2O_3 and (-) 10% SiO_2 cut-off has been intersected in four boreholes i.e. CMMD-01, 02, 03, and 05 in the central and north eastern part of the block.

Bauxite zone delineated at (+) 30% Al_2O_3 and irrespective of SiO_2 cutoff has been intersected in in all five boreholes i.e. CMMD-01, 02, 03,04, and 05 occupying most of the block area.

Aluminous laterite zone delineated at (+)20% Al_2O_3 and irrespective of SiO_2 cutoff has been intersected in all five boreholes i.e. CMMD-01, 02, 03,04, and 05 occupying most of the block area.

Hence, it seems that the bauxite is bouldery and present in pockets and irregular type of deposit and the thickness of bauxite zone gradually thin out from the central part of the plateau towards edge of the plateau. The colour of bauxite varies from grey to grayish white, white, and reddish white. Structurally and texturally, it may be pisolitic, oolitic, massive, vesicular, and breccia like.

The thickness of bauxite zone delineated at (+) 30% Al_2O_3 and (-) 7% SiO_2 cut-off as intersected in the boreholes varies from 1.30 m (CMMD-03) to 3.00m (CMMD-02), and the bauxite zone delineated at (+) 30% Al_2O_3 and (-) 10% SiO_2 cut-off as intersected in the boreholes varies from 1.30 m (CMMD-03) to 4.00m (CMMD-02).

5.5.4 LITHOMARGE

Lithomarge is exposed at the base of the laterite/bauxite scarps of the area. It is generally a variegated to pink clay. As approved by NMET, all boreholes drilled in G-3 stage of exploration i.e. from CMMD-01 to CMMD-05 are drilled upto lithomarge clay. Range of intersection of lithomarge in drilled boreholes within the block varies from thickness of 3.20m (CMMD-05) to 5.0m (CMMD-01, CMMD-02 & CMMD-03). It has a gradational contact with the overlying ferruginous laterite and the underlying altered trap rock.

5.5.5 GRANITE GNEISS (PARENT ROCK)

Discontinuous exposures of granite gneiss occur along the margins of the plateau of Maduapat-I. The granitic gneiss rock is whitish to greyish in colour, medium grained, massive and hard/compact.

5.6.0 STRUCTURE OF THE BLOCK

Almost 80% of the area is concealed under soil and only few outcrops of bauxite/aluminous laterite/laterite are exposed. These litho units are by-products of weathering and the structures preserved in the parental rock (granite gneiss) are completely obliterated. Therefore, presently no structures are observed in the field area.

5.7.0 PETROGRAPHIC STUDIES

5.7.1 A total of five nos. of specimens of different lithology of bauxite profile samples were collected from different field area during preliminary exploration and petrographic study was carried out. The results are furnished as Annexure VII.

5.7.2 The petrographic studies of the bauxite samples show that the major minerals (>5%) in the rocks are Cliachite, Clay Minerals, Gibbsite and Ferruginous matter. Gibbsite and Boehmite are present as fine to very fine aggregates replacing Cliachitic patches from both core and periphery and also along fractures. Ferruginous matter has been found to be present as reddish patches and stains, mostly acting as cementing ingredient. Clay minerals have been found to be associated with Cliachite in areas. Chert occurs as fine patchy fillings, comprising very fine micro-crystalline aggregates. The minor minerals (>1% to <5%) are Boehmite, Clay minerals and opaque. Whereas accessory minerals (<1%) are mostly Boehmite, Chert and opaques.

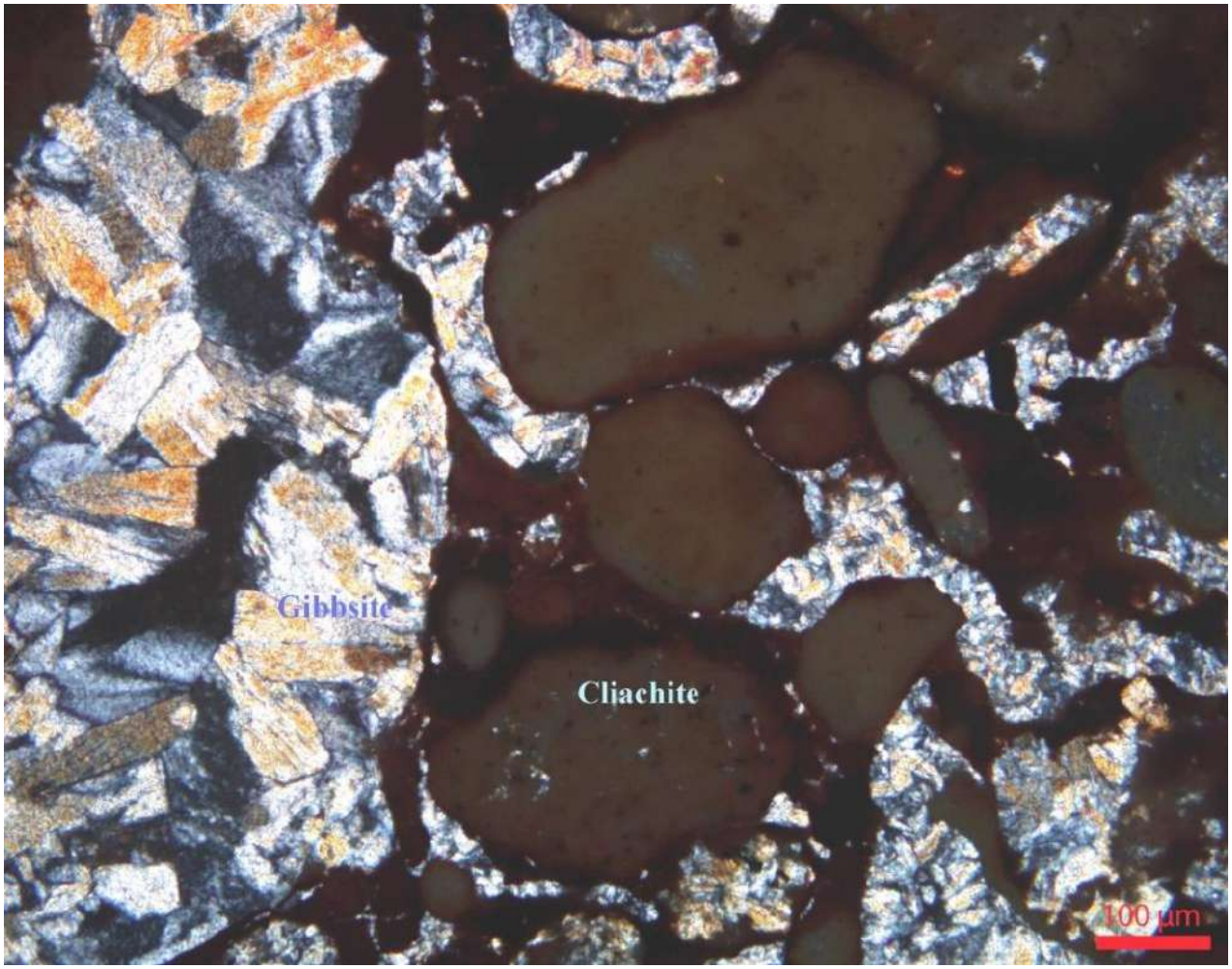
5.7.3 The different outcrop samples/specimen studied under microscope has been identified as Bauxite, Bauxite bearing laterite, Laterite etc.

Photograph – 5

Photomicrograph showing fine to very fine prismatic aggregates of gibbsite replacing cliachitic pisolites as seen under crossed nicols, Madua Pat-I, Dist – Lohardaga, Jharkhand.

Specimen No.: PS-10

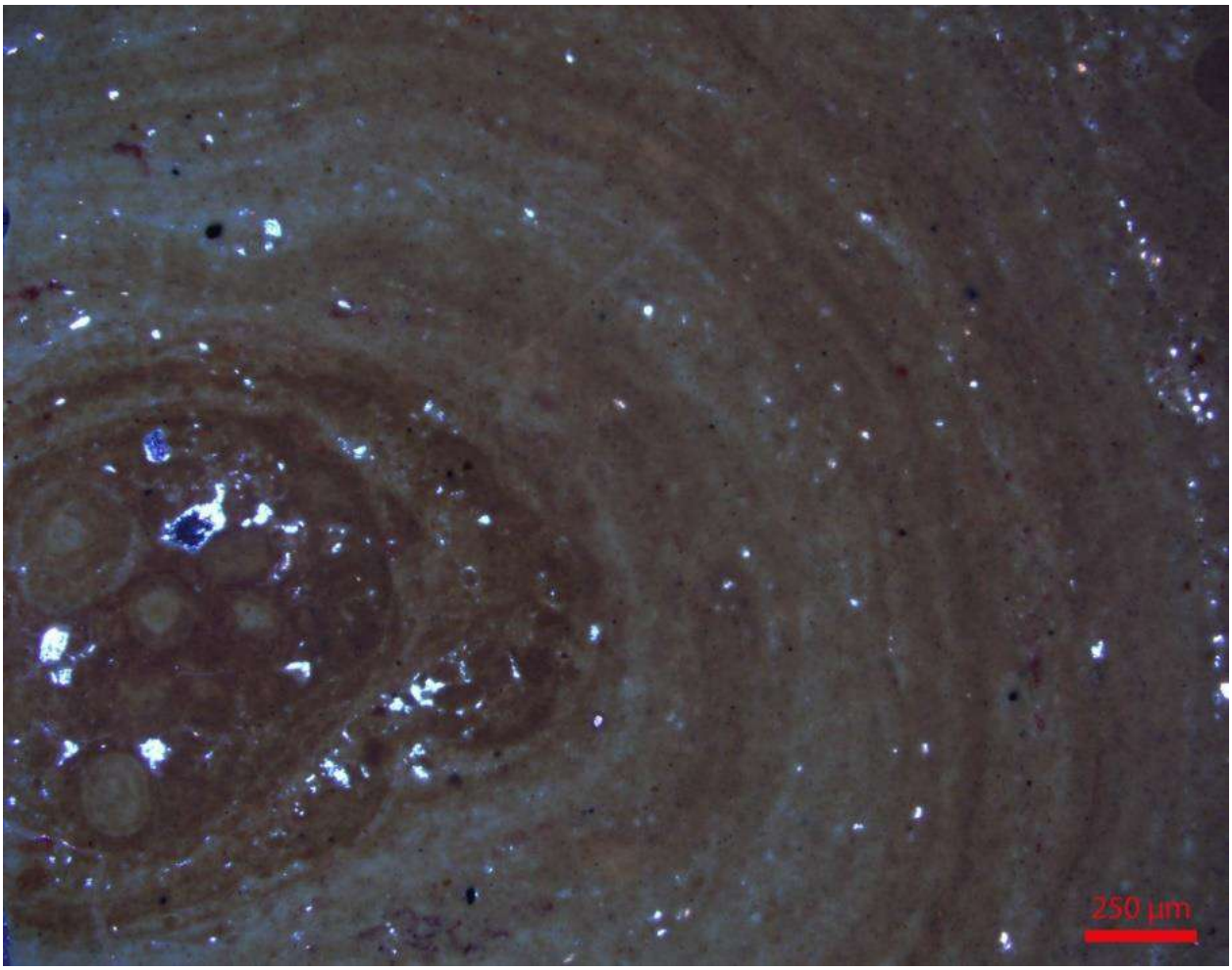
Magnification: 100X



Photograph – 6
Photomicrograph showing cliachitic pisolite/ oolite with multiple concentric rings as seen under cross nicols. Madua Pat-I, Dist – Lohardaga, Jharkhand.

Specimen No.: PS-08

Magnification: 100X

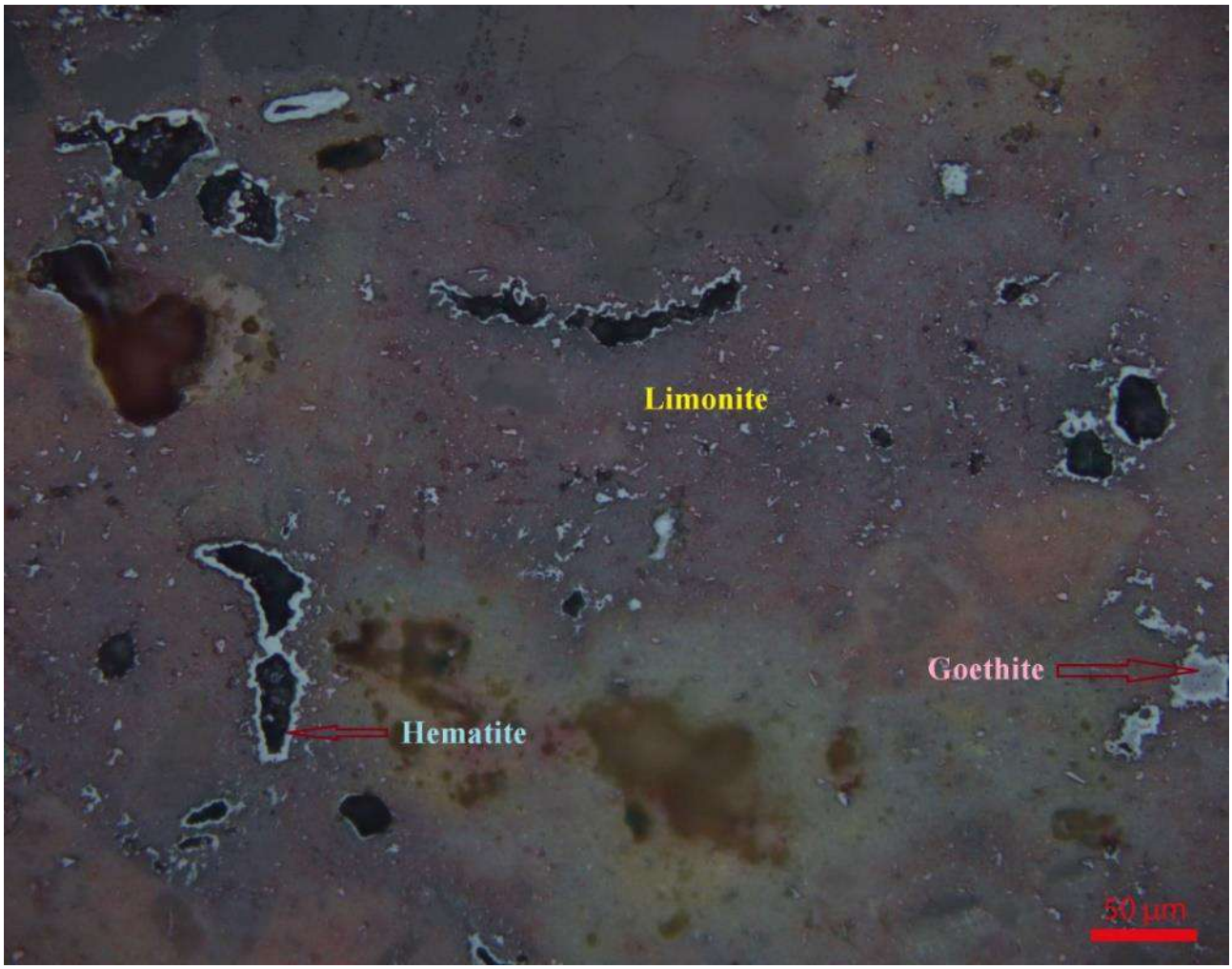


Photograph – 7

Photomicrograph showing hematite and goethite fillings, where goethite is replacing hematite and reddish limonitic patches comprising very fine amorphous aggregates as seen under reflected light, Madua Pat-I, Dist – Lohardaga, Jharkhand.

Specimen No.: PS-02

Magnification: 200X

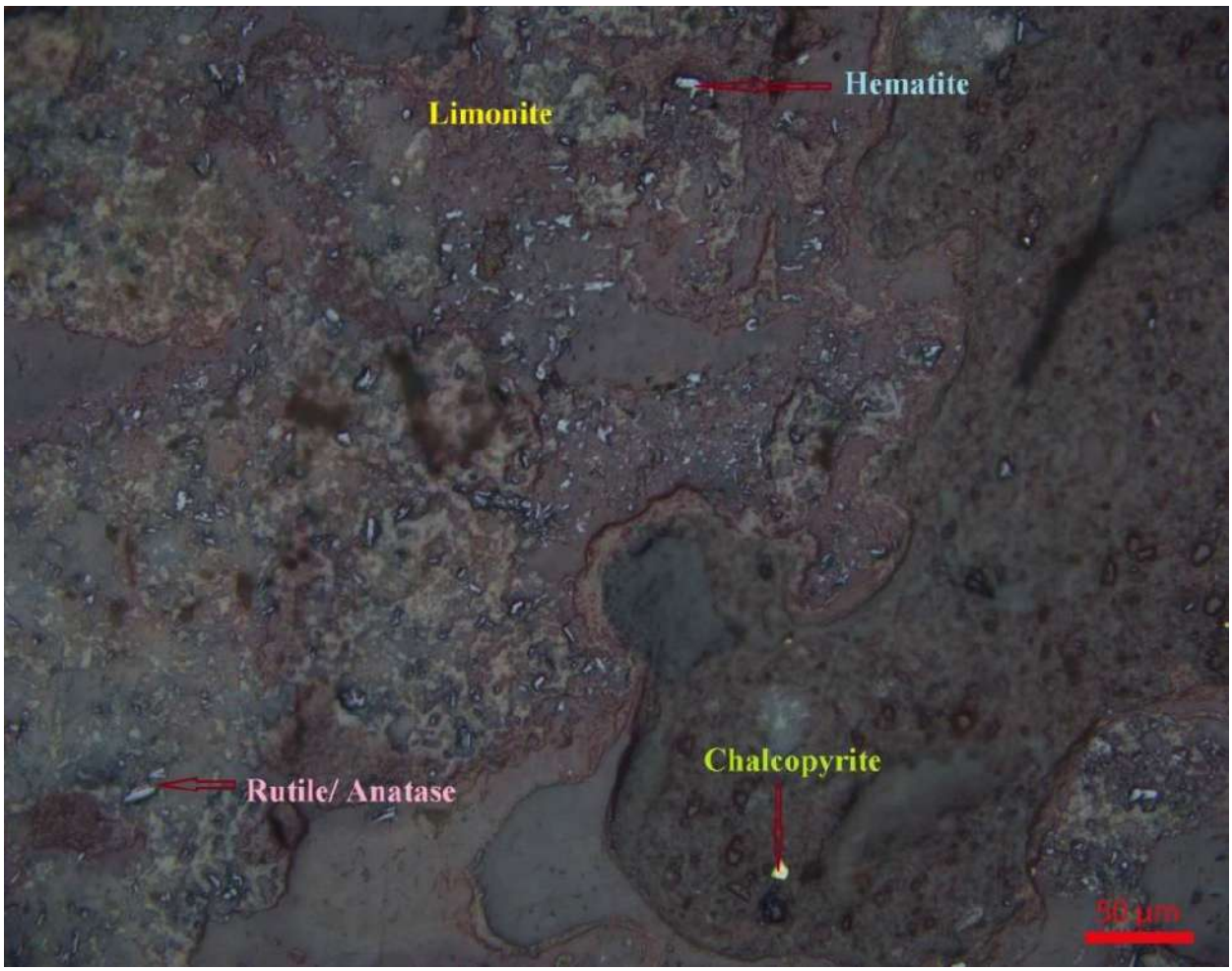


Photograph – 7

Photomicrograph showing very fine specks of hematite, rutile/ anatase and chalcopyrite and reddish limonitic patches and fillings as seen under reflected light., Madua Pat-I, Dist – Lohardaga, Jharkhand.

Specimen No.: PS-03

Magnification: 200X



5.8.0 TOTAL AVAILABLE ALUMINA AND REACTIVE SILICA

The total available alumina and reactive silica studies on 4 composite samples have been carried out in Chemical Laboratory. The samples are selected in such a manner that the entire bauxite bearing areas are represented by them.

The silica is not desired in worthy bauxite samples. This is because silica being abrasive in nature causes erosion in components of various process equipment like heat exchangers, pipelines,valves, pump casing and impellers etc. Thus, higher silica content in bauxite causes higher initial investment for the plant and higher repair maintenance cost for routine replacement of critical components of equipment. Available alumina or Al_2O_3 in bauxite sample refers to the amount of extractable alumina, which is present in a form that allows it to be extracted by a refining plant. The higher the number, the more alumina is available for refining and the better the result.

Reactive silica is the amount of silica within a bauxite, which is not free and is locked up in the form of clays and kaolinite, which are detrimental in the Bayer bauxite processing method. Quartz is free silica. During digestion of bauxite in Bayer process, 'reactive' silica reacts with caustic soda and comparable amounts of alumina to form insoluble sodium aluminum silicate, which is lost as refinery plant residue. Hence, lower the percentage of reactive silica, easier and better for the extraction of alumina. The results are given in Table No. 6 and 7 respectively.

Table – 6

Result of Total Available Alumina in borehole samples, Madua Pat-I, Dist – Lohardaga, Jharkhand.

SL. NO.	BH.NO/SAMPLE NO	DEPTH (m)			Total Available alumina (%)
		FROM	TO	THICK	
1	CMMD 2/9+2/10	11.50	13.00	1.50	30.40
2	CMMD 2/13+2/14+2/15	14.00	16.50	2.50	36.58
3	CMMD 5/10+5/11+5/12	12.70	16.10	3.40	29.73
4	CMMD 3/5+3/6	8.00	10.00	2.00	50.67

Table – 7

Result of Reactive silica in borehole samples in borehole samples, Madua Pat-I, Dist – Lohardaga, Jharkhand.

SL. NO.	BH.NO/ SAMPLENO	DEPTH (m)			Total silica as per analysis (%)	Reactive silica (%)	Percentage of reactive silica in total silica.
		FROM	TO	THICK			
1	CMMD 2/9+2/10	11.50	13.00	1.50	7.24	6.79	93.78
2	CMMD 2/13+2/14+2/15	14.00	16.50	2.50	7.06	6.61	93.63
3	CMMD 5/10+5/11+5/12	12.70	16.10	3.40	8.66	8.23	95.03
4	CMMD 3/5+3/6	8.00	10.00	2.00	10.17	8.58	84.37

The above result shows the total available alumina ranges between 29.73% to 50.67% with an average of 36.845% in Madua Pat-I block. Furthermore, the results at the reactive silica ranges between 6.61% to 8.58% with an average of 7.55 % and the percentage of reactive silica out of the total silica analyzed ranges between 84.37 to 95.03%. The result is relatively satisfactory for utilization of alumina in beneficiation plant.

5.9.0 LABORATORY PROCEDURE

The analysis of lithomarge by ICP-MS typically involves a series of steps, including sample preparation, digestion, and analysis. Here is a general overview of the methodology:

Sample preparation: The first step in the analysis of lithomarge by ICP-MS is sample preparation. The sample is typically ground to a fine powder and then sieved to remove any large particles or debris that could interfere with the analysis. The sample may then be dried or ignited to remove any water or organic matter.

Digestion: The next step is digestion, which involves dissolving the sample in a suitable acid or mixture of acids to release the elements of interest. Depending on the nature of the lithomarge and the elements being analyzed, different digestion methods may be used. Commonly used acids for lithomarge digestion include hydrofluoric acid (HF), nitric acid (HNO_3), hydrochloric acid (HCl), and aqua regia (a mixture of nitric and hydrochloric acids).

Analysis: After digestion, the sample solution is typically analyzed using ICP-MS. ICP-MS is a highly sensitive analytical technique that can detect and quantify a wide range of elements, including both major and trace elements. The sample solution is introduced into the ICP-MS instrument, where it is vaporized, atomized, and ionized. The resulting ions are then separated based on their mass-to-charge ratio and detected by a mass spectrometer. The resulting data can be used to determine the concentrations of various elements in the lithomarge sample.

5.10.0 X-RAY DIFFRACTION STUDIES

A total of four nos. of composite samples representing the bauxite deposit havebeen studied by X-ray diffraction techniques to know mineralogical distribution of the different minerals present in the bauxite ore of the block. The result of the X-Ray Diffraction study of four nos. composite samples have been presented in Annexure-VII/A.

X-ray diffraction studies of 4 composite samples of bauxite have revealed the presence of Gibbsite, Goethite and Haematite as major constituent (≥5 %); of which, Gibbsite is predominantly present;Kaolinite, Anatase, Dickite, Boehmite, Haematite, Albite, Chlorite, Calcite, Phlogopite, Ilmenite and Magnetite occur as minor minerals (≤5 to ≥1%). Muscovite, Rutile, Pyrite, Dickite, Ilmenite, Biotite, Albite, Anorthite, Magnetite, Boehmite and Phlogopite occur as accessory or trace minerals (≤1 %).

5.11.0 REE STUDIES AND TRACE ELEMENT STUDIES

A total 19 nos. of primary clay samples from all the 5 boreholes have been studied. The results of 19 nos. of primary clay are presented as Annexure-VI.

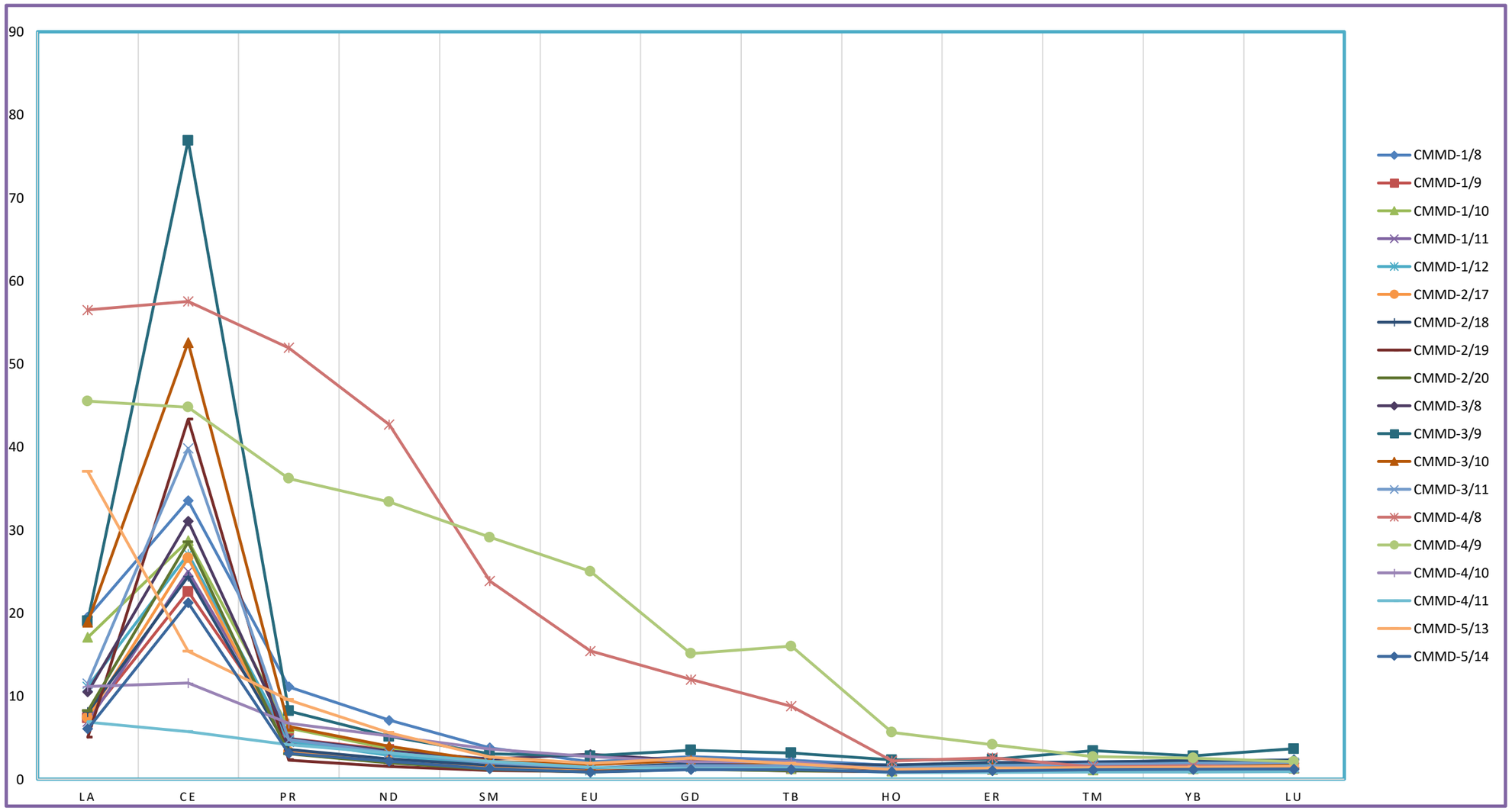
The elemental ranges of REE in 19 no. of primary clay samples in respect La, Ce and Nd are as follows:

- La ranges from 1.85 ppm to 20.74 ppm, Ce ranges from 5.46 ppm to 73.61 ppm, Nd ranges from 1.09 ppm to 30.35 ppm.
- Other elements such as Eu, Gd and Dy range from 0.07 ppm to 2.18 ppm, 0.35 to 4.63 ppm and 0.34 to 3.69 ppm respectively.

Below Chondrite normalized REE plots (spider plots) (Photograph – 8) of clay samples is showing plots generated from 19 nos. of primary clay samples. The combined plots do not show similarity and are hardly correlated to arrive at any conclusion. Although it is clear that LREE in the area are richer than HREE. The presence of clay minerals and anatase may hold the LREE content in the lithomarge clay.

Photograph - 8

Chondrite Normalization distribution of REE in primary clay/Saprolite samples, Madua Pat-I, Dist – Lohardaga, Jharkhand.



Twenty (20) nos of samples from 1 entire borehole have been studied for 33 suite of elements (REE, Trace Elements) ICP-AES/ICPMS. The detailed results are tabulated in annexure – VI/B. Among trace elements,

- Ga ranges from 33ppm to 96ppm.
- Zr ranges from 397ppm to 851ppm.

The concentration of Li is very low and <5ppm. Among REE,

- Ce ranges from 21.50ppm to 155.25 ppm.
- La ranges from 55.01ppm to 103.75ppm.

5.12.0 BOND WORK INDEX:

3 numbers of samples have been considered for determination of Bond Work Index (BWI) and the results are tabulated in Annexure VII/E.

5.13.0 MINERALIZATION IN THE BLOCK

The Madua Pat-1 bauxite is of pocket/bouldery type deposit and derived from the granitic gneisses of the Chotanagpur Gneissic Complex. The bauxite is considered to be the in-situ weathering product formed under tropical-subtropical climatic conditions. Bauxite occurs as irregular discontinuous boulders, lenses in the laterite capping (Photograph - 03 and 04). The bauxite lenses are sandwiched between laterite with gradational contacts. Bauxitization is more predominant in the upper horizon of the hard laterite zone. The intensity of bauxite boulders frequently changes laterally in all directions. The bauxite boulders are predominantly grey, greyish white or reddish white in colour with ferruginous coating on surface. Massive, Pisolitic, and Oolitic textures are seen. The pisolites are either wholly aluminous, or partly aluminous and partly ferruginous. The litho-contact of laterite/bauxite with lithomarge horizon is gradational and roughly follows the surface topography.

5.14.0 MINERAGRAPHIC STUDY:

5 samples were subjected to Mineragraphic Studies and the observations are tabulated in Annexure- VII/C.

5.15.0 GEOPHYSICAL STUDIES

Geophysical studies was beyond the scope of the work in the said block.

CHAPTER – 6

6.1.0 EXPLORATION BY SYSTEMATIC DRILLING:

The bauxite deposit of Madua Pat-I is well within the bauxite province of Chotanagpur plateau of Eastern India. The bauxite deposits of Eastern India, in general, are invariably associated with Laterite and are confined typically to the areas of physiographical relief such as plateaus and flat topped hills isolated by narrow valleys. Massive, compact and bouldery bauxite occurs within the horizontal capping of Laterite found on the high plateaus. Based on geological province, a scheme of exploration was formulated keeping in view to evaluate shape, size, mineralogical composition and grade of bauxite.

The prospecting in Madua Pat-I block by the GSI, Jharkhand during 1963 revealed that principal rock formation of this area comprises what has been earlier designated as the Chotanagpur granite gneiss comprising the various types of granite gneiss, porphyritic granite, granulitic rocks, quartz veins and pegmatites, which is a composite rock type. Laterite and bauxite in this area rest on the older Chotanagpur granite gneiss rock. Bauxite occurs as pockets of various dimensions and is Massive, compact and bouldery in nature.

The present Preliminary level exploration in the Madua Pat-I block has been taken up further to substantiate the existence of bauxite both along the longer and the shorter axis of the plateau and to decipher the quality and the thickness of Bauxite horizons in order to assess the bauxite resource in the block.

Exploratory core drilling for commenced after receipt of sanction order from NMET. The drilling, survey of the block completed on dated 24.06.2022 and Geological Mapping work was completed on October, 2022. Thus total 107.00m have been drilled in 5 boreholes at G-3 level exploration. Due to non-approachability factor, few boreholes have been slightly shifted to accessible location.

The bauxite deposits are invariably associated with Laterite and are confined typically to the areas of physiographical relief such as plateaus and flat topped hills isolated by narrow valleys. Massive, compact and bouldery bauxite occurs within the horizontal capping of Laterite found on the high plateaus. Bauxite is exposed at places where the Laterite capping is absent, as on scarp faces. Thus as part of exploration programe, the 5 boreholes were drilled by using dry conventional core drilling method with the help of RD-60 (MEC-348), skid mounted drill machine. As the bauxite column of the area was not known, five boreholes at G-3 stage had gone deeper

sufficiently to confirm the floor of the bauxite zone by exposing the variegated clay horizon or lithomarge. The considering the possibility of higher REE rich zone, the drilling in lithomarge zone was also approved by NMET for taking up for necessary analysis. So, the boreholes drilled during Preliminary Exploration were closed after drilling 2-4m in lithomarge horizon.

All the boreholes drilled were vertical. The drilling was carried out by HQ rod and NX casing. The inner diameter of the drill holes was 77.8 mm and that of core was 63.5 mm. The depths of the boreholes ranged between 19.00. to 24.30m. The boreholes drilled by CMPDI in Madua Pat-I block are of CMMD series. The details of boreholes drilled are given in Annexure-IA. The locations of these boreholes are marked on Borehole location plan and Geological plan and are presented as Plate No-5 and 3.

The short runs were drilled at an interval of 0.50m or 1.00m so that optimum core recovery is maintained and achieved. Systematic logging of litho-units was done and run-wise lithologs are given as Annexure- II.

Photograph – 9

RD-60 (MEC-348) Rig used for Drilling work at Madua Pat-I Block,

District – Lohardaga, Jharkhand.



6.2.0 DRILLED CORE LOGGING

The material obtained by way of dry core drilling, generally a mixture of core, powder and fragments, was stored run-wise in GI steel core boxes lined with a polythene. The core/fragments/cuttings were carefully examined and the details like

their color, lithology, physical nature, ore mineral constituents/ assemblage and type of occurrence of ore minerals along with the visual estimate of Al_2O_3% were recorded and grouped into different rock types/ore types depending upon the above mentioned characteristics. Due to poor recovery at places, extrapolations of various drilled depth were done on proportionate basis considering the physical characteristics of individual units recovered.

The run-wise litho logs of boreholes CMMD-01 to 05 (G3 stage) drilled by CMPDI are given as Annexure-II. The consolidated lithologs were prepared to show the lithounits as graphic representation. The grouping of litho units was done as Soil, Laterite, Ferruginous Laterite, Aluminous Laterite, Bauxite, Lateritic Bauxite, Lithomargic/Saprolitic Clays and Granitic Gneiss. Graphic lithologs (on 1:100) of the boreholes along with assay data for 5 radicals and zone details are presented as Plate no. 6.

Photograph – 10

Drilled Cores of borehole no – CMMD-03 at Madua Pat-I Block,

District – Lohardaga, Jharkhand.



6.3.0 DRILLED CORE PACKING AND TRANSPORTATION

On completion of Lithological logging the drilled boreholes, run wise samples of each borehole have been packed using polythene with proper leveling for necessary trapsporation to sample preparation unit for preparation of samples so that further analysis can be taken up.

Photograph – 11

Packet Drilled Cores for transportation to Laboratory of Borehole no-CMMD-02 at Madua Pat-I Block, District – Lohardaga, Jharkhand.



6.3.1 LIMITATIONS

The entire assessment of the deposit is based on surface and subsurface data obtained from geological mapping, drilling, outcrop, drill core sampling during preliminary exploration. Thus, limitations as regards to logical interpretation and analytical data of samples are deciphered.

6.3.2 Due to bouldery, lensoidal, erratic nature of the bauxite, the boreholes drilled at ramdom grids as approved by NMET may not give highest level of confidence about resources.

6.3.3 The entire interpretations of the exploration data is based on the assumptions of gradual change between the points of observations and rule of influence of nearest points.

6.4 ERECTION OF BOREHOLE PILLARS:

On completion of all the 5 drilled boreholes, Borehole Pillars indicating the Point number has been erected at the Drill site for future reference and identification.

Photograph – 12

Borehole Pillar (CMMD-02) erected at drill site, Madua Pat-I Block, District – Lohardaga, Jharkhand.



Photograph – 13

Borehole Pillar (CMMD-04) erected at drill site, Madua Pat-I Block, District – Lohardaga, Jharkhand.



6.5.0 SAMPLING AND ANALYSIS

6.5.1 PRIMARY AND CHECK SAMPLES

Considering the nature of the bauxite, aluminous Laterite and Laterite over considerable thickness in all boreholes and after study of lithological data generated during course of exploration, in general the sample length is taken as 0.50 m-1m. The drilled cores thus obtained were dried and samples drawn. The length of the sample may vary because of variation in lithology and concentration of mineralization. At the top as well as in the bottom of the ore zones, length of samples has been adjusted according to the length of overburden in the form of soil / Laterite. However, the zones of different grade as per the concentration of ore minerals like gibbsite/bohemite depending upon colour variation have been sampled separately. In case of a non-bauxite zone encountered within an ore zone, similar adjustments have been made to sample the non-ore zone separately. Except the visually confirmed overburden in the form of soil or alluvium at the top, nearly the entire drilled cores were divided in to two parts. One part was crushed to (-) 60 mesh initially and thereafter the crushed materials has been reduced to a 500 gm representative homogenous sample by gradual coning and quartering with repeated mixing. These samples were further powdered to(-) 120 mesh and reduced by the said methodology to prepare three representative samples weighing 100gm, 100 gm & 300 gm respectively. A packet of primary representative sample weighing about 100 gm was sent to chemical laboratory in an airtight double polythene bag packing for quantitative determination of five radicals i.e. Al_2O_3, SiO_2, Fe_2O_3, TiO_2 & LOI at Chemical Laboratory. Another packet of 100gm has been used for check analysis while the third packet has been preserved for preparation of composite samples. The left-out sample of (-) 60 mesh has been stored as duplicate sample.

The above mentioned 5 radicals are the basis and most essential chemical constituents to ensure the grade of bauxite zone in view of its utilization aspects in metallurgical and chemical industries. About 10% of primary samples (2 Nos.) were sent to DGM Lab, Jharkhand (external labs) as external check samples for analysis of 5 radicals i.e., Al_2O_3, SiO_2, Fe_2O_3, TiO_2 & LOI for ascertain reproducibility and reliability of grade estimation.

The results of primary analysis of borehole core samples for 5 radicals of G3 stage are furnished as Annexure III/A. Analytical results of primary samples versus check (Internal & External) samples are given as Annexure V.

6.5.2 COMPOSITE SAMPLE ANALYSIS

After receiving the chemical analysis for primary samples of the boreholes, the thickness of bauxite zone was delineated for each of the bauxite horizons intersected

in the boreholes at (+)30% Al_2O_3 & (-) 7% SiO_2, (+)30% Al_2O_3 & (-)10% SiO_2, (+)30% to (-)12% SiO_2, composite samples were prepared by combining all the constituent primary samples of the zone in proportion of their lengths they represent. A total of 4 composite samples have been generated by taking powdered material (-) 120 mesh of each primary samples in volume proportion and thoroughly mixed and followed by coning and quartering and all samples are analyzed for 7 radicals (Al_2O_3 , SiO_2 , Fe_2O_3 , TiO_2 V_2O_5, P_2O_5 & LOI) and 14 radicals (Al_2O_3, SiO_2, Fe_2O_3, TiO_2, CaO, MgO, SO_3, P_2O_5, V_2O_5, Mn_2O_3, K_2O, Na_2O, LOI & Organic). The details of analytical results of 7 & 14 radicals of composite samples G-3 level are given as Annexure-III/B and III/C.

6.5.3 PIT DIGGGING AND MEASUREMENT OF BULK DENSITY:

2 nos of Pits (1meter x 1meter x 1meter) have been executed in the said block area during Preliminary stage of Exploration work for expediting the measurement of Bulk density. (refer Annexure-VIII/B).

Photograph – 14

Stages of Pit digging at Exploratory block, Madua Pat-I Block, District – Lohardaga, Jharkhand.






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6.6.0 EVALUATION OF BAUXITE, ALUMINOUS LATERITE AND LATERITE ZONE

6.6.1 The investigation carried out by way of exploratory drilling with associated geological and laboratory studies which have generated the exploration data to establish the shape, size and geometry of the deposit. This formed the basis for evaluation of various deposit parameters of the block. The salient features of the evaluation are discussed below:

6.6.2 The grades of the bauxite intersected in the boreholes drilled in Madua Pat-I block has been assessed on the basis of compositional variations of main chemical constituent's viz. Al_2O_3, SiO_2, Fe_2O_3, TiO_2 and LOI of primary chemical analysis. The cut-off limit for compositional variations of main chemical constituents fixed for different grades of bauxite have been adopted in the report with marginal variations as per **END USE GRADE** classification recommended by IBM. The above grade wise compositional variation holds good in best way with the specification of bauxite utilized in different industries.

6.6.3 As per Ministry of Mines Notification dated 25th April 2018, the threshold value of bauxite mineral has been classified into the following two categories:

(i) For **Aluminous Laterite**: Al_2O_3– 20% (min.)

(ii) For **Bauxite**: Al_2O_3– 30% (min.) and SiO_2 (Total) –7% (max.)

Further, Laterite with Fe_2O_3 : Al_2O_3 ratio more than 1.00, and SiO_2 :Fe_2O_3 ratio less than 1.33 is termed as **Ferruginous Laterite**, while that having Fe_2O_3 :Al_2O_3 ratio less than 1.00 and SiO_2 :Al_2O_3 ratio less than 1.33 is termed as **Aluminous Laterite**.

6.7.0 BOREHOLE WISE EVALUATION

The depth of intersection details of Bauxite and Aluminous Bauxite are given below:

Table – 8

Borehole wise encounter details of Bauxite Zone, Madua Pat-I, Dist – Lohardaga, Jharkhand.

BH No.	Depth (in meter)		Lithology
	From	To	
CMDD-02	12.50	15.50	Bauxite
CMMD-03	18.00	19.30	Bauxite

Table– 9

Borehole wise encounter details of Aluminous Laterite Zone, Madua Pat-I, Dist – Lohardaga, Jharkhand.

BH No	From	To	Lithology
CMMD-01	1.00	4.00	Aluminous Laterite
CMMD-01	8.50	9.00	Aluminous Laterite
CMDD-02	2.50	3.00	Aluminous Laterite
CMDD-02	9.50	12.50	Aluminous Laterite
CMDD-02	15.50	16.50	Aluminous Laterite
CMMD-03	3.50	10.00	Aluminous Laterite
CMDD-04	4.00	10.40	Aluminous Laterite
CMDD-05	0.30	3.00	Aluminous Laterite
CMDD-05	3.60	5.40	Aluminous Laterite
CMDD-05	6.10	10.30	Aluminous Laterite
CMDD-05	12.70	16.10	Aluminous Laterite

Table – 10

Borehole wise encounter details of Variegated Clay Zone, Madua Pat-I, Dist. – Lohardaga, Jharkhand.

BH No	From	To	Lithology
CMMD-01	15.00	20.00	Variegated Clay
CMMD-02	18.00	22.50	Variegated Clay
CMDD-03	19.30	24.30	Variegated Clay

BH No	From	To	Lithology
CMDD-04	14.80	19.00	Variegated Clay
CMDD-05	17.50	20.70	Variegated Clay

Table – 11

Borehole wise encounter details of Ferruginous Laterite Zone, Madua Pat-I, Dist. – Lohardaga, Jharkhand.

BH No	From	To	Lithology
CMMD-01	4.00	5.50	Ferruginous Laterite
CMMD-01	7.50	8.50	Ferruginous Laterite
CMMD-02	3.00	4.00	Ferruginous Laterite
CMMD-02	6.00	9.50	Ferruginous Laterite
CMMD-02	13.00	14.00	Ferruginous Laterite
CMMD-02	16.50	17.50	Ferruginous Laterite
CMMD-04	12.60	13.40	Ferruginous Laterite

6.8.0 DETAILS OF ORE ZONES OF THE BOREHOLES DRILLED AND THEIR CORRELATION

The general rock types in the area as intersected in the boreholes with thin layer of soil cover varies from 0.30 m. (CMMD-05) to 2.50 m. (CMMD-03). The Bauxite profile varies in thickness from 1.30m (CMMD-03) to 3.00m (CMMD-02). The thickness of Aluminous Laterite profile varies in thickness from 0.50m (CMMD-01, 02, 03) to 6.40m (CMMD-04).

It is inferred from the surface and sub-surface data acquired during the G-3 and course of exploration, the Laterite-bauxite profile thickness ranges from 15.00 meter to 19.30 meter with average thickness of 12.35m.

The exploratory data reveals that bauxite deposit is of highly irregular in nature or bouldery/pockety type. The details of borehole wise bauxite zones are given

in the Table – 12 and 13.

Table – 12

Borehole-wise intersection of Bauxite zones delineated at (+) 30% Al_2O_3 and 7 %SiO_2 (max) Cut Off in boreholes drilled by CMPDI in Madua Pat-I Block, Dist: Lohardaga Jharkhand.

BH No	Depth Details (meter)		Thickness (meter)	Lithology	SiO_2%	Al_2O_3%	Fe_2O_3%	TiO_2%	LOI%
	From	To							
CMMD-02	12.50	13.00	0.50	Bauxite	5.68	33.78	30.66	8.99	20.25
CMMD-02	14.00	15.00	1.00	Bauxite	6.16	34.73	29.68	9.12	19.69
CMMD-02	15.00	15.50	0.50	Bauxite	4.88	39.86	24.55	8.02	22.03
CMMD-03	18.00	19.30	1.30	Bauxite	5.25	35.69	31.55	7.71	19.13

Table – 13

Borehole-wise intersection of Bauxite zones delineated at (+) 30% Al_2O_3 and 10 %SiO_2 (max) Cut Off in boreholes drilled by CMPDI in Madua Pat-I Block, Dist: Lohardaga Jharkhand.

BH No	Depth Details (meter)		Thickness (meter)	Lithology	SiO_2%	Al_2O_3%	Fe_2O_3%	TiO_2%	LOI%
	From	To							
CMMD-02	11.50	13.00	0.50	Bauxite	5.68	33.78	30.66	8.99	20.25
CMMD-02	14.00	15.00	1.00	Bauxite	6.16	34.73	29.68	9.12	19.69
CMMD-02	15.00	15.50	0.50	Bauxite	4.88	39.86	24.55	8.02	22.03
CMMD-03	18.00	19.30	1.30	Bauxite	5.25	35.69	31.55	7.71	19.13

Table – 14

Borehole-wise intersection of Aluminous Laterite zones delineated at (+) 20% Al_2O_3 Cut Off, Fe_2O_3 : Al_2O_3 ratio less than 1.00 and SiO_2 : Al_2O_3 ratio less than 1.33 in boreholes drilled by CMPDI in Madua Pat-I Block, Dist: Lohardaga Jharkhand.

BH No	From	To	Thickness	Lithology	SiO_2%	Al_2O_3%	Fe_2O_3%	TiO_2%	LOI%
CMMD-01	1.00	2.00	1.00	Aluminous Laterite	23.46	33.79	21.03	4.26	16.64
CMMD-01	2.00	3.00	1.00	Aluminous Laterite	15.49	38.01	20.40	5.40	20.02
CMMD-01	3.00	4.00	1.00	Aluminous Laterite	21.80	36.11	18.33	4.79	18.21
CMMD-01	8.50	9.00	0.50	Aluminous Laterite	8.90	35.95	31.44	7.02	16.02
CMMD-02	2.50	3.00	0.50	Aluminous laterite	28.03	28.77	26.56	2.20	13.73
CMMD-02	9.50	10.50	1.00	Aluminous laterite	11.79	35.16	30.23	5.98	16.14
CMMD-02	10.50	11.50	1.00	Aluminous laterite	11.48	27.32	39.51	4.92	16.03
CMMD-02	15.50	16.50	1.00	Aluminous Laterite	9.68	34.42	30.09	6.15	19.03
CMMD-03	3.50	4.50	1.00	Aluminous Laterite	33.60	29.34	20.53	2.55	12.90
CMMD-03	4.50	6.10	1.60	Aluminous Laterite	30.88	28.93	23.15	2.96	13.24
CMMD-03	6.10	7.10	1.00	Aluminous Laterite	28.40	28.13	25.82	3.59	13.32
CMMD-03	7.10	8.00	0.90	Aluminous Laterite	22.98	32.30	22.68	4.83	16.47
CMMD-03	8.00	9.20	1.20	Aluminous Laterite	9.86	45.91	13.27	7.68	22.68
CMMD-03	9.20	10.00	0.80	Aluminous Laterite	10.58	46.11	14.42	6.95	21.20
CMMD-04	4.00	5.50	1.50	Aluminous Laterite	30.65	27.07	25.89	3.21	12.45
CMMD-04	5.50	6.60	1.10	Aluminous Laterite	22.81	33.97	22.56	5.96	14.09
CMMD-04	6.60	8.00	1.40	Aluminous Laterite	14.49	39.30	20.08	8.15	17.26
CMMD-04	8.00	9.00	1.00	Aluminous Laterite	18.40	40.49	13.57	8.46	18.35
CMMD-04	9.00	9.80	0.80	Aluminous Laterite	14.43	40.72	16.40	10.04	17.79
CMMD-04	9.80	10.40	0.60	Aluminous Laterite	11.50	40.74	18.70	10.58	17.68
CMMD-05	0.30	1.50	1.20	Aluminous Laterite	34.00	30.13	18.88	2.41	13.59

BH No	From	To	Thickness	Lithology	SiO_2%	Al_2O_3%	Fe_2O_3%	TiO_2%	LOI%
CMMD-05	1.50	2.50	1.00	Aluminous Laterite	28.97	28.61	25.80	2.78	13.23
CMMD-05	2.50	3.00	0.50	Aluminous Laterite	27.86	29.00	26.21	3.14	13.03
CMMD-05	3.60	4.80	1.20	Aluminous Laterite	21.12	30.22	28.88	5.02	14.13
CMMD-05	4.80	5.40	0.60	Aluminous Laterite	15.89	38.31	21.37	6.48	17.25
CMMD-05	6.10	6.70	0.60	Aluminous Laterite	20.26	36.07	20.58	5.75	16.68
CMMD-05	6.70	7.90	1.20	Aluminous Laterite	14.74	40.48	18.07	7.54	18.39
CMMD-05	7.90	9.10	1.20	Aluminous Laterite	13.23	38.55	22.61	8.18	16.79
CMMD-05	9.10	10.30	1.20	Aluminous Laterite	12.14	33.00	31.63	6.26	16.35
CMMD-05	12.70	13.90	1.20	Aluminous Laterite	10.73	33.81	27.14	7.47	20.16
CMMD-05	13.90	15.10	1.20	Aluminous Laterite	7.73	35.80	25.28	9.25	21.20
CMMD-05	15.10	16.10	1.00	Aluminous Laterite	8.34	32.16	33.77	5.96	19.03

6.9.0 UTILISATION ASPECTS AND MINEABILITY (END USE GRADECLASSIFICATION AS PER INDIAN BUREAU OF MINES)

Table – 15

The specifications of Bauxite for various industrial uses

Industry	Al_2O_3	Fe_2O_3	TiO_2	SiO_2	P_2O_5	V_2O_5	CaO	MnO	LOI	Remarks
Chemical Industry	58% Min	2.5% Max								
Refractory Industry	58% Min	3% Max	3% Max	1-3% Max			0.5-0.6% Max			
Metallurgical Industry	40% Min	30 % Max.		4% Max	0.20% Max	0.2% Max			20% Max	Grade-I
	47% Min	30 % Max.		4% Max	0.20% Max	0.2% Max			20% Max	Grade-II
Abrasive Industry	50% Min.	3.5 - 12% Max	3-4% Max	2.5-3% Max						
Absorbent Industry	58% Min	2.0% Max	4.0% Max	3% Max	0.3% Max		2.0% Max (incl MgO)	0.1% Max	32% Max	
Steel Industry	55% Min			7% Max	Not Allowable					

CHAPTER – 7

7.0.0 RESOURCES AND GRADE

7.1.0 GENERAL

7.1.1 MEMC rules, January, 22 (as amended up to 14th December, 2021) suggests that for deposits specified in Schedule II (that includes Bauxite), 3 bore holes drilled so as to form a polygon in blocks of less than 100 hectares and 5 bore holes in blocks more than 100 hectares may be sufficient however for limestone, bauxite, potash and salt beds the grid spacing of bore holes may be 200m or closer for regular habit and 100m or closer for irregular habit.

Considering the same on receiving of approval from NMET, CMPDI carried out exploration to establish occurrence, depth continuity and thickness of bauxite horizon in Madua Pat-I block by drilling 5 vertical scout boreholes (preliminary level) in an area of 1.50 sq. km. with the objective to make qualitative and quantitative assessment of bauxite and to estimate inferred (333) category resource.

7.1.2 In Madua Pat-I block, bauxite zones have been delineated and resources are categorized as below:

- Bauxite grade- (+) 30% Al_2O_3 & (-) 10% SiO_2 (max.) cut-off.
- Bauxite grade- (+) 30% Al_2O_3 & (-) 7% SiO_2 (max.) cut-off.
- Aluminous Laterite - (+) 20.00% Al_2O_3 (min.) cut-off.

The bauxite resources have been computed polygon-wise as on above given parameters.

7.2.0 BASIC PARAMETERS AND ASSUMPTIONS

In order to estimate the resources and grades of bauxite following parameters have been considered

7.2.1 The basic parameters considered for the estimation of ore resources comprise the area extent, thickness and the tonnage factor of the ore. Data from preliminary level of exploration combines, all 5 boreholes and analysis of 49 primary samples and 4 no. of composite samples along with the data of geological mapping forms the basis of ore resource estimation.

7.2.2 The aerial extent/physical lateral continuity of the Laterite/bauxite capping as well as the non- bauxite portions within the deposit has been delineated with the help of surface geological map on a scale of 1:4000.

7.2.3 The data generated by drilling of exploratory boreholes has been accounted for the influence of the thickness and grade of bauxite zone in each borehole is assumed to

extend halfway to the adjacent borehole using the “rule of nearest point”. The influence of peripheral boreholes is assumed to extend up to the bauxite contact as demarcated based on geological mapping/ or up to the block boundary/ Plateau boundary.

7.2.4 The resource estimated is of In-Situ nature and are “Geological Resource”. However, an attempt has been made to delineate the bauxite zones based on (+) 20% Al_2O_3 irrespective of SiO_2 for estimation of Aluminous Laterite resource.

7.2.5 Computation of grade of the ore zone at a given exploration point is done by weighing the thickness of the individual samples in the column with their grades. As the thickness of the bauxite profile is very limited and it is to be mined by one bench as practiced in the adjacent area, the non-ore values within the bauxite column have been absorbed within the ore as far as possible. The thin partings of non-ore patches in the ore zone up to 2meters thickness are considered a part of the ore zone if it does not become non-ore by compounding the two or more sub zonesof ore and parting keeping to the cut-off limits for the purpose of estimating the grade also.

7.2.6 In case the non-ore parting is more than 2.00m thick, the ore zone has been separated into two or more parts. As such, the computations have taken care of possible dilution at the stage of mining.

7.2.7 Primary samples analysis has been considered for demarcation/delineation of zones of bauxite at (+) 30% Al_2O_3 and (+) 20% Al_2O_3 separately irrespective of SiO_2 and zone with TiO_2 content ≥5% with in Bauxite and Aluminous Zone.

7.2.8 All the boreholes at G-3 stage are closed in the lithomarge.

7.2.9 The thin and impersistent/localized zone (up to 2.00 m. thickness) intersected in the boreholes established on the basis of primary analysis were merged with the thick / most persistent zones within which they were occurring and average composite analysis have been calculated for these merged zones.

7.2.10 The area of non-development of bauxite zone has been marked on the polygonal map for each zone and these areas has been excluded from the resource estimation. The area considered for resource estimation of bauxite zone is indicated on polygonal map in Plate no. 7, 8.

7.2.11 The resource has been estimated by polygonal method by incorporating data generated during Preliminary stage of Exploration.

7.2.12 The entire bauxite resources have been kept under inferred category (333) as per exploration norms for irregular/boulder type bauxite enumerated in the Minerals (Evidence of Mineral Contents) Rule 2015.

7.2.13 The specific gravity of bauxite has been determined ranging from 2.25 to 2.41 arrived from mean of specific gravity determined for 50 bauxite samples (Annexure VIII/A) at CMPDI Laboratory. **However, for calculation of resources the average specific gravity 2.33 has been considered.** Further, the bulk density of the rock measured from Pit are ranging from 2.003 g/mL to 2.015 g/mL (Annexure VIII/B).

7.2.14 The average grade of zone-wise bauxite resource and total resource of the block has beenestimated by averaging grades of different area segment with different grades.

7.2.15 In view of bouldery/pockety and discontinuous nature of the bauxite occurrence in pockets in Madua Pat-I block, a recovery factor of 60% is considered for arriving at net in-situ resource from gross in situ resource. In the mining of this type of bauxite deposit, the recovery of ore about 60% holds good.

7.3.0 SPECIFIC GRAVITY

7.3.1 A total of 5 nos. samples collected from outcrop have been subjected to specific gravity determinations at CMPDI laboratory, Ranchi. The average specific gravity as determined is 2.33 is considered for ore resources calculation. The study details are given as Annexure- VIII/A. (Page-100).

7.4.0 OVERBURDEN

7.4.1 Bauxite in Madua Pat-I block is mostly concealed in lateritic cover. The thickness of over burdens of prime importance in planning the economics of an open cast mine. Normally with the increasing thickness of overburden, more mechanization is bound to be introduced as large quantity of overburden required to be handled for the same production of Bauxite. The range of overburden thickness in this Bauxite block is given below:

Table – 16

Range of overburden thickness above Bauxite Zone in Madua Pat-I Block, Dist: Lohardaga Jharkhand.

Minimum (m)	Maximum (m)
11.50 meter	18.00 meter
CMMD-02 (Bore Hole)	CMMD-03 (Bore Hole)

7.5.0 METHODOLOGY

7.5.1 The resources of bauxite have been estimated by polygon method. The methodology adopted, keeping the above assumptions in view, for resource estimation is as follows:

7.5.2 POLYGON METHOD:

Computation of resource and grade of bauxite in Madua Pat-I block is done by the

"Polygonal Method" (Average factor and Area method). For the bouldery and pockety nature of the bauxite deposits, the "polygonal" method of resource estimation appears to be more reliable than Iso- chore and Cross-sectional method. In Iso-chore" and "cross sectional" methods, drives the thought of gradual change of bauxite thickness and grade from one point to the adjacent point. This may not hold good for this type of deposit and hence, may lead to inaccurate resource estimation.

In this method, the thickness and grade of the bauxite zone at a given sample point was considered to hold good, up to the half distance between the point and the adjacent ones. This method, utilizes the 'Rule of the nearest points'. Each sample point has an area of influence in the form of rectangle, or a square or any other geometrical shape depending on the pattern of drilling grid used for exploration. In this way, the area of influence of a given point was delineated on the plan (Plate no. 7, 8). In case of peripheral borehole, the plateau boundary has been considered as influence area. The volume of the ore available in that area was arrived by the formula.

$$R = P_A \times Th \times Sp.\ Gravity$$

Where, **R**= Resource/ Tonnage.

$\mathbf{P_A}$ = Area of Polygon.

Th= Thickness of Bauxite.

Sp. Gravity= Specific Gravity (Taken as 2.33).

As such Madua Pat-I block has been divided into a series of 5 polygons (P-1 to P-5) by the perpendicular bisectors of lines drawn to connect the boreholes, at the specified cut-off considered. The plan area of the polygon/rectangle is measured on 1:4000 scale by computerusing AutoCAD Map software. The area of influence multiplied by the zone thickness encountered within the segment gives the volume of mineralised portion and this volume when multiplied by the average specific gravity (which is uniformly taken as 2.33) gives the gross tonnage of bauxite in each polygon/segment. The sum of tonnage in all the segments gives the total gross tonnage of bauxite at a particular cut off. Borehole-wise polygons and area for resource of bauxite zones are presented in Plate no. 7.

If one borehole encountered the bauxitic zone, but the other borehole just immediate neighbors in same grid line not encountered bauxite zone. In that case on 50% of influence area of positive borehole has been considered towards negative borehole for resource calculation of bauxite.

Due to boulder, pocket and discontinuous nature of the bauxite pockets in Madua Pat-I block, a recovery factor of 60% is considered for arriving at net in-situ resource from gross in situ resource. (Net resources have been reduced to 60% of the gross

resources).

The average grade of the ore tonnages is calculated by weighing the tonnage in the constituent polygons/segments against their average grades. The average thickness of the ore body is arrived at by dividing the total volume of the ore body by the total area in respective cutoffs.

7.6.0 CATEGORISATION OF RESOURCES

Resources of bauxite have been categorized under Inferred Mineral Resources (333) as per the United Nations Framework Classification (UNFC).

7.7.0 RESOURCES

7.7.1 The resource of Bauxite and Aluminous Laterite has been estimated by Polygonal method are furnished as Annexure-IXA to IXB (polygon wise, different zone wise and gross and net).

7.7.2 All resources belong to inferred category (333) for pocket/boulder type bauxite deposit**.** Specific gravity considered for calculation is 2.33 gm/cc.

7.8.0 SUMMARY OF RESOURCES:

7.8.1 A total of 1.397 **million tonnes net in-situ bauxite resource** with an average grade of 33.46% Al_2O_3, 5.62% SiO_2, 32.74% Fe_2O_3, 8.01% TiO_2 & 19.53% LOI have been estimated in the block in an area of 0.38 sq. km under the cut-off value of 7% SiO_2**.** A total of **1.814 million tonnes net in-situ bauxite resource** with an average grade of 33.14% Al_2O_3, 6.33% SiO_2, 32.59% Fe_2O_3, 7.90% TiO_2 & 19.41% LOI have been estimated in the block in an area of 0.38 sq. km under the cut-off value of 10% SiO_2. All the resource belongs to **"Inferred Category" (333)**. The Bauxite resource estimated by polygonal method is given in Annexure-IXA.

7.8.2 A total of **11.666 million tonnes net in-situ aluminous Laterite resource** with an average grade of 34.56% Al_2O_3, 18.15% SiO_2 have been estimated in the block in an area of 1.33 sq. km**.** All the resource belongs to **"Inferred Category" (333)**. The aluminous Laterite resource estimated by polygonal method is given in Annexure- IX/B.

7.8.3 Ore resources based on TiO_2 content has been taken up **(≥5% TiO_2)** with-in potential 'Bauxite zone' as well as 'Aluminous Laterite zone'. The resources remains same as of Bauxite in the 'Bauxite zone' whereas, in Aluminous Laterite zone a total of **9.551 million tonnes** net in-situ ore containing **TiO_2 ≥5%** containing 35.47% Al_2O_3, 15.48% SiO_2, 24.10% Fe_2O_3, 7.04% TiO_2 & 17.22% LOI have been estimated in the block and belongs to **"Inferred Category" (333).**

FIGURE: POLYGONAL MAPS (BAUXITE ZONE)

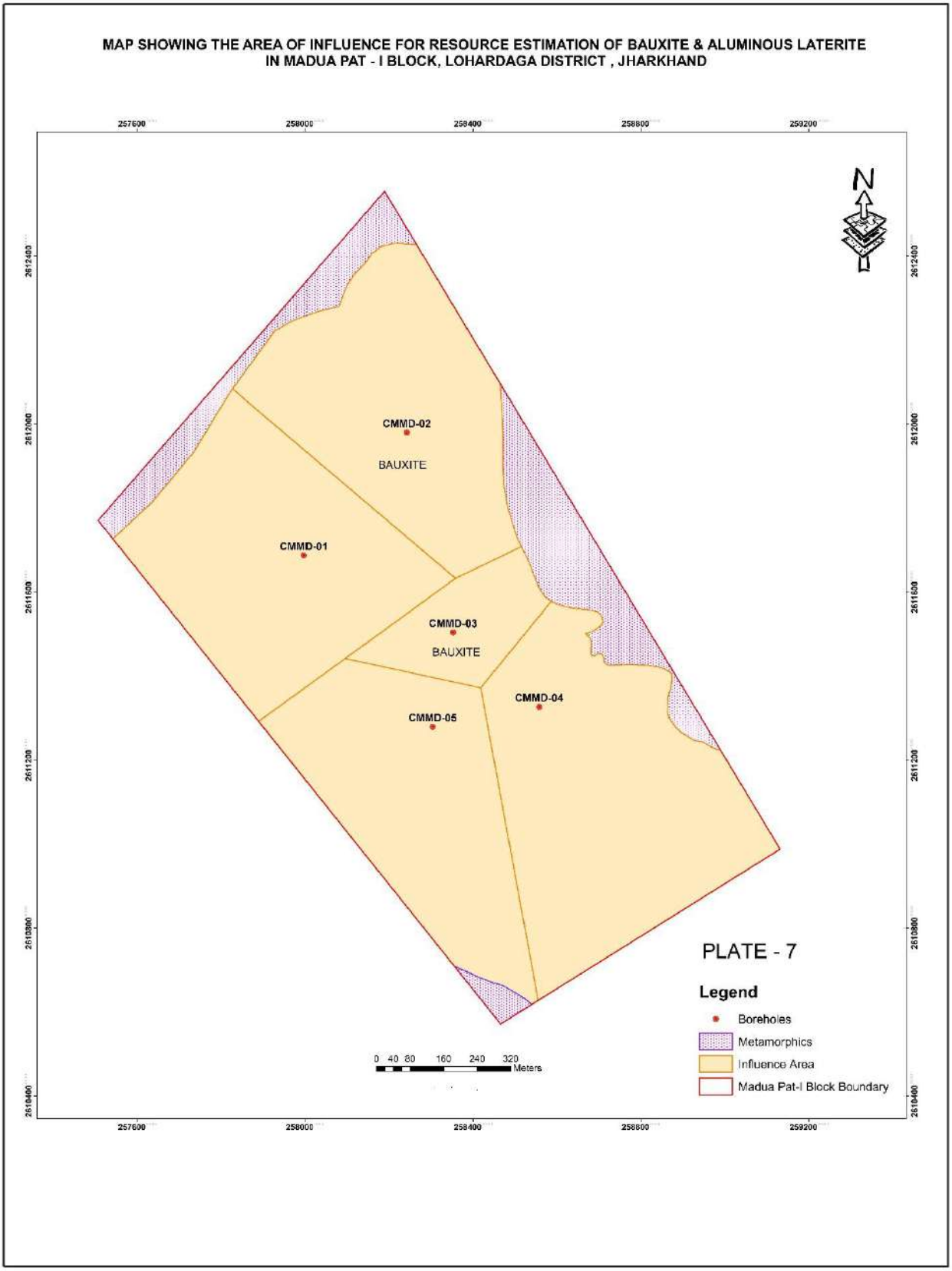
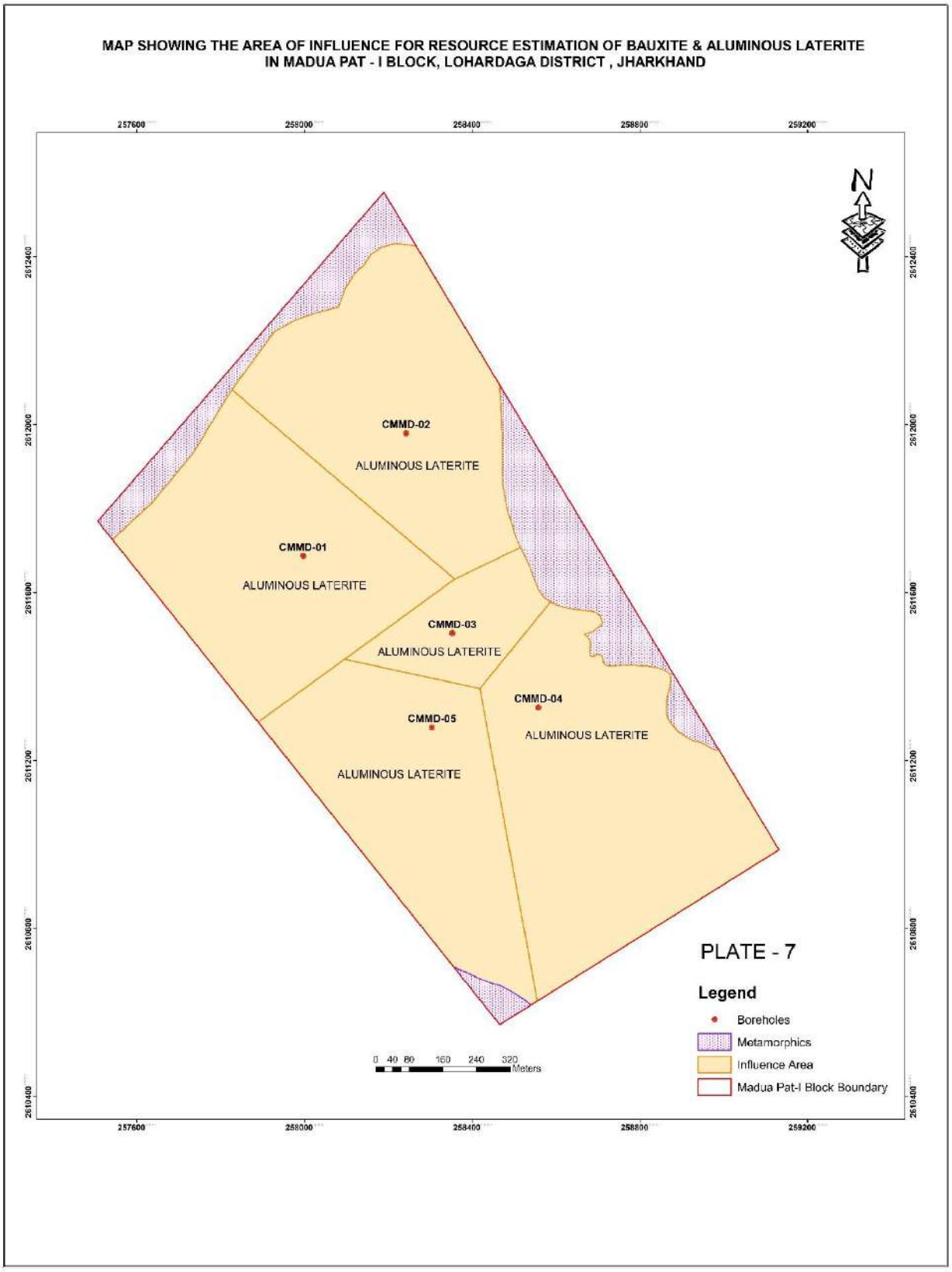


FIGURE: POLYGONAL MAPS (ALLUMINOUS LATERITE ZONE)



CHAPTER – 8

8.0.0 CONCLUSIONS AND RECOMMENDATIONS

8.1 CONCLUSION

On enactment of MMDR Amendment Act- 2015, Mineral (Evidence of Mineral Contents) Rule- 2015 and Mineral Auction Rule - 2015, Govt. of India directed State Governments to speed up exploration work for different Mineral Commodities Especially Notified Minerals i.e., Iron ore, Manganese, Bauxite and Limestone in the respective states for auctioning of the blocks. An MOU has been signed between Department of Mines and Geology (DMG), Government of Jharkhand and Central Mine Planning & Design Institute Limited (CMPDI) for exploration of various mineral blocks by CMPDI in the state of Jharkhand. DMG, Government of Jharkhand requested CMPDI to take up exploration work of Bauxite in 2 Blocks of Lohardaga district in Jharkhand. On request of CMPDI, DMG, Jharkhand provided information required for preparation of exploration proposals pertaining to Bauxite deposit vide letter No भूिनअंवे– 13/200-983 dt. 25/06/2021. Accordingly, CMPDI prepared the proposal for preliminary stage of exploration (G3) for Bauxite in the proposed block viz. Madupat -1, District - Lohardaga, Jharkhand and put up for consent of DMG, Jharkhand for taking up exploration work under NMET funding.

Bauxite deposits of the area around Lohardaga, Ranchi district, Bihar, were investigated during the field-season 1961-62 with a view to estimate the reserves of various grades of ore in the unleased areas. All the bauxite bearing areas in the vicinity were also investigated and plotted on a Degree Sheet (scale 1:2,53,440) with a view to assess the regional reserves of bauxite. As a part of this program, Madua Pat was also investigated and found that about 1.5m- 3m of bauxite of various grades is exposed in the quarries worked by Minerals & Minerals Ltd. The bauxite is pinkish or grey in colour and massive or bouldery in nature. Sometimes the bauxite is found to grade gradually into aluminous Laterite and Laterite. Quarries on the northern scarp show development of the best quality bauxite.

The Exploration Program for G3 stage of Exploration approved by TCC of NMET during its 33rd Meeting on dated 24.09.2021. Later the executive committee of NMET in its 22nd meeting held on 22.12.2021 approved the proposal for upgrading to G-3 level. The administrative / financial sanction for exploration for Madua Pat-I block at estimated cost of 42.98 lakhs INR was received from Ministry of Mines, Govt of India, National Mineral Exploration Trust, New Delhi vide F.No.23/228/2022-NMET/17, dated, 02.05.2022.

The principal rock formation of this area comprises what has been earlier designated

as the Chotanagpur granite gneiss, which is a composite rock type. Laterite and bauxite in this area rest on the older Chotanagpur granite gneiss comprising the various types of granite gneiss, porphyritic granite, granulitic rocks, quartz veins and pegmatites. During Geological Mapping carried out in 1:4000 scale, different lithological units like Bauxite, Aluminous Laterite, Massive Laterite, Laterite, Saprolite Zone and Granitic Gneiss have been identified on field and plotted in Geological Map.

The area under investigation is in lateritised zone with irregular bauxite lenses overlies the basalt, the boreholes were drilled by using combination of dry core drilling method with the help of RD-60 (MEC-351), skid mounted drill machines. As the bauxite column of the area was not known, five boreholes at G-3 stage has gone sufficiently deeper to confirm the floor of the bauxite zone up to lithomarge. The boreholes drilled were closed after drilling 2-4 meters in lithomarge.

Bauxite and laterite occur as fine grained, massive dense rock. Pisolitic variety are also common. The pisolites range in size up to 1 cm in diameter. Laterite is brown to red-brown in color. Greyish tinge in the lower part of the laterite crust is noted due to the presence of goethite.

A total of 1.397 million tonnes net in-situ bauxite resource with an average grade of 33.46% Al_2O_3, 5.62% SiO_2, 32.74% Fe_2O_3, 8.01% TiO_2 & 19.53% LOI have been estimated in the block in an area of 0.38 sq. km under the cut-off value of 7% SiO2. A total of 1.814 million tonnes net in-situ bauxite resource with an average grade of 33.14% Al_2O_3, 6.33% SiO_2, 32.59% Fe_2O_3, 7.90% TiO_2 & 19.41% LOI have been estimated in the block in an area of 0.38 sq. km under the cut-off value of 10% SiO2.

A total of 11.666 million tonnes net in-situ 'Aluminous Laterite' resource with an average grade of 34.56% Al_2O3, 18.15% SiO_2 have been estimated in the block in an area of 1.33 sq. km.

An attempt of measurement of ore resources based on TiO_2 content has been taken up (≥5% TiO_2) with-in potential 'Bauxite zone' as well as 'Aluminous Laterite zone'. The resources remains same as of Bauxite in the 'Bauxite zone' whereas, in Aluminous Laterite zone a total of 9.551 million tonnes net in-situ ore containing TiO_2 ≥5% containing 35.47% Al_2O_3, 15.48% SiO_2, 24.10% Fe_2O_3, 7.04% TiO_2 & 17.22% LOI have been estimated in the block.

Ore resource is classified as "Inferred Category" as per the United Nations Frame Work Classification (UNFC) and assigned code 333.

8.2.0 LIMITATIONS

8.2.1 Contours beyond TLS surveyed boundary (up to Patch boundary) generated with the help of elevation taken from google earth and using open source software like GPS Visualizer, TCX converter and QGIS. The limitations of these open source software is +/- 2 meters in elevation.

8.2.2 Sufficient boreholes in 100*100 Grid considering the irregular habit of Bauxite occurrence as per MEMC Rule, 2022, could have helped in determination of resources more precisely considering the boulder and patchy nature of occurrence of Bauxite.

8.3.0 RECOMMENDATIONS

8.3.1 The exploration is carried out as per Mineral (Evidence of Mineral Contents) Rule 2015, Mineral Auction Rule-2015 and MMDR Amendment Act – 2015.

8.3.2 This level of exploration will facilitate the state government of Jharkhand for auctioning of block under Composite License.

8.3.3 Bauxite mineralized zones occur as irregular, discontinuous lenses and pockets and are bouldery in nature. The continuity of bauxite zone from one exploration point to adjacent point is not certain even for a short distance. It is, therefore, recommended to upgrade the deposit at G-2 stage by close spaced drilling at 200x200 m grid and further 100mX100m or less spaced drilling may be done in throughout the block to know actual behavior of bauxite horizon and to get a more confidence level of resource estimation (332 indicated resources of MEMC).

8.3.4 A few deep pits and trenching along longer and shorter dimension of the plateau may be carried out for assessing the lateral and vertical configuration of bauxite profile to ascertain its continuity along three dimensions.

BIBLIOGRAPHY

- Report on the Bauxite Deposits near Lohardaga area, Ranchi District, Bihar, Field-Season 1961-62, March, 1963, by N. Majumdar, Assistant Geologist, Geological Survey of India.
- Report on compilation of one-inch scale geological maps of the bauxite belt in Bihar and Orissa, in the jurisdiction of eastern region, GSI, Field Season 1987-88 by T. K. Sen, Geologist (Sr.), GSI.

ANNEXURE-I/A

DETAILS OF CO-ORDINATES (IN WGS-84) OF DRILLED BOREHOLES WITH REDUCED LEVEL, DRILLED DEPTHS, DATE OF COMMENCMENT AND CLOSURE DETAILS OF MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

BH No	Longitude (Easting)	Latitude (Northing)	RL	Total depth (meter)	Commencement Date	Closure date
CMMD-01	84°37'42.44506"E	23°35'51.51938"N	1010.944	20	31.03.22	13.04.22
CMMD-02	84°37'50.92877"E	23°36'01.14581"N	1010.358	23	13.04.22	24.04.22
CMMD-03	84°37'55.09375"E	23°35'45.75457"N	1015.364	24.3	24.04.22	02.05.22
CMMD-04	84°38'02.43638"E	23°35'40.10449"N	1015.111	19	03.05.22	09.05.22
CMMD-05	84°37'53.50819"E	23°35'38.42704"N	1012.233	20.7	09.05.22	13.05.22

ANNEXURE-I/B

CO-ORDINATES OF CARDINAL POINTS (IN WGS-84) OF BLOCK BOUNDARY PILLARS, MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

BLOCK BOUNDARY	CO-ORDINATES		REDUCED LEVEL
POINT	NORTHING	EASTING	(In meter).
A	2610786.617	258810.537	1064.79
B	2611769.969	257506.474	870.432
C	2610570.943	258465.214	965.887
D	2610987.141	259131.602	1067.42
E	2612553.658	258189.146	997.324

ANNEXURE - II

STATEMENT SHOWING RUN-WISE LITHOLOGS OF BOREHOLES DRILLED BY CMPDI, MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

NORTHING	23°35'51.51938"N	DEPTH DRILLED (M)	20 Meter.
EASTING	84°37'42.44506"E	DATE OF COMMENCEMENT	31.03.2022
REDUCED LEVEL (M):	1010.944 Meter.	DATE OF CLOSURE	13.04.2022

BH No	From (meter)	To (meter)	Thickness (meter)	Recovery (meter)	Lithology	Color	Natural Evidence/ Features
CMMD-01	0.00	0.50	0.50	0.50	Soil	Whitish Brown	Earthen
CMMD-01	0.50	1.00	0.50	0.50	Laterite	Brown	Cavernous and Pisiolitic
CMMD-01	1.00	1.50	0.50	0.50	Aluminous Laterite	Whitish Brown	Cavernous and Pisiolitic
CMMD-01	1.50	2.00	0.50	0.50	Aluminous Laterite	Whitish Brown	Cavernous and Pisiolitic
CMMD-01	2.00	2.50	0.50	0.50	Aluminous Laterite	Whitish Brown	Cavernous and Pisiolitic
CMMD-01	2.50	3.00	0.50	0.50	Aluminous Laterite	Pinkish white	Pisiolitic
CMMD-01	3.00	3.50	0.50	0.50	Aluminous Laterite	Pinkish white	Cavernous and Pisiolitic
CMMD-01	3.50	4.00	0.50	0.50	Aluminous Laterite	Brown	Cavernous and Pisiolitic
CMMD-01	4.00	4.50	0.50	0.50	Ferruginous Laterite	Brown	Cavernous and Pisiolitic
CMMD-01	4.50	5.00	0.50	0.50	Ferruginous Laterite	Brown	Cavernous and Pisiolitic
CMMD-01	5.00	5.50	0.50	0.50	Ferruginous Laterite	Brown	Cavernous and Pisiolitic
CMMD-01	5.50	6.00	0.50	0.50	Laterite	Brown	Cavernous and Pisiolitic
CMMD-01	6.00	6.50	0.50	0.50	Laterite	Brown	Cavernous and Pisiolitic
CMMD-01	6.50	7.00	0.50	0.50	Laterite	Brown	Cavernous and Pisiolitic

NORTHING	23°35'51.51938"N	DEPTH DRILLED (M)	20 Meter.
EASTING	84°37'42.44506"E	DATE OF COMMENCEMENT	31.03.2022
REDUCED LEVEL (M):	1010.944 Meter.	DATE OF CLOSURE	13.04.2022

BH No	From (meter)	To (meter)	Thickness (meter)	Recovery (meter)	Lithology	Color	Natural Evidence/ Features
CMMD-01	7.00	7.50	0.50	0.50	Laterite	Brown	Cavernous and Pisiolitic
CMMD-01	7.50	8.00	0.50	0.50	Ferruginous Laterite	Brown	Cavernous and Pisiolitic
CMMD-01	8.00	8.50	0.50	0.50	Ferruginous Laterite	Brown	Cavernous and Pisiolitic
CMMD-01	8.50	9.00	0.50	0.50	Aluminous Laterite	Whitish Pink	Pisiolitic
CMMD-01	9.00	9.50	0.50	0.50	Laterite	Reddish Brown	Pisiolitic
CMMD-01	9.50	10.00	0.50	0.50	Laterite	Reddish Brown	Cavernous
CMMD-01	10.00	10.50	0.50	0.50	Laterite	Deep Brown	Cavernous
CMMD-01	10.50	11.00	0.50	0.50	Laterite	Deep Brown	Cavernous
CMMD-01	11.00	11.50	0.50	0.50	Laterite	Brown	Cavernous
CMMD-01	11.50	12.00	0.50	0.50	Laterite	Brown	Cavernous
CMMD-01	12.00	12.50	0.50	0.50	Laterite	Brown	Cavernous
CMMD-01	12.50	13.00	0.50	0.50	Laterite	Yellowish Brown	Cavernous
CMMD-01	13.00	13.50	0.50	0.50	Laterite	Yellowish Brown	Cavernous
CMMD-01	13.50	14.00	0.50	0.50	Laterite	Whitish Brown	Cavernous
CMMD-01	14.00	14.50	0.50	0.50	Laterite with Clay	Whitish Brown	Cavernous
CMMD-01	14.50	15.00	0.50	0.50	Laterite with Clay	Whitish Brown	Cavernous
CMMD-01	15.00	15.50	0.50	0.50	Variegated Clay	Yellowish Brown	Finely grained, soily, soft

NORTHING	23°35'51.51938"N	DEPTH DRILLED (M)	20 Meter.
EASTING	84°37'42.44506"E	DATE OF COMMENCEMENT	31.03.2022
REDUCED LEVEL (M):	1010.944 Meter.	DATE OF CLOSURE	13.04.2022

BH No	From (meter)	To (meter)	Thickness (meter)	Recovery (meter)	Lithology	Color	Natural Evidence/ Features
CMMD-01	15.50	16.00	0.50	0.50	Variegated Clay	Yellowish Brown	Finely grained, soily, soft
CMMD-01	16.00	16.50	0.50	0.50	Variegated Clay	Yellowish Brown	Finely grained, soily, soft
CMMD-01	16.50	17.00	0.50	0.50	Variegated Clay	Yellowish Brown	Finely grained, soily, soft
CMMD-01	17.00	17.50	0.50	0.50	Variegated Clay	Yellowish Brown	Finely grained, soily, soft
CMMD-01	17.50	18.00	0.50	0.50	Variegated Clay	Yellowish Brown	Finely grained, soily, soft
CMMD-01	18.00	18.50	0.50	0.50	Variegated Clay	Yellowish Brown	Finely grained, soily, soft
CMMD-01	18.50	19.00	0.50	0.50	Variegated Clay	Yellowish Brown	Finely grained, soily, soft
CMMD-01	19.00	19.50	0.50	0.50	Variegated Clay	Yellowish Brown	Finely grained, soily, soft
CMMD-01	19.50	20.00	0.50	0.50	Variegated Clay	Yellowish Brown	Finely grained, soily, soft

NORTHING	23°36'01.14581"N	DEPTH DRILLED (M)	23 Meter.
EASTING	84°37'50.92877"E	DATE OF COMMENCEMENT	13.04.2022
REDUCED LEVEL (M):	1010.358Meter.	DATE OF CLOSURE	24.04.2022

BH No	From (meter)	To (meter)	Thickness (meter)	Recovery (meter)	Lithology	Color	Natural Evidence/ Features
CMDD-02	0.00	0.60	0.60	0.55	Soil/Laterite	Whitish Brown	Earthen
CMDD-02	0.60	1.00	0.40	0.40	Laterite	Brown	Cavernous, oolotic
CMDD-02	1.00	1.50	0.50	0.45	Laterite	Brown	Cavernous, oolotic
CMDD-02	1.50	2.00	0.50	0.50	Laterite	Brown	Cavernous, oolotic
CMDD-02	2.00	2.50	0.50	0.50	Laterite	Brown	Cavernous, oolotic
CMDD-02	2.50	3.00	0.50	0.50	Aluminous laterite	Light Brown	Cavernous and Pisiolitic
CMDD-02	3.00	3.50	0.50	0.50	Ferruginous Laterite	Brown	Cavernous and Pisiolitic
CMDD-02	3.50	4.00	0.50	0.50	Ferruginous Laterite	Brown	Cavernous and Pisiolitic
CMDD-02	4.00	4.50	0.50	0.50	laterite	Brown	Cavernous
CMDD-02	4.50	5.00	0.50	0.50	Laterite	Brown	Cavernous
CMDD-02	5.00	5.50	0.50	0.50	Laterite, cavernous	Brown	Cavernous
CMDD-02	5.50	6.00	0.50	0.50	Laterite, cavernous	Brown	Cavernous
CMDD-02	6.00	6.50	0.50	0.50	Ferruginous Laterite	Brown	Cavernous and Pisiolitic
CMDD-02	6.50	7.00	0.50	0.50	Ferruginous Laterite	Brown	Cavernous and Pisiolitic
CMDD-02	7.00	7.50	0.50	0.50	Ferruginous Laterite	Brown	Cavernous and Pisiolitic
CMDD-02	7.50	8.00	0.50	0.50	Ferruginous Laterite	Brown	Cavernous and Pisiolitic
CMDD-02	8.00	8.50	0.50	0.50	Ferruginous Laterite	Brown	Cavernous and Pisiolitic

NORTHING	23°36'01.14581"N	DEPTH DRILLED (M)	23 Meter.
EASTING	84°37'50.92877"E	DATE OF COMMENCEMENT	13.04.2022
REDUCED LEVEL (M):	1010.358Meter.	DATE OF CLOSURE	24.04.2022

BH No	From (meter)	To (meter)	Thickness (meter)	Recovery (meter)	Lithology	Color	Natural Evidence/ Features
CMDD-02	8.50	9.00	0.50	0.50	Ferruginous Laterite	Brown	Cavernous and Pisiolitic
CMDD-02	9.00	9.50	0.50	0.50	Ferruginous Laterite	Brown	Cavernous and Pisiolitic
CMDD-02	9.50	10.00	0.50	0.50	Aluminous laterite	Whitish Pink	Cavernous and Pisiolitic
CMDD-02	10.00	10.50	0.50	0.50	Aluminous laterite	Whitish Pink	Cavernous and Pisiolitic
CMDD-02	10.50	11.00	0.50	0.50	Aluminous laterite	Whitish Pink	Cavernous and Pisiolitic
CMDD-02	11.00	11.50	0.50	0.50	Aluminous laterite	Whitish Pink	Cavernous and Pisiolitic
CMDD-02	11.50	12.00	0.50	0.50	Siliceous Bauxite	White	Oolitic, Cavernous
CMDD-02	12.00	12.50	0.50	0.50	Siliceous Bauxite	White	Oolitic, Cavernous
CMDD-02	12.50	13.00	0.50	0.50	Bauxite	White	Oolitic, Cavernous
CMDD-02	13.00	13.50	0.50	0.50	Ferruginous Bauxite	Brown	Cavernous and Pisiolitic
CMDD-02	13.50	14.00	0.50	0.50	Ferruginous Laterite	Brown	Cavernous and Pisiolitic
CMDD-02	14.00	14.50	0.50	0.50	Bauxite	White	Oolitic, Cavernous
CMDD-02	14.50	15.00	0.50	0.50	Bauxite	White	Oolitic, Cavernous
CMDD-02	15.00	15.50	0.50	0.50	Bauxite	White	Oolitic, Cavernous
CMDD-02	15.50	16.00	0.50	0.50	Aluminous laterite	Whitish Pink	Cavernous and Pisiolitic
CMDD-02	16.00	16.50	0.50	0.50	Aluminous laterite	Whitish Pink	Cavernous and Pisiolitic
CMDD-02	16.50	17.00	0.50	0.50	Ferruginous Laterite	Brown	Cavernous and Pisiolitic

NORTHING	23°36'01.14581"N	DEPTH DRILLED (M)	23 Meter.
EASTING	84°37'50.92877"E	DATE OF COMMENCEMENT	13.04.2022
REDUCED LEVEL (M):	1010.358Meter.	DATE OF CLOSURE	24.04.2022

BH No	From (meter)	To (meter)	Thickness (meter)	Recovery (meter)	Lithology	Color	Natural Evidence/ Features
CMDD-02	17.00	17.50	0.50	0.50	Ferruginous Laterite	Brown	Cavernous and Pisiolitic
CMDD-02	17.50	18.00	0.50	0.50	Laterite with clay	Deep Brown	Cavernous
CMDD-02	18.00	18.50	0.50	0.50	Variegated clay	Pinkish brown	Finely grained, soily, soft
CMDD-02	18.50	19.20	0.70	0.70	Variegated clay	Pinkish brown	Finely grained, soily, soft
CMDD-02	19.20	20.50	1.30	1.30	Variegated clay	Fleshy	Finely grained, soily, soft
CMDD-02	20.50	20.80	0.30	0.30	Variegated clay	Light brown	Finely grained, soily, soft
CMDD-02	20.80	21.50	0.70	0.70	Variegated clay	Yellowish Brown	Finely grained, soily, soft
CMDD-02	21.50	22.00	0.50	0.50	Variegated clay	Yellowish Brown	Finely grained, soily, soft
CMDD-02	22.00	22.50	0.50	0.50	Variegated clay	Yellowish Brown	Finely grained, soily, soft
CMDD-02	22.50	23.00	0.50	0.50	Variegated clay	Yellowish Brown	Finely grained, soily, soft

NORTHING	23°35'45.75457"N	DEPTH DRILLED (M)	24.3 Meter.
EASTING	84°37'55.09375"E	DATE OF COMMENCEMENT	24.04.2022
REDUCED LEVEL (M):	1015.364 meter	DATE OF CLOSURE	02.05.2022

BH No	From (meter)	To (meter)	Thickness (meter)	Recovery (meter)	Lithology	Color	Natural Evidence/ Features
CMMD-03	0.00	0.30	0.30	0.30	Soil	Earthy Brown	Earthen
CMMD-03	0.30	1.00	0.70	0.70	Soil	Earthy Brown	Earthen
CMMD-03	1.00	1.50	0.50	0.50	Soil	Earthy Brown	Earthen
CMMD-03	1.50	2.50	1.00	1.00	Soil with Morems	Earthy Brown	Nodular
CMMD-03	2.50	3.00	0.50	0.42	Laterite	Reddish Brown	Cavernous
CMMD-03	3.00	3.50	0.50	0.45	Laterite	Reddish Brown	Cavernous
CMMD-03	3.50	4.00	0.50	0.37	Aluminous Laterite	Pinkish Brown	Cavernous
CMMD-03	4.00	4.50	0.50	0.50	Aluminous Laterite	Pinkish Brown	Oolitic, Cavernous
CMMD-03	4.50	5.20	0.70	0.65	Aluminous Laterite	Whitish Brown	Oolitic, Cavernous
CMMD-03	5.20	6.10	0.90	0.88	Aluminous Laterite	Whitish Brown	Oolitic, Cavernous
CMMD-03	6.10	7.10	1.00	0.95	Aluminous Laterite	Whitish Brown	Oolitic, Cavernous
CMMD-03	7.10	7.50	0.40	0.40	Aluminous Laterite	Pinkish Brown	Cavernous
CMMD-03	7.50	8.00	0.50	0.35	Aluminous Laterite	Pinkish Brown	Cavernous
CMMD-03	8.00	8.80	0.80	0.65	Aluminous Laterite	Pinkish Brown	Cavernous
CMMD-03	8.80	9.20	0.40	0.40	Aluminous Laterite	Pinkish Brown	Cavernous
CMMD-03	9.20	10.00	0.80	0.65	Aluminous Laterite	Pinkish Brown	Cavernous
CMMD-03	10.00	10.50	0.50	0.50	Laterite	Brown	Cavernous

NORTHING	23°35'45.75457"N	DEPTH DRILLED (M)	24.3 Meter.
EASTING	84°37'55.09375"E	DATE OF COMMENCEMENT	24.04.2022
REDUCED LEVEL (M):	1015.364 meter	DATE OF CLOSURE	02.05.2022

BH No	From (meter)	To (meter)	Thickness (meter)	Recovery (meter)	Lithology	Color	Natural Evidence/ Features
CMMD-03	10.50	11.00	0.50	0.50	Laterite	Brown	Cavernous
CMMD-03	11.00	11.40	0.40	0.36	Laterite	Reddish Brown	Cavernous
CMMD-03	11.40	12.00	0.60	0.60	Laterite	Brown	Cavernous
CMMD-03	12.00	12.50	0.50	0.50	Laterite	Brown	Cavernous
CMMD-03	12.50	13.00	0.50	0.50	Laterite	Brown	Cavernous
CMMD-03	13.00	13.50	0.50	0.48	Laterite	Brown	Cavernous
CMMD-03	13.50	14.00	0.50	0.46	Laterite	Brown	Cavernous
CMMD-03	14.00	14.50	0.50	0.50	Laterite	Brown	Cavernous
CMMD-03	14.50	15.00	0.50	0.50	Laterite	Brown	Cavernous
CMMD-03	15.00	15.60	0.60	0.50	Laterite	Brown	Cavernous
CMMD-03	15.60	16.10	0.50	0.50	Laterite	Brown	Cavernous
CMMD-03	16.10	16.60	0.50	0.50	Sideritic Laterite	Brown	Cavernous
CMMD-03	16.60	17.30	0.70	0.70	Siliceous Laterite	Brown	Cavernous, Shinning
CMMD-03	17.30	18.00	0.70	0.70	Laterite	Brown	Cavernous
CMMD-03	18.00	18.50	0.50	0.40	Bauxite	White	Cavernous
CMMD-03	18.50	19.00	0.50	0.50	Bauxite	White	Cavernous, Spongy
CMMD-03	19.00	19.30	0.30	0.30	Bauxite	White	Cavernous

NORTHING	23°35'45.75457"N	DEPTH DRILLED (M)	24.3 Meter.
EASTING	84°37'55.09375"E	DATE OF COMMENCEMENT	24.04.2022
REDUCED LEVEL (M):	1015.364 meter	DATE OF CLOSURE	02.05.2022

BH No	From (meter)	To (meter)	Thickness (meter)	Recovery (meter)	Lithology	Color	Natural Evidence/ Features
CMMD-03	19.30	20.20	0.90	0.80	Variegated Clay, Sideritic rich	Yellow	Finely grained, soily, soft
CMMD-03	20.20	20.70	0.50	0.50	Variegated Clay	Pinkish White	Finely grained, soily, soft
CMMD-03	20.70	21.50	0.80	0.50	Variegated Clay	Pinkish White	Finely grained, soily, soft
CMMD-03	21.50	22.10	0.60	0.50	Variegated Clay	Pinkish White	Finely grained, soily, soft
CMMD-03	22.10	22.80	0.70	0.60	Variegated Clay	Pinkish White	Finely grained, soily, soft
CMMD-03	22.80	23.30	0.50	0.50	Variegated Clay	Brownish White	Finely grained, soily, soft
CMDD-03	23.30	24.30	1.00	0.82	Variegated Clay	Yellowish White	Finely grained, soily, soft

NORTHING	23°35'40.10449"N	DEPTH DRILLED (M)	19 Meter.
EASTING	84°38'02.43638"E.	DATE OF COMMENCEMENT	03.05.2022
REDUCED LEVEL (M):	1015.111 meter	DATE OF CLOSURE	09.05.2022

BH No	From (meter)	To (meter)	Thickness (meter)	Recovery (meter)	Lithology	Color	Natural Evidence/ Features
CMDD-04	0.00	0.50	0.50	0.50	Soil	Brown	Earthen
CMDD-04	0.50	1.00	0.50	0.50	Soil with Morram	Brown	Earthen
CMDD-04	1.00	1.50	0.50	0.50	Soil with Morram	Brown	Earthen
CMDD-04	1.50	2.00	0.50	0.50	Soil with Morram	Brown	Cavernous, Oolitic
CMDD-04	2.00	2.40	0.40	0.40	Laterite	Brown	Cavernous, Oolitic
CMDD-04	2.40	3.00	0.60	0.60	Laterite	Brown	Cavernous, Oolitic
CMDD-04	3.00	3.50	0.50	0.50	Laterite	Brown	Cavernous, Oolitic
CMDD-04	3.50	4.00	0.50	0.50	Laterite	Brown	Cavernous, Oolitic
CMDD-04	4.00	4.80	0.80	0.80	Aluminous laterite	Light Brown	Cavernous, Oolitic
CMDD-04	4.80	5.50	0.70	0.70	Aluminous laterite	Light Brown	Cavernous, Oolitic
CMDD-04	5.50	6.00	0.50	0.50	Aluminous laterite	Light Brown	Cavernous, Oolitic
CMDD-04	6.00	6.60	0.60	0.60	Aluminous laterite	Light Brown	Cavernous, Oolitic
CMDD-04	6.60	7.40	0.80	0.80	Aluminous laterite	Reddish Brown	Cavernous
CMDD-04	7.40	8.00	0.60	0.60	Aluminous laterite	Whitish Brown	Cavernous
CMDD-04	8.00	8.60	0.60	0.60	Aluminous laterite	Light Whitish Brown	Powdery
CMDD-04	8.60	9.00	0.40	0.40	Aluminous laterite	Light Whitish Brown	Powdery
CMDD-04	9.00	9.50	0.50	0.50	Aluminous laterite	Brown	Cavernous

NORTHING	23°35'40.10449"N	DEPTH DRILLED (M)	19 Meter.
EASTING	84°38'02.43638"E.	DATE OF COMMENCEMENT	03.05.2022
REDUCED LEVEL (M):	1015.111 meter	DATE OF CLOSURE	09.05.2022

BH No	From (meter)	To (meter)	Thickness (meter)	Recovery (meter)	Lithology	Color	Natural Evidence/ Features
CMDD-04	9.50	9.80	0.30	0.30	Aluminous laterite	Brown	Cavernous, Oolitic
CMDD-04	9.80	10.40	0.60	0.60	Aluminous laterite	Brown	Cavernous
CMDD-04	10.40	11.00	0.60	0.60	Laterite, Sideritic in places	Reddish Brown	Cavernous
CMDD-04	11.00	11.50	0.50	0.50	Laterite, Sideritic in places	Brown	Cavernous
CMDD-04	11.50	12.00	0.50	0.50	Laterite, Sideritic in places	Brown	Cavernous
CMDD-04	12.00	12.60	0.60	0.60	Laterite, Sideritic in places	Brown	Cavernous
CMDD-04	12.60	13.40	0.80	0.80	Ferruginous Laterite	Brown	Cavernous, Oolitic
CMDD-04	13.40	14.20	0.80	0.80	Laterite	Brown	Cavernous
CMDD-04	14.20	14.80	0.60	0.60	Laterite	Brown	Cavernous
CMDD-04	14.80	15.40	0.60	0.60	Variegated Clay, Soft	Yellow	Finely grained, soily, soft
CMDD-04	15.40	16.00	0.60	0.60	Variegated Clay, Soft	Yellow	Finely grained, soily, soft
CMDD-04	16.00	16.50	0.50	0.50	Variegated Clay, Soft	Yellow	Finely grained, soily, soft
CMDD-04	16.50	16.80	0.30	0.30	Variegated Clay, Soft	Yellow	Finely grained, soily, soft
CMDD-04	16.80	17.40	0.60	0.60	Variegated Clay, Soft	Yellow	Finely grained, soily, soft

NORTHING	23°35'40.10449"N	DEPTH DRILLED (M)	19 Meter.
EASTING	84°38'02.43638"E.	DATE OF COMMENCEMENT	03.05.2022
REDUCED LEVEL (M):	1015.111 meter	DATE OF CLOSURE	09.05.2022

BH No	From (meter)	To (meter)	Thickness (meter)	Recovery (meter)	Lithology	Color	Natural Evidence/ Features
CMDD-04	17.40	18.00	0.60	0.60	Variegated Clay, Soft	Yellow	Finely grained, soily, soft
CMDD-04	18.00	18.60	0.60	0.60	Variegated Clay, Soft	Yellow	Finely grained, soily, soft
CMDD-04	18.60	19.00	0.40	0.40	Variegated Clay, Soft	Yellow	Finely grained, soily, soft
CMDD-04	0.00	0.50	0.50	0.50	Soil	Brown	Earthen
CMDD-04	0.50	1.00	0.50	0.50	Soil with Morram	Brown	Earthen
CMDD-04	1.00	1.50	0.50	0.50	Soil with Morram	Brown	Earthen
CMDD-04	1.50	2.00	0.50	0.50	Soil with Morram	Brown	Cavernous, Oolitic
CMDD-04	2.00	2.40	0.40	0.40	Laterite	Brown	Cavernous, Oolitic
CMDD-04	2.40	3.00	0.60	0.60	Laterite	Brown	Cavernous, Oolitic
CMDD-04	3.00	3.50	0.50	0.50	Laterite	Brown	Cavernous, Oolitic

NORTHING	23°35'38.42704"N	DEPTH DRILLED (M)	20.7 Meter.
EASTING	84°37'53.50819"E	DATE OF COMMENCEMENT	09.05.2022
REDUCED LEVEL (M):	1012.233 meter	DATE OF CLOSURE	13.05.2022

BH No	From (meter)	To (meter)	Thickness (meter)	Recovery (meter)	Lithology	Color	Natural Evidence/ Features
CMDD-05	0.00	0.30	0.30	0.3	Soil	Brown	Earthen
CMDD-05	0.30	0.70	0.40	0.4	Aluminous Laterite	Reddish Brown	Pisiolitic
CMDD-05	0.70	1.00	0.30	0.3	Aluminous Laterite	Reddish Brown	Pisiolitic
CMDD-05	1.00	1.50	0.50	0.5	Aluminous Laterite	Reddish Brown	Pisiolitic, Limonite
CMDD-05	1.50	2.00	0.50	0.5	Aluminous Laterite	Reddish Brown	Cavernous and Pisiolitic
CMDD-05	2.00	2.50	0.50	0.5	Aluminous Laterite	Reddish Brown	Cavernous and Pisiolitic
CMDD-05	2.50	3.00	0.50	0.5	Aluminous Laterite	Reddish Brown	Cavernous and Pisiolitic
CMDD-05	3.00	3.60	0.60	0.6	Laterite	Dark Reddish Brown	Pisiolitic
CMDD-05	3.60	4.20	0.60	0.6	Aluminous Laterite	Light Pink	Cavernous and Pisiolitic
CMDD-05	4.20	4.80	0.60	0.6	Aluminous Laterite	Reddish Brown	Cavernous and Pisiolitic
CMDD-05	4.80	5.40	0.60	0.6	Aluminous Laterite	Brown	Cavernous and Pisiolitic
CMDD-05	5.40	6.10	0.70	0.7	Laterite	Dark Reddish Brown	Pisiolitic
CMDD-05	6.10	6.70	0.60	0.6	Aluminous Laterite	Pinkish Brown	Cavernous and Pisiolitic
CMDD-05	6.70	7.30	0.60	0.6	Aluminous Laterite	Brown	Cavernous and Pisiolitic
CMDD-05	7.30	7.90	0.60	0.6	Aluminous Laterite	Reddish Brown	Cavernous and Pisiolitic

NORTHING	23°35'38.42704"N	DEPTH DRILLED (M)	20.7 Meter.
EASTING	84°37'53.50819"E	DATE OF COMMENCEMENT	09.05.2022
REDUCED LEVEL (M):	1012.233 meter	DATE OF CLOSURE	13.05.2022

BH No	From (meter)	To (meter)	Thickness (meter)	Recovery (meter)	Lithology	Color	Natural Evidence/ Features
CMDD-05	7.90	8.50	0.60	0.6	Aluminous Laterite	Reddish Brown	Cavernous and Pisiolitic
CMDD-05	8.50	9.10	0.60	0.6	Aluminous Laterite	Reddish Brown	Cavernous and Pisiolitic
CMDD-05	9.10	9.70	0.60	0.6	Aluminous Laterite	Reddish Brown	Cavernous and Pisiolitic
CMDD-05	9.70	10.30	0.60	0.6	Aluminous Laterite	Yellowish Brown	Cavernous and Pisiolitic
CMDD-05	10.30	10.90	0.60	0.6	Laterite	Pinkish Brown	Pisiolitic
CMDD-05	10.90	11.50	0.60	0.6	Laterite	Reddish Brown	Pisiolitic
CMDD-05	11.50	12.10	0.60	0.6	Laterite	Reddish Brown	Pisiolitic
CMDD-05	12.10	12.70	0.60	0.6	Laterite	Reddish Brown	Pisiolitic
CMDD-05	12.70	13.30	0.60	0.6	Aluminous Laterite	Dark Reddish Brown	Cavernous and Pisiolitic
CMDD-05	13.30	13.90	0.60	0.6	Aluminous Laterite	Dark Reddish Brown	Cavernous and Pisiolitic
CMDD-05	13.90	14.50	0.60	0.6	Aluminous Laterite	Dark Brown	Cavernous and Pisiolitic
CMDD-05	14.50	15.10	0.60	0.6	Aluminous Laterite	Dark Brown	Cavernous and Pisiolitic
CMDD-05	15.10	15.70	0.60	0.6	Aluminous Laterite	Yellowish Brown at Bottom	Cavernous and Pisiolitic
CMDD-05	15.70	16.10	0.40	0.4	Aluminous Laterite	Brown	Cavernous and Pisiolitic
CMDD-05	16.10	16.70	0.60	0.6	Laterite	Brown	Pisiolitic

NORTHING	23°35'38.42704"N	DEPTH DRILLED (M)	20.7 Meter.
EASTING	84°37'53.50819"E	DATE OF COMMENCEMENT	09.05.2022
REDUCED LEVEL (M):	1012.233 meter	DATE OF CLOSURE	13.05.2022

BH No	From (meter)	To (meter)	Thickness (meter)	Recovery (meter)	Lithology	Color	Natural Evidence/ Features
CMDD-05	16.70	17.50	0.80	0.8	Laterite	Brown	Pisiolitic
CMDD-05	17.50	18.40	0.90	0.9	Variegated Clay, Soft	Yellow	Finely grained, soily, soft
CMDD-05	18.40	19.60	1.20	1.2	Variegated Clay, Soft	Yellow	Finely grained, soily, soft
CMDD-05	19.60	20.70	1.10	1.1	Variegated Clay, Soft	Yellow	Finely grained, soily, soft
CMDD-05	0.00	0.30	0.30	0.3	Soil	Brown	Earthen
CMDD-05	0.30	0.70	0.40	0.4	Aluminous Laterite	Reddish Brown	Pisiolitic
CMDD-05	0.70	1.00	0.30	0.3	Aluminous Laterite	Reddish Brown	Pisiolitic
CMDD-05	1.00	1.50	0.50	0.5	Aluminous Laterite	Reddish Brown	Pisiolitic, Limonite
CMDD-05	1.50	2.00	0.50	0.5	Aluminous Laterite	Reddish Brown	Cavernous and Pisiolitic
CMDD-05	2.00	2.50	0.50	0.5	Aluminous Laterite	Reddish Brown	Cavernous and Pisiolitic
CMDD-05	2.50	3.00	0.50	0.5	Aluminous Laterite	Reddish Brown	Cavernous and Pisiolitic

ANNEXURE – III/A

STATEMENT SHOWING PRIMARY ANALYSIS (FOR 5 RADICALS, Al_2O_3, SiO2, Fe_2O_3, TiO_2 & LOI) OF BOREHOLE CORE SAMPLES GENERATED BY CMPDI, MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

BH No	Depth Details (meter)		Thickness (meter)	Lithology	Sample No	5 Primary Oxides				
	From	To				SiO_2%	Al_2O_3%	Fe_2O_3%	TiO_2%	LOI%
CMMD-01	1.00	2.00	1.00	Aluminous Laterite	CMMD1/1	23.46	33.79	21.03	4.26	16.64
CMMD-01	2.00	3.00	1.00	Aluminous Laterite	CMMD1/2	15.49	38.01	20.40	5.40	20.02
CMMD-01	3.00	4.00	1.00	Aluminous Laterite	CMMD1/3	21.80	36.11	18.33	4.79	18.21
CMMD-01	4.00	5.00	1.00	Ferruginous Laterite	CMMD1/4	14.51	29.54	34.52	4.27	16.49
CMMD-01	5.00	5.50	0.50	Ferruginous Laterite	CMMD1/5	10.71	21.72	50.16	3.18	13.58
CMMD-01	7.50	8.50	1.00	Ferruginous Laterite	CMMD1/6	9.77	28.75	40.80	5.32	14.68
CMMD-01	8.50	9.00	0.50	Aluminous Laterite	CMMD1/7	8.90	35.95	31.44	7.02	16.02
CMMD-02	2.50	3.00	0.50	Aluminous laterite	CMMD-2/1	28.03	28.77	26.56	2.20	13.73
CMMD-02	3.00	4.00	1.00	Ferruginous Laterite	CMMD-2/2	23.67	27.13	31.97	2.77	13.84
CMMD-02	6.00	6.50	0.50	Ferruginous Laterite	CMMD-2/3	11.79	29.25	39.37	4.87	14.04
CMMD-02	6.50	7.50	1.00	Ferruginous Laterite	CMMD-2/4	12.84	31.69	34.00	5.58	15.25
CMMD-02	7.50	8.50	1.00	Ferruginous Laterite	CMMD-2/5	12.69	29.58	37.69	5.18	14.13
CMMD-02	8.50	9.50	1.00	Ferruginous Laterite	CMMD-2/6	12.11	29.96	37.38	5.12	14.80
CMMD-02	9.50	10.50	1.00	Aluminous laterite	CMMD-2/7	11.79	35.16	30.23	5.98	16.14
CMMD-02	10.50	11.50	1.00	Aluminous laterite	CMMD-2/8	11.48	27.32	39.51	4.92	16.03
CMMD-02	11.50	12.50	1.00	Siliceous bauxite	CMMD-2/9	8.69	32.07	32.07	7.53	19.02
CMMD-02	12.50	13.00	0.50	Bauxite	CMMD-2/10	5.68	33.78	30.66	8.99	20.25
CMMD-02	13.00	13.50	0.50	Ferruginous Bauxite	CMMD-2/11	5.79	29.19	38.32	7.01	19.02
CMMD-02	13.50	14.00	0.50	Ferruginous Bauxite10.	CMMD-2/12	5.32	26.88	44.39	5.99	16.79
CMMD-02	14.00	15.00	1.00	Bauxite	CMMD-2/13	6.16	34.73	29.68	9.12	19.69
CMMD-02	15.00	15.50	0.50	Bauxite	CMMD-2/14	4.88	39.86	24.55	8.02	22.03
CMMD-02	15.50	16.50	1.00	Aluminous Laterite	CMMD-2/15	9.68	34.42	30.09	6.15	19.03
CMMD-02	16.50	17.50	1.00	Ferruginous Laterite	CMMD-2/16	7.90	27.47	42.57	4.39	17.01
CMMD-03	3.50	4.50	1.00	Aluminous Laterite	CMMD-3/1	33.60	29.34	20.53	2.55	12.90
CMMD-03	4.50	6.10	1.60	Aluminous Laterite	CMMD-3/2	30.88	28.93	23.15	2.96	13.24
CMMD-03	6.10	7.10	1.00	Aluminous Laterite	CMMD-3/3	28.40	28.13	25.82	3.59	13.32

BH No	Depth Details (meter)		Thickness (meter)	Lithology	Sample No	5 Primary Oxides				
	From	To				SiO_2%	Al_2O_3%	Fe_2O_3%	TiO_2%	LOI%
CMMD-03	7.10	8.00	0.90	Aluminous Laterite	CMMD-3/4	22.98	32.30	22.68	4.83	16.47
CMMD-03	8.00	9.20	1.20	Aluminous Laterite	CMMD-3/5	9.86	45.91	13.27	7.68	22.68
CMMD-03	9.20	10.00	0.80	Aluminous Laterite	CMMD-3/6	10.58	46.11	14.42	6.95	21.20
CMMD-03	18.00	19.30	1.30	Bauxite	CMMD-3/7	5.25	35.69	31.55	7.71	19.13
CMMD-04	4.00	5.50	1.50	Aluminous Laterite	CMMD-4/1	30.65	27.07	25.89	3.21	12.45
CMMD-04	5.50	6.60	1.10	Aluminous Laterite	CMMD-4/2	22.81	33.97	22.56	5.96	14.09
CMMD-04	6.60	8.00	1.40	Aluminous Laterite	CMMD-4/3	14.49	39.30	20.08	8.15	17.26
CMMD-04	8.00	9.00	1.00	Aluminous Laterite	CMMD-4/4	18.40	40.49	13.57	8.46	18.35
CMMD-04	9.00	9.80	0.80	Aluminous Laterite	CMMD-4/5	14.43	40.72	16.40	10.04	17.79
CMMD-04	9.80	10.40	0.60	Aluminous Laterite	CMMD-4/6	11.50	40.74	18.70	10.58	17.68
CMMD-04	12.60	13.40	0.80	Ferruginous Laterite	CMMD-4/7	10.66	30.35	34.10	8.04	16.14
CMMD-05	0.30	1.50	1.20	Aluminous Laterite	CMMD-5/1	34.00	30.13	18.88	2.41	13.59
CMMD-05	1.50	2.50	1.00	Aluminous Laterite	CMMD-5/2	28.97	28.61	25.80	2.78	13.23
CMMD-05	2.50	3.00	0.50	Aluminous Laterite	CMMD-5/3	27.86	29.00	26.21	3.14	13.03
CMMD-05	3.60	4.80	1.20	Aluminous Laterite	CMMD-5/4	21.12	30.22	28.88	5.02	14.13
CMMD-05	4.80	5.40	0.60	Aluminous Laterite	CMMD-5/5	15.89	38.31	21.37	6.48	17.25
CMMD-05	6.10	6.70	0.60	Aluminous Laterite	CMMD-5/6	20.26	36.07	20.58	5.75	16.68
CMMD-05	6.70	7.90	1.20	Aluminous Laterite	CMMD-5/7	14.74	40.48	18.07	7.54	18.39
CMMD-05	7.90	9.10	1.20	Aluminous Laterite	CMMD-5/8	13.23	38.55	22.61	8.18	16.79
CMMD-05	9.10	10.30	1.20	Aluminous Laterite	CMMD-5/9	12.14	33.00	31.63	6.26	16.35
CMMD-05	12.70	13.90	1.20	Aluminous Laterite	CMMD-5/10	10.73	33.81	27.14	7.47	20.16
CMMD-05	13.90	15.10	1.20	Aluminous Laterite	CMMD-5/11	7.73	35.80	25.28	9.25	21.20
CMMD-05	15.10	16.10	1.00	Aluminous Laterite	CMMD-5/12	8.34	32.16	33.77	5.96	19.03

ANNEXURE – III/B

STATEMENT SHOWING DETAILS OF COMPOSITE CORE SAMPLE ANALYSIS FOR 14 RADICALS VIZ. Al_2O_3, SiO_2, Fe_2O_3, TiO_2, SO_3, CaO, MgO, MnO, P_2O_5, V_2O_5, Na_2O, K_2O, Organic C % & LOI OF BOREHOLES DRILLED BY CMPDI, MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

BH NO.	Sample No.	Al_2O_3%	SiO_2%	Fe_2O_3%	TiO_2%	V_2O_5%	P_2O_5%	MnO%	CaO%	MgO%	K_2O%	Na_2O%	SO_3%	Organic C %	LOI%
CMMD-02	CMMD 2/9+2/10	32.21	7.24	32.19	8.01	0.13	0.16	0.03	0.02	0.02	0.02	0.02	0.05	0.14	19.46
CMMD-02	CMMD 2/13+2/14+2/15	35.59	7.06	28.54	8.40	0.12	0.10	0.04	0.03	0.02	0.03	0.01	0.07	0.17	19.61
CMMD-05	CMMD 5/10+5/11+5/12	34.76	8.66	27.94	7.78	0.10	0.12	0.06	0.04	0.03	0.05	0.01	0.06	0.16	20.06
CMMD-03	CMMD 3/5+3/6	45.91	10.17	14.27	7.42	0.09	0.08	0.02	0.03	0.05	0.11	0.02	0.05	0.21	21.36

ANNEXURE – III/C

STATEMENT SHOWING DETAILS OF COMPOSITE CORE SAMPLE ANALYSIS FOR 7 RADICALS VIZ. Al_2O_3, SiO_2, Fe_2O_3, TiO_2, P_2O_5, V_2O_5 & LOI OF BOREHOLES DRILLED BY CMPDI, MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

Borehole No.	Sample No	Al_2O_3%	SiO_2%	Fe_2O_3%	TiO_2%	V_2O_5%	P_2O_5%	LOI%
CMMD-02	CMMD 2/9+2/10	32.21	7.24	32.19	8.01	0.13	0.16	19.46
CMMD-02	CMMD 2/13+2/14+2/15	35.59	7.06	28.54	8.40	0.12	0.10	19.61
CMMD-05	CMMD 5/10+5/11+5/12	34.76	8.66	27.94	7.78	0.10	0.12	20.06
CMPD-72	CMMD 3/5+3/6	45.91	10.17	14.27	7.42	0.09	0.08	21.36

ANNEXURE IV/A

STATEMENT SHOWING BOREHOLE-WISE INTERSECTIONS OF BAUXITE ZONES AT VARIOUS CUT- OFF AS END USE OF BAUXITE RECOMMENDED BY IBM, MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

Cut off Value – Max 10% SiO_2, >=30% Al_2O_3.

BH No.	Depth Details		Thickness	Lithology	Sample No	5 Radicals Primary Oxide percentage details				
	From	To				SiO_2%	Al_2O_3%	Fe_2O_3%	TiO_2%	LOI%
CMMD-02	11.50	12.50	1.00	Siliceous Bauxite	CMMD-2/9	8.69	32.07	32.07	7.53	19.02
CMMD-02	12.50	13.00	0.50	Bauxite	CMMD-2/10	5.68	33.78	30.66	8.99	20.25
CMMD-02	13.00	13.50	0.50	Fe Bauxite	CMMD-2/11	5.79	29.19	38.32	7.01	19.02
CMMD-02	13.50	14.00	0.50	Fe Bauxite	CMMD-2/12	5.32	26.88	44.39	5.99	16.79
CMMD-02	14.00	15.00	1.00	Bauxite	CMMD-2/13	6.16	34.73	29.68	9.12	19.69
CMMD-02	15.00	15.50	0.50	Bauxite	CMMD-2/14	4.88	39.86	24.55	8.02	22.03
CMMD-03	18.00	19.30	1.30	Bauxite	CMMD-3/7	5.25	35.69	31.55	7.71	19.13

Cut off Value – Max 7% SiO_2, >=30% Al_2O_3.

BH No.	Depth Details		Thickness	Lithology	Sample No	5 Radicals Primary Oxide percentage details				
	From	To				SiO_2%	Al_2O_3%	Fe_2O_3%	TiO_2%	LOI%
CMMD-02	12.50	13.00	0.50	Bauxite	CMMD-2/10	5.68	33.78	30.66	8.99	20.25
CMMD-02	13.00	13.50	0.50	Fe Bauxite	CMMD-2/11	5.79	29.19	38.32	7.01	19.02
CMMD-02	13.50	14.00	0.50	Fe Bauxite	CMMD-2/12	5.32	26.88	44.39	5.99	16.79
CMMD-02	14.00	15.00	1.00	Bauxite	CMMD-2/13	6.16	34.73	29.68	9.12	19.69
CMMD-02	15.00	15.50	0.50	Bauxite	CMMD-2/14	4.88	39.86	24.55	8.02	22.03

BH No.	Depth Details		Thickness	Lithology	Sample No	5 Radicals Primary Oxide percentage details				
	From	To				SiO_2%	Al_2O_3%	Fe_2O_3%	TiO_2%	LOI%
CMMD-03	18.00	19.30	1.30	Bauxite	CMMD-3/7	5.25	35.69	31.55	7.71	19.13

ANNEXURE IV/B

STATEMENT SHOWING BOREHOLE-WISE INTERSECTIONS OF ALUMINOUS LATERITE ZONES AT VARIOUS CUT- OFF AS END USE OF ALUMINOUS LATERITE RECOMMENDED BY IBM, MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

Cut off value – Al2O3– 20% (min.), Fe_2O_3 : Al_2O_3 ratio less than 1.00 and SiO_2 : Al_2O_3 ratio less than 1.33.

BH No.	Depth Details		Thickness	Lithology	Sample No	5 Radicals Primary Oxide percentage details				
	From	To				SiO_2%	Al_2O_3%	Fe_2O_3%	TiO_2%	LOI%
CMMD-01	1.00	2.00	1.00	Aluminous Laterite	CMMD1/1	23.46	33.79	21.03	4.26	16.64
CMMD-01	2.00	3.00	1.00	Aluminous Laterite	CMMD1/2	15.49	38.01	20.40	5.40	20.02
CMMD-01	3.00	4.00	1.00	Aluminous Laterite	CMMD1/3	21.80	36.11	18.33	4.79	18.21
CMMD-01	8.50	9.00	0.50	Aluminous Laterite	CMMD1/7	8.90	35.95	31.44	7.02	16.02
CMMD-02	2.50	3.00	0.50	Aluminous laterite	CMMD-2/1	28.03	28.77	26.56	2.20	13.73
CMMD-02	9.50	10.50	1.00	Aluminous laterite	CMMD-2/7	11.79	35.16	30.23	5.98	16.14
CMMD-02	10.50	11.50	1.00	Aluminous laterite	CMMD-2/8	11.48	27.32	39.51	4.92	16.03
CMMD-02	15.50	16.50	1.00	Aluminous Laterite	CMMD-2/15	9.68	34.42	30.09	6.15	19.03
CMMD-03	3.50	4.50	1.00	Aluminous Laterite	CMMD-3/1	33.60	29.34	20.53	2.55	12.90
CMMD-03	4.50	6.10	1.60	Aluminous Laterite	CMMD-3/2	30.88	28.93	23.15	2.96	13.24
CMMD-03	6.10	7.10	1.00	Aluminous Laterite	CMMD-3/3	28.40	28.13	25.82	3.59	13.32
CMMD-03	7.10	8.00	0.90	Aluminous Laterite	CMMD-3/4	22.98	32.30	22.68	4.83	16.47
CMMD-03	8.00	9.20	1.20	Aluminous Laterite	CMMD-3/5	9.86	45.91	13.27	7.68	22.68

BH No.	Depth Details		Thickness	Lithology	Sample No	5 Radicals Primary Oxide percentage details				
	From	To				SiO_2%	Al_2O_3%	Fe_2O_3%	TiO_2%	LOI%
CMMD-03	9.20	10.00	0.80	Aluminous Laterite	CMMD-3/6	10.58	46.11	14.42	6.95	21.20
CMMD-04	4.00	5.50	1.50	Aluminous Laterite	CMMD-4/1	30.65	27.07	25.89	3.21	12.45
CMMD-04	5.50	6.60	1.10	Aluminous Laterite	CMMD-4/2	22.81	33.97	22.56	5.96	14.09
CMMD-04	6.60	8.00	1.40	Aluminous Laterite	CMMD-4/3	14.49	39.30	20.08	8.15	17.26
CMMD-04	8.00	9.00	1.00	Aluminous Laterite	CMMD-4/4	18.40	40.49	13.57	8.46	18.35
CMMD-04	9.00	9.80	0.80	Aluminous Laterite	CMMD-4/5	14.43	40.72	16.40	10.04	17.79
CMMD-04	9.80	10.40	0.60	Aluminous Laterite	CMMD-4/6	11.50	40.74	18.70	10.58	17.68
CMMD-05	0.30	1.50	1.20	Aluminous Laterite	CMMD-5/1	34.00	30.13	18.88	2.41	13.59
CMMD-05	1.50	2.50	1.00	Aluminous Laterite	CMMD-5/2	28.97	28.61	25.80	2.78	13.23
CMMD-05	2.50	3.00	0.50	Aluminous Laterite	CMMD-5/3	27.86	29.00	26.21	3.14	13.03
CMMD-05	3.60	4.80	1.20	Aluminous Laterite	CMMD-5/4	21.12	30.22	28.88	5.02	14.13
CMMD-05	4.80	5.40	0.60	Aluminous Laterite	CMMD-5/5	15.89	38.31	21.37	6.48	17.25
CMMD-05	6.10	6.70	0.60	Aluminous Laterite	CMMD-5/6	20.26	36.07	20.58	5.75	16.68
CMMD-05	6.70	7.90	1.20	Aluminous Laterite	CMMD-5/7	14.74	40.48	18.07	7.54	18.39
CMMD-05	7.90	9.10	1.20	Aluminous Laterite	CMMD-5/8	13.23	38.55	22.61	8.18	16.79
CMMD-05	9.10	10.30	1.20	Aluminous Laterite	CMMD-5/9	12.14	33.00	31.63	6.26	16.35
CMMD-05	12.70	13.90	1.20	Aluminous Laterite	CMMD-5/10	10.73	33.81	27.14	7.47	20.16

BH No.	Depth Details		Thickness	Lithology	Sample No	5 Radicals Primary Oxide percentage details				
	From	To				SiO_2%	Al_2O_3%	Fe_2O_3%	TiO_2%	LOI%
CMMD-05	13.90	15.10	1.20	Aluminous Laterite	CMMD-5/11	7.73	35.80	25.28	9.25	21.20
CMMD-05	15.10	16.10	1.00	Aluminous Laterite	CMMD-5/12	8.34	32.16	33.77	5.96	19.03

ANNEXURE – V

STATEMENT SHOWING PRIMARY AGAINST CHECK SAMPLES OF CORE SAMPLES GENERATED IN BOREHOLES DRILLED AT MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

Internal Check Samples:

BH No	Depth Details (meter)		Thickness (meter)	Lithology	Sample No	Primary Analysis					Check Samples Analysis					Remarks
	From	To				SiO_2%	Al_2O_3%	Fe_2O_3%	TiO_2%	LOI%	SiO_2%	Al_2O_3%	Fe_2O_3%	TiO_2%	LOI%	
CMMD-02	15.50	16.50	1.00	Aluminous Laterite	CMMD-2/15	9.68	34.42	30.09	6.15	19.03	9.85	34.07	30.65	6.08	18.48	Internal Check sample

External Check Samples:

BH No	Depth Details (meter)		Thickness (meter)	Lithology	Sample No	Primary Analysis					Check Samples Analysis					Remarks
	From	To				SiO_2%	Al_2O_3%	Fe_2O_3%	TiO_2%	LOI%	SiO_2%	Al_2O_3%	Fe_2O_3%	TiO_2%	LOI%	
CMMD-05	13.90	15.10	1.20	Aluminous Laterite	CMMD-5/11	7.73	35.80	25.28	9.25	21.20	7.60	39.57	32.17	4.78	15.28	External Check sample
CMMD-01	8.50	9.00	0.50	Bauxite	CMMD-1/7	8.90	35.95	31.44	7.02	16.02	6.56	41.96	24.13	6.62	20.61	External Check sample

ANNEXURE – VI/A

STATEMENT SHOWING RESULTS OF REE ANALYSIS OF PRIMARY SAPROLITE CLAY SAMPLES USING ICP-MS METHODS AT MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

Sr. No	BH No.	Sample No	REE Concentration (in ppm)															
			Sc	Y	La	Ce	Pr	Nd	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
1	CMMD-1	CMMD-1/8	18.07	2.85	7.16	32.09	1.52	5.04	0.87	0.18	0.83	0.13	0.66	0.14	0.47	0.08	0.54	0.09
2	CMMD-1	CMMD-1/9	17.73	2.27	2.70	21.63	0.68	2.52	0.51	0.11	0.51	0.09	0.57	0.12	0.40	0.06	0.46	0.07
3	CMMD-1	CMMD-1/10	17.40	1.53	6.27	27.49	0.84	2.61	0.45	0.11	0.48	0.07	0.42	0.08	0.29	0.04	0.31	0.05
4	CMMD-1	CMMD-1/11	16.52	1.82	2.54	23.95	0.61	2.28	0.46	0.10	0.47	0.08	0.46	0.09	0.29	0.05	0.34	0.05
5	CMMD-1	CMMD-1/12	11.94	3.05	4.09	25.94	0.63	2.03	0.38	0.09	0.47	0.09	0.56	0.12	0.42	0.07	0.47	0.07
6	CMMD-2	CMMD-2/17	22.17	2.31	2.72	25.48	0.46	1.62	0.34	0.09	0.44	0.07	0.45	0.10	0.34	0.05	0.41	0.06
7	CMMD-2	CMMD-2/18	26.06	3.60	2.96	23.36	0.49	1.72	0.38	0.11	0.59	0.11	0.67	0.15	0.50	0.07	0.57	0.09
8	CMMD-2	CMMD-2/19	22.03	1.69	1.85	41.47	0.31	1.09	0.24	0.08	0.40	0.06	0.36	0.08	0.28	0.04	0.34	0.05
9	CMMD-2	CMMD-2/20	22.14	2.02	3.01	27.30	0.41	1.38	0.27	0.07	0.36	0.06	0.36	0.08	0.28	0.04	0.33	0.05
10	CMMD-3	CMMD-3/8	17.27	1.82	3.85	29.74	0.66	2.40	0.56	0.26	0.67	0.10	0.53	0.10	0.31	0.04	0.31	0.05
11	CMMD-3	CMMD-3/9	11.84	4.19	6.99	73.61	1.12	3.65	0.71	0.24	1.07	0.18	0.81	0.20	0.59	0.12	0.70	0.14
12	CMMD-3	CMMD-3/10	20.02	2.53	6.93	50.34	0.87	2.82	0.50	0.15	0.69	0.10	0.50	0.11	0.37	0.06	0.44	0.07
13	CMMD-3	CMMD-3/11	15.74	2.86	4.23	38.13	0.65	2.26	0.45	0.12	0.50	0.09	0.53	0.12	0.40	0.07	0.46	0.08
14	CMMD-4	CMMD-4/8	11.90	2.77	20.74	55.06	7.12	30.35	5.51	1.34	3.67	0.51	1.39	0.19	0.64	0.05	0.41	0.05
15	CMMD-4	CMMD-4/9	10.38	6.82	16.71	42.86	4.96	23.74	6.73	2.18	4.63	0.93	3.69	0.48	1.04	0.10	0.63	0.08
16	CMMD-4	CMMD-4/10	11.45	1.71	4.09	11.08	0.92	3.69	0.84	0.24	0.65	0.12	0.55	0.09	0.29	0.04	0.30	0.05
17	CMMD-4	CMMD-4/11	17.43	1.16	2.52	5.46	0.57	2.24	0.49	0.12	0.40	0.08	0.36	0.07	0.20	0.03	0.21	0.03
18	CMMD-5	CMMD-5/13	13.00	1.96	13.59	14.72	1.31	3.97	0.62	0.17	0.77	0.11	0.48	0.10	0.33	0.05	0.36	0.06
19	CMMD-5	CMMD-5/14	17.77	1.54	2.22	20.33	0.43	1.53	0.29	0.07	0.35	0.07	0.34	0.07	0.25	0.04	0.29	0.05
	Minimum Value		10.38	1.16	1.85	5.46	0.31	1.09	0.24	0.07	0.35	0.06	0.34	0.07	0.20	0.03	0.21	0.03

Sr. No	BH No.	Sample No	REE Concentration (in ppm)														
			Sc	Y	La	Ce	Pr	Nd	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
	Maximum Value		26.06	6.82	20.74	73.61	7.12	30.35	6.73	2.18	4.63	0.93	3.69	0.48	1.04	0.12	0.70	0.14

***BDL= BELOW DETECTION LIMIT.**

ANNEXURE – VI/B

STATEMENT SHOWING RESULTS OF ANALYSIS OF PACKAGE BY 34 ELEMENTS BY ICP-AES/ICPMS AT MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

Method of Analysis: **Acid Decomposition & ICP-MS.**

Sample Sl. No.	Ref. No.	All Values in "PPM" (mg/Kg)			
		Li	Mo	In	W
1	CMMD-2/1	10.52	5.40	0.24	7.54
2	CMMD-2/2	7.62	5.40	0.35	7.98
3	CMMD-2/3	<5.00	4.73	0.43	7.42
4	CMMD-2/4	<5.00	4.53	0.32	4.56
5	CMMD-2/5	<5.00	3.23	0.27	3.61
6	CMMD-2/6	<5.00	3.15	0.27	3.35
7	CMMD-2/7	<5.00	2.06	0.26	3.28
8	CMMD-2/8	<5.00	2.35	0.25	3.65
9	CMMD-2/9	<5.00	1.81	0.28	3.57
10	CMMD-2/10	<5.00	1.68	0.29	1.37
11	CMMD-2/11	<5.00	1.75	0.29	3.38
12	CMMD-2/12	<5.00	1.59	0.27	1.67
13	CMMD-2/13	<5.00	1.88	0.28	2.86
14	CMMD-2/14	<5.00	1.42	0.30	0.95
15	CMMD-2/15	<5.00	1.63	0.29	3.45
16	CMMD-2/16	<5.00	0.55	0.26	3.30
17	CMMD-2/17	<5.00	1.05	0.12	2.80
18	CMMD-2/18	<5.00	1.01	0.10	2.81
19	CMMD-2/19	<5.00	0.87	0.10	2.92
20	CMMD-2/20	<5.00	0.91	0.08	1.72

Method of Analysis: ICP-MS Fusion Method (GSI/SR/HYD/SOP-ICPMS/01/1D)

PR No.	Sample Ref. No.	Be	Ge	La	Ce	Pr	Nd	Eu	Sm	Tb	Gd	Dy	Ho	Er	Tm	Yb	Lu	Ta	U
		Concentrations in ppm (mg/Kg)																	
CMPDI-1-1	CMMD-2/1	1.41	2.13	86.36	110.93	16.76	53.46	0.90	6.84	0.85	4.99	4.88	0.90	3.29	0.45	3.58	0.62	2.47	20.64
CMPDI-1-2	CMMD-2/2	1.26	2.20	58.96	74.00	7.70	24.05	1.01	4.63	0.64	3.70	4.31	0.91	3.01	0.54	3.55	0.53	0.44	17.57
CMPDI-1-3	CMMD-2/3	1.28	1.30	67.14	54.64	4.43	17.95	0.54	3.09	0.41	2.08	2.55	0.46	1.39	0.29	1.99	0.22	1.30	11.42
CMPDI-1-4	CMMD-2/4	1.75	2.03	65.69	36.26	5.72	16.31	0.49	2.93	0.57	5.05	3.28	0.58	1.80	0.25	1.56	0.31	2.46	17.78
CMPDI-1-5	CMMD-2/5	0.88	1.53	70.62	40.70	5.94	20.59	0.58	2.38	0.39	2.31	2.60	0.43	1.39	0.21	1.51	0.20	2.08	19.86
CMPDI-1-6	CMMD-2/6	0.70	1.50	73.74	43.79	6.79	21.67	0.47	2.65	0.36	2.10	2.23	0.45	1.05	0.19	1.64	0.29	2.29	16.41
CMPDI-1-7	CMMD-2/7	0.74	0.59	62.53	44.83	4.22	14.64	0.43	3.04	0.41	1.87	1.75	0.41	1.35	0.27	1.68	0.28	0.87	11.47
CMPDI-1-8	CMMD-2/8	2.26	1.05	103.75	95.96	12.29	43.18	1.54	8.12	1.17	5.78	7.10	1.27	3.89	0.61	4.49	0.74	2.64	16.72
CMPDI-1-9	CMMD-2/9	1.58	0.58	68.94	27.19	13.01	38.55	0.70	2.69	0.52	2.47	2.81	0.62	2.04	0.26	1.98	0.21	2.11	14.55
CMPDI-1-10	CMMD-2/10	0.88	0.59	56.03	25.44	3.47	12.12	0.23	1.98	0.31	1.43	2.16	0.48	1.27	0.18	1.89	0.29	2.68	14.87
CMPDI-1-11	CMMD-2/11	1.37	0.64	55.01	21.50	4.35	14.81	0.54	2.51	0.44	2.24	1.92	0.44	1.26	0.20	1.73	0.26	2.36	16.11
CMPDI-1-12	CMMD-2/12	2.12	0.29	55.95	25.15	4.21	16.65	0.55	2.62	0.27	1.83	2.21	0.37	1.33	0.25	2.13	0.22	2.21	16.80
CMPDI-1-13	CMMD-2/13	1.15	0.31	66.51	32.38	5.24	18.54	0.45	2.23	0.37	2.08	2.15	0.37	1.43	0.28	1.89	0.32	2.95	13.40
CMPDI-1-14	CMMD-2/14	1.04	0.68	60.57	40.93	4.50	13.25	0.65	2.74	0.38	4.92	2.38	0.39	1.66	0.26	1.63	0.27	2.50	11.81
CMPDI-1-15	CMMD-2/15	0.85	1.07	98.17	57.32	4.59	16.83	0.72	3.11	0.67	2.86	4.20	0.80	3.20	0.50	3.36	0.57	1.29	9.74
CMPDI-1-16	CMMD-2/16	0.67	0.57	67.79	26.44	5.10	17.47	0.53	2.66	0.38	2.70	2.89	0.54	2.01	0.32	2.40	0.39	2.01	13.71
CMPDI-1-17	CMMD-2/17	1.18	0.65	72.44	81.31	4.74	17.29	0.49	2.33	0.44	2.36	3.00	0.73	1.88	0.36	3.14	0.63	1.88	12.54
CMPDI-1-18	CMMD-2/18	2.24	1.03	57.78	109.69	4.86	16.87	0.41	1.93	0.76	3.64	5.31	1.08	4.01	0.53	4.05	0.64	1.73	12.39
CMPDI-1-19	CMMD-2/19	1.14	1.25	71.63	155.25	4.25	17.78	0.37	2.72	0.48	2.32	2.63	0.68	2.18	0.43	2.44	0.47	1.90	13.64
CMPDI-1-20	CMMD-2/20	0.40	1.14	89.49	95.66	4.70	16.10	0.67	2.10	0.45	1.96	3.06	0.64	2.06	0.30	2.79	0.45	1.71	11.97

Method of Analysis: **XRF Pressed Pellet Method (GSI/SR-HYD/CL/SOP/XRF/1A)**

PR.No.	Ref.No.	Ba	Ga	Sc	Th	Co	Rb	Sr	Y	Zr	Nb	Cr
		(ppm) or (mg/kg)										
CMPDI-1-1	CMMD-2/1	<50	73	53	25	<1	<5	28	25	397	27	518
CMPDI-1-2	CMMD-2/2	<50	96	69	24	<1	<5	23	22	454	25	720
CMPDI-1-3	CMMD-2/3	<50	83	71	16	<1	<5	18	14	584	28	496
CMPDI-1-4	CMMD-2/4	<50	88	69	17	<1	<5	19	15	537	36	359
CMPDI-1-5	CMMD-2/5	<50	70	80	17	<1	<5	18	14	538	31	348
CMPDI-1-6	CMMD-2/6	<50	67	75	15	<1	<5	16	11	536	29	334
CMPDI-1-7	CMMD-2/7	<50	69	60	14	<1	<5	16	11	524	41	252
CMPDI-1-8	CMMD-2/8	<50	56	74	13	<1	<5	15	15	474	23	287
CMPDI-1-9	CMMD-2/9	<50	72	97	12	5	<5	13	10	584	47	267
CMPDI-1-10	CMMD-2/10	<50	84	98	10	<1	<5	10	8	657	59	281
CMPDI-1-11	CMMD-2/11	<50	81	123	9	<1	<5	12	10	566	41	392
CMPDI-1-12	CMMD-2/12	<50	68	120	10	<1	<5	12	10	489	27	391
CMPDI-1-13	CMMD-2/13	<50	85	91	11	4	<5	13	10	692	66	298
CMPDI-1-14	CMMD-2/14	<50	91	71	16	16	<5	13	11	695	64	287
CMPDI-1-15	CMMD-2/15	<50	76	54	21	<1	<5	15	20	851	56	363
CMPDI-1-16	CMMD-2/16	<50	53	67	16	<1	<5	12	16	600	27	315
CMPDI-1-17	CMMD-2/17	70	36	38	24	<1	<5	10	18	532	39	105
CMPDI-1-18	CMMD-2/18	71	34	39	29	6	<5	9	25	612	36	64
CMPDI-1-19	CMMD-2/19	163	35	25	29	5	<5	10	18	718	38	39
CMPDI-1-20	CMMD-2/20	106	33	22	24	2	<5	8	15	646	35	17

ANNEXURE – VII/A

STATEMENT SHOWING X-RAY DIFFRACTION ANALYSIS (XRD) RESULTS OF COMPOSITE BAUXITE SAMPLES AT MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

Sl. No	BH. No	Sample No	Major (≥ 5 %)	Minor (≤ 5 % to ≥ 1 %)	Trace, Faint Trace (≤ 1 %)
1	CMMD 2	CMMD 2/9 + 2/10	Gibbsite, Goethite, Hematite	Anatase, Kaolinite, Calcite, Phlogopite	Pyrite, Anorthite, Magnetite, Dickite, Ilmenite, biotite
2	CMMD 2	CMMD 2/13 + 2/14 + 2/15	Gibbsite, Hematite, Goethite	Anatase, Ilmenite, Kaolinite	Dickite, Rutile
3	CMMD 5	CMMD 5/10 + 5/11 + 5/12	Gibbsite, Ilmenite	Anatase, Dickite, Hematite, Magnetite	Albite, Phlogopite, Biotite, Rutile, Illite
4	CMMD 3	CMMD 3/5 + 3/6	Gibbsite, Kaolinite, Hematite, Goethite	Kyanite, Calcite, Boehmite, Chlorite, Albite	Rutile, Muscovite, Ilmenite
Note: Only crystalline phases of the samples are recorded. Amorphous phases are out of the scope of XRD and hence excluded.					

ANNEXURE – VII/B

STATEMENT SHOWING PETEROGRAPHIC STUDIES OF SAMPLES OF MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

Sl. No.	Sample No	Texture	Mineral Composition			Description
			Major >5%	Minor <5%->1%	Accessory <1%	
1	PS-10	It is a brick red coloured weathered and altered rock showing whitish pisolites, pores and cavities.	Cliachite Clay minerals Gibbsite Ferruginous matter	Opaques	Boehmite	The specimen is mostly made up of medium to moderately coarse clay mixed cliachitic pisolites set in reddish ferruginous cement. Gibbsite and boehmite are seen present as fine to very fine aggregates replacing cliachitic patches from both core and periphery and also along fractures. Gibbsite is also present as thin to moderately thick cavity fillings and as patches comprising fine prismatic aggregates and showing comb structure. Opaques are noted as anhedral patchy and streaky relicts within reddish ferruginous matter. The specimen is a **Bauxite.**
2	PS-05	It is a reddish brown coloured weathered and altered rock showing opaques patches, very fine whitish fillings, pores and cavities.	Ferruginous matter Opaques Gibbsite/ Boehmite	Clay minerals	….	The specimen is made up of reddish ferruginous (possibly limonitic) patches throughout the specimen showing pores and cavities in-between. Opaques occur as fine to medium anhedral grains and patches set within ferruginous patches and showing colloform texture. Gibbsite/ boehmite occurs as very fine to fine aggregates, mostly intruded as fine fillings and are stained by reddish ferruginous matter in areas. Dirty clayey patches are seen associated with ferruginous patches in areas. The specimen is a **Laterite.**
3	PS-07	It is a reddish brown coloured weathered and altered rock showing fine	Opaques Ferruginous	Clay minerals	….	Opaques occur as coarse anhedral patches supported by reddish ferruginous patches and fillings. Gibbsite is present as thin cavity fillings and as pisolitic patches comprising fine

Sl. No.	Sample No	Texture	Mineral Composition			Description
			Major >5%	Minor <5%->1%	Accessory <1%	
		whitish pisolites, fillings, pores and cavities.	matter Gibbsite Cliachite			prismatic aggregates. Cliachite occurs as fine to medium pisolites, often replaced by gibbsite. Dirty and very fine clay particles are seen associated with ferruginous and cliachitic patches in areas. The specimen is a **Bauxite bearing Laterite.**
4	PS-08	It is a whitish grey coloured weathered and altered rock showing pisolites, reddish stains, pores and cavities.	Cliachite Gibbsite Ferruginous matter	Boehmite Clay minerals	Opaques	The specimen is composed of fine to moderately coarse clichitic pisolites, oolites and patches, often showing multiple concentric rings. Gibbsite occurs as fine fillings replacing cliachite along cracks. Ferruginous matter is present as reddish patches and stains, mostly acting as cementing ingredient in the specimen. Boehmite occurs as fine to medium patches in pockets in association with gibbsite, comprising very fine granular aggregates. Dirty clay particles are seen present being intermixed with cliachite. Opaques are noted as very fine specks in accessories. The specimen is a **Bauxite.**
5	PS-09	It is a whitish grey coloured weathered and altered rock showing pisolites, reddish stains, pores and cavities.	Cliachite Gibbsite Ferruginous matter	Clay minerals Boehmite	Chert Opaques	Cliachite occurs as fine to moderately coarse pisolites and as anhedral patches being replaced by very fine to fine aggregates of gibbsite and boehmite. Gibbsite is also seen present as thin cavity fillings, comprising fine prismatic aggregates. Ferruginous matter is present as reddish patches, stains and filings. Clay minerals are seen associated with cliachite in areas. Chert occurs as fine patchy fillings, comprising very fine micro-crystalline aggregates. Opaques are noted as very fine specks in accessories.

Sl. No.	Sample No	Texture	Mineral Composition			Description
			Major >5%	Minor <5%->1%	Accessory <1%	
						The specimen is a **Bauxite.**

ANNEXURE – VII/C

STATEMENT SHOWING MINEROGRAPHIC STUDIES OF SAMPLES OF MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

Sl. No.	Sample No	% of ore minerals in polished section	ORE MINERAL COMPOSITION				Description
			Major >5%	Minor <5% - >1%	Accessory <1% - >0.1%	Traces <0.1%	
1.	PS-09	*	Hematite Rutile/ Anatase Limonite	….	….	….	Hematite is present as very fine to fine specks, blades, streaky fillings and as skeletal grains. Rutile/ anatase occur as very fine disseminated specks and as microcrystalline aggregates in association with hematite. Limonite is seen present as reddish patches, stains and as fillings.
2.	PS-02	*	Hematite Rutile/ Anatase Limonite Goethite	….	….	….	Hematite occurs as very fine specks, fine blades, anhedral grains, streaks and as cavity fillings. Rutile/ anatase are present as very fine micro-crystalline grains in association with hematite. Limonite is present as reddish patches comprising very fine amorphous aggregates and also occurs as cavity fillings. Goethite is noted as fillings in association with and replacing hematite filings in a few areas.
3.	PS-06	*	Hematite Rutile/ Anatase Limonite Goethite	….	….	….	Hematite occurs as fine fillings, often being intermixed with goethite and are showing compositional zoning & colloform texture. Hematite also occurs as very fine specks, anhedral, bladed and skeletal grains and as streaks. Rutile/ anatase are present as very fine dispersed micro-crystalline particles. Limonite is noted as reddish patches and fillings comprising very fine amorphous aggregates.
4.	PS-01	*	Hematite Rutile/ Anatase Limonite Goethite	….	….	….	Hematite occurs as fine anhedral grains, very fine specks, blades and as fine fillings and patches, often showing compositional zoning being intermixed with goethite. Rutile/ anatase are present as very fine micro-crystalline grains in pockets. Limonite is seen present as reddish patches and fillings.
5.	PS-03	*	Hematite Rutile/ Anatase	….	….	….	Hematite occurs as very fine disseminated blades, specks, fine anhedral grains and as fillings along pores and cavities. Rutile/ anatase are present as very fine blades, bladed aggregates and

Sl. No.	Sample No	% of ore minerals in polished section	ORE MINERAL COMPOSITION				Description
			Major >5%	Minor <5% - >1%	Accessory <1% - >0.1%	Traces <0.1%	
			Limonite Goethite Chalcopyrite				as micro-crystalline segregations. Limonite is present as reddish stains, fillings and patches. Goethite fillings and patches are seen associated with hematite fillings and replacing it at places. Chalcopyrite is noted as very fine isolated specks in traces.

N.B.: * Those specimens are extremely weathered/ lateritic in nature and occurring ore minerals are mostly very finely dispersed as amorphous/ micro-crystalline matter &/ or present as stains throughout the specimen, thus conventional modal calculation may mislead. Hence, % of ore minerals calculation is avoided. Although, it can be inferred that the presence of ore minerals in all the above studied samples are in the range of 1% to 5%.

Please note that bauxite minerals are not identifiable under reflected light. Thus, ore minerals referred here is excluding of any bauxite minerals.

ANNEXURE – VII/D

STATEMENT SHOWING RESULT OF DETERMINATION OF MONO-HYDRATED ALUMINA (MHA), TRI-HYDRATED ALUMINA (THA) AND REACTIVE SILICA OF SAMPLES OF MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

Sl. No	Borehole No	Sample No	THA%	MHA%	Reactive Silica%
1	CMMD-02	CMMD 2/9+2/10	29.57	0.83	6.79
2	CMMD-02	CMMD 2/13+2/14+2/15	35.43	1.15	6.61
3	CMMD-05	CMMD 5/10+5/11+5/12	28.89	0.84	8.23
4	CMMD-03	CMMD 3/5+3/6	47.90	2.77	8.58

ANNEXURE – VII/E

STATEMENT SHOWING RESULT OF DETERMINATION OF BOND WORK INDEX OF BAUXITE SAMPLES OF MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

Sample Code	Test Sieve Size (micron)	80% Passing size of feed (micron)	80% passing size of circuit product (micron)	Mass of undersize per mill revolution (g/rev)	Bond work index (kWh/t)
BW-1	150	1800	113	1.53	14.07
BW-2	150	1800	109	1.33	15.41
BW-3	150	1680	117	1.68	13.5

ANNEXURE VIII/A

STATEMENT SHOWING SPECIFIC GRAVITY OF BAXUITE SAMPLES MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

Sl. No	Sample No.	Borehole No	Depth (meter)		Specific Gravity (gm/cc).
			From	To	
2	SG-01	CMMD-01	8.50	9.00	2.33
3	SG-02	CMMD-02	12.50	13.00	2.32
4	SG-03	CMMD-03	18.00	19.30	2.41
5	SG-04	CMMD-04	9.80	10.40	2.35
6	SG-05	CMMD-05	9.10	10.30	2.25

Specific Gravity varies from 2.25 gm/cc to 2.41 gm/cc but the average has been taken as 2.33 gm/cc.

ANNEXURE VIII/B

STATEMENT SHOWING IN-SITU BULK DENSITY USING PIT ROCK SAMPLES, MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

Pit No	Pit Location		Bulk density
	Longitude (Easting)	Latitude (Northing)	
Pit-01	84° 37' 42.445" E	23° 35' 51.500" N	2.003 g/mL
Pit-02	84° 38' 5.496" E	23° 35' 42.915" N	2.015 g/mL

Pit specification – 1meter x 1meter x 1meter.

ANNEXURE IX/A

STATEMENT SHOWING RESOURCES OF BAUXITE ZONES AT VARIOUS CUT- OFF AS END USE OF BAUXITE RECOMMENDED BY IBM, MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

Cut off Value – Max 7% SiO_2, >=30% Al_2O_3.

BH No	Area (in Sq. meter)	From (meter)	To (meter)	Thickness (meter)	Specific Gravity (gm/cc)	Resource (Tonne)	Resource (million tonne)	Net Reserve (million tonne)	Weighted SiO_2%	Weighted Al_2O_3%	Weighted Fe_2O_3%	Weighted TiO_2%	Weighted LOI%
						Gross Reserve	Gross Reserve						
CMMD-02	298155.997	12.50	15.50	3.00	2.33	2084110.419	2.084	1.250	5.67	33.20	32.88	8.04	19.58
CMMD-03	80464.632	18.00	19.30	1.30	2.33	243727.369	0.244	0.146	5.25	35.69	31.55	7.71	19.13
							2.328	**1.397**	**5.62**	**33.46**	**32.74**	**8.01**	**19.53**

Cut off Value – Max 10% SiO_2, >=30% Al_2O_3.

BH No	Area (in Sq. meter)	From (meter)	To (meter)	Thickness (meter)	Specific Gravity (gm/cc)	Resource (Tonne)	Resource (million tonne)	Net Reserve (million tonne)	Weighted SiO_2%	Weighted Al_2O_3%	Weighted Fe_2O_3%	Weighted TiO_2%	Weighted LOI%
						Gross Reserve	Gross Reserve						
CMMD-02	298155.997	11.50	15.50	4.00	2.33	2778813.892	2.779	1.667	6.42	32.91	32.68	7.91	19.44
CMMD-03	80464.632	18.00	19.30	1.30	2.33	243727.369	0.244	0.146	5.25	35.69	31.55	7.71	19.13
							3.023	**1.814**	**6.33**	**33.14**	**32.59**	**7.90**	**19.41**

Note: The ore resources containing TiO_2 of within Bauxite zone based on TiO_2 content ≥5% remains same as above.

ANNEXURE IX/B

STATEMENT SHOWING RESOURCES OF ALUMINOUS LATERITE ZONES CUT- OFF AS END USE OF LATERITE RECOMMENDED BY IBM, MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

BH No (Polygon No)	Area (in Sq. meter)	From (meter)	To (meter)	Thickness (meter)	Specific Gravity (gm/cc)	Resource (Tonne)	Resource (million	Net Reserve (million tonne)	Weighted SiO_2%	Weighted Al_2O_3%	Weighted Fe_2O_3%	Weighted TiO_2%	Weighted LOI%
						Gross Reserve	Gross Reserve						
CMMD-01	313418.24	1.00	4.00	3.00	2.33	2190793.470	2.1908	1.3145	20.25	35.97	19.92	4.82	18.29
CMMD-01	313418.24	8.50	9.00	0.50	2.33	365132.2451	0.3651	0.2191	8.9	35.95	31.44	7.02	16.02
Sub Total in CMMD-01 Polygon							**2.5559**	**1.5336**	**18.63**	**35.97**	**21.57**	**5.13**	**17.97**
CMMD-02	298156.00	2.50	3.00	0.50	2.33	347351.737	0.3474	0.2084	28.03	28.77	26.56	2.20	13.73
CMMD-02	298156.00	9.50	11.50	2.00	2.33	1389406.946	1.3894	0.8336	11.635	31.24	34.87	5.45	16.09
CMMD-02	298156.00	15.50	16.50	1.00	2.33	694703.473	0.6947	0.4168	9.68	34.42	30.09	6.15	19.03
Sub Total in CMMD-02 Polygon							**2.4315**	**1.4589**	**13.42**	**31.80**	**32.32**	**10.46**	**16.59**
CMMD-03	80464.63	3.00	10.00	6.50	2.33	1218636.847	1.2186	0.7312	23.44	34.59	20.19	4.62	16.37
Sub Total in CMMD-03 Polygon							**1.2186**	**0.7312**	**23.44**	**34.59**	**20.19**	**4.62**	**16.37**
CMMD-04	371560.03	4.00	10.40	6.40	2.33	5540703.094	5.5407	3.3244	20.03	36.02	20.26	7.13	15.86
Sub Total in CMMD-04 Polygon							**5.5407**	**3.3244**	**20.03**	**36.02**	**20.26**	**7.13**	**15.86**
CMMD-05	272993.00	0.30	3.00	2.70	2.33	1717398.963	1.7174	1.0304	31.00	29.36	22.80	2.68	13.35
CMMD-05	272993.00	3.60	5.40	1.80	2.33	1144932.642	1.1449	0.6870	19.38	32.92	26.38	5.51	15.17
CMMD-05	272993.00	6.10	10.30	4.20	2.33	2671509.498	2.6715	1.6029	14.35	37.16	23.60	7.10	17.11
CMMD-05	272993.00	12.70	16.10	3.40	2.33	2162650.546	2.1627	1.2976	8.97	34.03	28.43	7.65	20.19
Sub Total in CMMD-05 Polygon							**7.6965**	**4.6179**	**17.30**	**33.91**	**22.92**	**6.03**	**16.85**
							19.4432	**11.6659 (~11.666)**	**18.15**	**34.56**	**22.99**	**6.69**	**16.65**

ANNEXURE IX/C

STATEMENT SHOWING ORE RESOURCES CONTAINING TiO_2 CONTENT ≥5% IN ALUMINOUS LATERITE ZONES, MADUA PAT-I BLOCK, DIST – LOHARDAGA, STATE – JHARKHAND.

BH No (Polygon No)	Area (in Sq. meter)	From (meter)	To (meter)	Thickness (meter)	Specific Gravity (gm/cc)	Reserves (Tonne)	Reserves (million tonne)	Net Reserve (million tonne)	Weighted SiO2%	Weighted Al2O3%	Weighted Fe2O3%	Weighted TiO2%	Weighted LOI%
						Gross Reserve	Gross Reserve						
CMMD-01	313418.24	2.00	3.00	1.00	2.33	730264.4901	0.7303	0.4382	15.49	38.01	20.40	5.40	20.02
CMMD-01	313418.24	8.50	9.00	0.50	2.33	365132.2451	0.3651	0.2191	8.9	35.95	31.44	7.02	16.02
Sub Total in CMMD-01 Polygon							**1.0954**	**0.6572**	**13.29**	**37.32**	**24.08**	**5.94**	**18.68**
CMMD-02	298156.00	9.50	11.50	2.00	2.33	1389406.946	1.3894	0.8336	11.635	31.24	34.87	5.45	16.09
CMMD-02	298156.00	15.50	16.50	1.00	2.33	694703.473	0.6947	0.4168	9.68	34.42	30.09	6.15	19.03
Sub Total in CMMD-02 Polygon							**2.0841**	**1.2505**	**10.98**	**32.30**	**33.28**	**5.68**	**17.07**
CMMD-03	80464.63	6.10	10.00	6.50	2.33	1218636.847	1.2186	0.7312	17.79	38.25	18.90	5.82	18.54
Sub Total in CMMD-03 Polygon							**1.2186**	**0.7312**	**17.79**	**38.25**	**18.90**	**5.82**	**18.54**
CMMD-04	371560.03	4.00	10.40	6.40	2.33	5540703.094	5.5407	3.3244	20.03	36.02	20.26	7.13	15.86
Sub Total in CMMD-04 Polygon							**5.5407**	**3.3244**	**20.03**	**36.02**	**20.26**	**7.13**	**15.86**
CMMD-05	272993.00	3.60	5.40	1.80	2.33	1144932.642	1.1449	0.6870	19.38	32.92	26.38	5.51	15.17
CMMD-05	272993.00	6.10	10.30	4.20	2.33	2671509.498	2.6715	1.6029	14.35	37.16	23.60	7.10	17.11
CMMD-05	272993.00	12.70	16.10	3.40	2.33	2162650.546	2.1627	1.2976	8.97	34.03	28.43	7.65	20.19
Sub Total in CMMD-05 Polygon							**5.9791**	**3.5875**	**13.37**	**35.21**	**25.88**	**7.00**	**17.85**
							15.918	**9.551**	**15.48**	**35.47**	**24.10**	**7.04**	**17.22**